OPTIMALITY AND UNIQUENESS OF ROMIK'S AMBIDEXTROUS SOFA

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ABSTRACT. The ambidextrous moving sofa problem asks for the largest area of a compact connected planar shape that can be moved around the right-angled corner of a hallway of unit width that turns right, and also, from the same starting orientation, around one that turns left. Romik constructed in 2016 a shape of area $M = 1+4Y^2+\arctan Y = 1.64495\ldots$, where Y is the positive root of $4Y^3+3Y = 1$, and proposed it as a possible solution. We prove that Romik's sofa is optimal: every ambidextrous sofa has area at most M. It is also the only optimal shape: an ambidextrous sofa of area M is a rotated and translated copy of Romik's sofa, and each of the two motions that move it reaches a full quarter turn in the appropriate direction at some time. The proof reduces a sofa to two convex caps, one for each hallway, bounds a score of each cap by $M/2$, and classifies the caps of score $M/2$. It uses one result from the literature, Baek's theorem that Gerver's sofa is optimal for the ordinary problem.

1. INTRODUCTION

1.1. **The ambidextrous moving sofa problem.** Moser's moving sofa problem asks how large a planar shape can be if it can pass around the right-angled corner of a hallway of unit width [8]. The ambidextrous variant requires the same shape to pass both a right-turning and a left-turning corner, starting in a common orientation. Romik constructed an ambidextrous sofa in 2016 [10]. Its area is

$$M = 1 + 4Y^2 + \arctan Y = 1.6449552184\ldots, \qquad 4Y^3 + 3Y = 1, \quad Y > 0. \tag{1.1}$$

1.2. **Optimality and uniqueness.** We prove that this is the largest possible area and that Romik's sofa is the only shape attaining it, up to rotations and translations.

Theorem 1.1. *Let S be an ambidextrous sofa in the sense of Definition 2.3. Then $|S| \le M$, with equality if and only if S is a rotated and translated copy of Romik's sofa S_{R}. Moreover, for every pair of motions (θ_-, c_-) and*

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(θ_+, c_+) witnessing that an ambidextrous sofa has area M, there are times $t_-, t_+ \in [0,1]$ such that

$$\theta_-(t_-) = \theta_-(0) - \frac{\pi}{2}, \qquad \theta_+(t_+) = \theta_+(0) + \frac{\pi}{2}.$$

The motions in this statement are arbitrary continuous rigid motions. They may pause or reverse their rotation, and their final angles are not prescribed. Their initial translations may differ; their initial rotation maps must agree. The last conclusion says that each passage reaches a quarter turn in its appropriate direction, regardless of what the motion does afterwards.

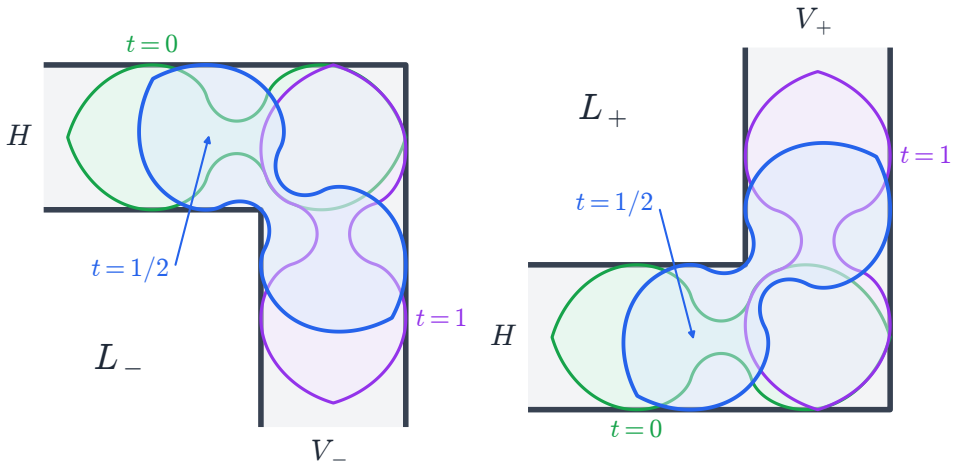


FIGURE 1. Romik's sofa passing a right-turning corner (left) and a left-turning corner (right), from the same placement in the incoming arm. The illustrated motions turn through $-\pi/2$ and $\pi/2$, respectively.

1.3. **Previous work.** Hammersley gave an early construction and the upper bound $2\sqrt{2}$ for the ordinary problem [6]. Gerver constructed a sofa of area $2.2195\ldots$ in 1992 [4], and Baek proved its optimality [1]. Requiring passage through both corners gives a different extremal problem. The ambidextrous variant appears in accounts of geometric problems [2, 11]; Gibbs obtained a numerical shape of area approximately 1.64495 [5] before Romik found its exact description.

Romik derived differential equations for the contacts between a sofa and the hallway walls, and solved them for the ambidextrous construction [10, Theorem 5]. Its boundary is piecewise algebraic, and its area can be expressed by radicals and an inverse tangent. The equations describe the construction within its contact pattern; they do not give an upper bound for all admissible shapes and motions. Kallus and Romik developed computer-assisted upper bounds for the moving sofa problem [7]. O'Keefe gives upper bounds for its

ambidextrous variant [9], and Georgiev, Gómez-Serrano, Tao and Wagner list the ambidextrous problem as Problem 6.63 [3].

We use Baek's theorem only to bound the areas of two auxiliary ordinary sofas that occur in strict exclusions. The form needed here is the rational upper bound 2.2199; Appendix A states this input and explains why it applies to those bodies.

1.4. **The strategy of the optimality proof.** Place the incoming sofa in the strip $\Sigma = \mathbb{R} \times [0, 1]$. Its convex hull determines two downward convex caps, U and V: the upper hull boundary is the roof of U, and reflecting the lower hull boundary in the midline gives the roof of V. Each passage cuts a niche out of that hull. We must bound the exterior roof area left after paying for the middle niche. At a maximum, moving a supporting wall balances those two areas; the balance then constrains the boundary curvature. This gives the following steps.

(1) *Pass from the actual motions to two cap scores.* Unless $|S| \le \sqrt{2}$, each passage supplies an effective turning angle in $(\pi/4, \pi/2]$. Let these angles be α and γ. The intermediate value theorem supplies every intervening hallway frame, even if the passage reverses its rotation. For a cap with projection I and middle half J, define

$$P_\alpha(U) = \int_{I \setminus J} f_U - \int_J N_\alpha(U).$$

Here f_U is its roof and $N_\alpha(U)$ is the lower barrier imposed by the turn. The score counts the roof over the exterior quarters and subtracts the barrier over the middle half. An exact partition of vertical fibers gives

$$|S| \le P_\alpha(U) + P_\gamma(V) \qquad \text{(Section 3)}.$$

(2) *Obtain geometric laws for every maximizing cap.* After a horizontal translation, competitive caps lie in a fixed rectangle, and the score is continuous under Hausdorff convergence. A maximizer therefore exists. Cutting its middle roof to the chord joining the two middle endpoints, then restoring height one, cannot decrease its score. We approximate each prescribed canonical maximizer by polygons selected with a penalty for moving away from it (Section 4). Moving one polygon side compares the exterior roof gained with the inner-wall region lost. Passing these inequalities to the prescribed limit gives curvature bounds and endpoint laws (Section 5).

(3) *Classify the full-turn maximizers.* The variation laws first exclude a tilted middle chord and force the middle roof to be horizontal (Section 6). They then give enough control of the two boundary arcs to identify the cap score with a functional of its support coordinates. A strict quadratic estimate and the equations of its stationary profiles show that this functional is at most $M/2$, with equality only at Romik's support profile (Section 7).

(4) *Exclude equality at every proper turning angle.* If $\alpha < \pi/2$, the final outgoing strip imposes its first inner wall over the whole projection. This wall is part of N_α, even where it rises above the swept niche. A variation bounds the length of the middle interval where this terminal wall is exposed. A weighted estimate for the opposite boundary arc forces that exposed interval to be longer, giving a contradiction (Section 8). Thus $P_\alpha(U) \le M/2$ for all angles under consideration, and equality requires a full turn.

The small-area alternative satisfies $\sqrt{2} < M$. In the other case, steps (1), (3), and (4) give

$$|S| \le P_\alpha(U) + P_\gamma(V) \le \frac{M}{2} + \frac{M}{2} = M. \tag{1.2}$$

The construction and exact area calculation in Appendix B show that the upper bound is attained.

1.5. **The strategy of the uniqueness proof.** Suppose $|S| = M$. Each inequality in (1.2) is then an equality. It follows that both original caps maximize their scores. To identify S, the cap classification must apply to these particular caps. The existence of a favorable maximizer would not suffice: selecting a different maximizer, or replacing a middle roof by its chord, can lose the information about the body that produced it.

This is why step (2) uses a penalty to select a prescribed cap. The variation laws hold for every canonical maximizer, so the strict support-function estimate identifies each one. The equality terms in the chord cut and height restoration then recover the cap before those operations. For a proper angle, the terminal-wall contradiction is likewise applied to the prescribed maximizing pair (α, U). Consequently equality in a cap score holds precisely when its angle is $\pi/2$ and its cap is a horizontal translate of K_{R}.

The two caps of S have the same projection. Their translates of K_{R} therefore align, and their intersection, after reflecting the second cap, is a translate of Romik's hull. The original passage constraints put S inside the corresponding translate of S_{R}. Equal area first gives equality up to a null set. Since S is closed and S_{R} is the closure of its interior, a missing point would leave a nonempty open subset of the interior missing as well, of positive area. Thus the two compact sets are equal (Sections 9 and 10). Applying this argument to any witnessing pair of motions also forces each of its effective angles to be $\pi/2$, proving the angular conclusion of Theorem 1.1.

1.6. **Organization.** Section 2 defines the hallways, passages, caps, and scores. Section 3 proves the reduction from the two actual motions. Sections 4 and 5 obtain the approximation and variation laws used by the analytic proof. Sections 6 and 7 classify full-turn maximizers, and Section 8 proves the universal cap bound and its strictness at proper angles. Section 9 proves optimality and determines both equality caps; Section 10 recovers the

original compact body and proves the quarter-turn conclusion. Section 11 gives further questions.

Appendix A records the ordinary-sofa result used in two exclusions. Appendix B describes Romik's sofa, proves its two explicit passages, and calculates its exact area. Appendix C proves the support-integral identities and contact-energy estimates used for a full turn; Appendix D supplies the low-height exclusion. Appendix E contains the initial-angle, circle, and reflected-support estimates for the terminal tail. Appendix F gives the width and curvature estimates for a horizontal middle, and Appendix G gives the preliminary width and tilt bounds. Finally, Appendix H excludes the remaining tilted configurations.

2. Hallways, motions, and cap scores

We use $|A|$ for two-dimensional Lebesgue measure and R_θ for counterclockwise rotation by θ. Put

$$u(\theta) = (\cos\theta, \sin\theta), \qquad u^\perp(\theta) = (-\sin\theta, \cos\theta), \qquad \rho(x, y) = (x, 1 - y).$$

The reflection ρ preserves the strip $\Sigma = \mathbb{R} \times [0, 1]$.

The strict geometric exclusions will only need a lower bound for the reference score. We obtain it by bracketing the cubic root and using the first two terms of the arctangent series. This lets the same comparison serve every exclusion. The reference root can be enclosed as follows. Let $f(y) = 4y^3 + 3y - 1$, start with $\ell_0 = 1/4$, $u_0 = 1/3$, and put

$$\ell_{j+1} = u_j - \frac{f(u_j)}{f'(\ell_j)}, \qquad u_{j+1} = u_j - \frac{f(u_j)}{f'(u_j)}. \tag{2.1}$$

These are lower and upper Newton enclosures. We retain their defining formulas throughout the estimates.

Lemma 2.1 (Elementary bounds for the reference constant)**.** *The root and constant in Equation (1.1) satisfy*

$$\ell_2 \le Y \le u_2, \qquad \sqrt{2} < M.$$

The formula-generated reference budget

$$\underline{p} = \frac{1 + 4\ell_2^2 + \ell_2 - \ell_2^3/3}{2} \quad \textit{satisfies} \quad \underline{p} < \frac{M}{2}. \tag{2.2}$$

Proof. The polynomial f is strictly increasing, and its signs at $1/4$ and $1/3$ bracket the root. For $0 \le \ell \le Y \le u$, the derivative $f'(y) = 3 + 12y^2$ increases, so

$$f'(\ell)(u - Y) \le f(u) - f(Y) \le f'(u)(u - Y).$$

Dividing by the positive derivatives proves that Equation (2.1) retains the bracket. Both new lower endpoints are positive by substitution. Hence $\ell_2 \le Y$. For $y \ge 0$, $\arctan y \ge y - y^3/3$, since the derivative of $\arctan y - y + y^3/3$ is $y^4/(1 + y^2) \ge 0$. Monotonicity of the square and arctangent on the positive half-line therefore gives $\underline{p} < M/2$: substitution into the two Newton steps

gives $f(\ell_2) < 0$, hence $\ell_2 < Y$. The same substitutions give $\sqrt{2} < 2\underline{p}$, by squaring positive quantities. This proves the remaining area comparison. □

2.1. **The two passages.** The incoming and outgoing arms are

$$H = (-\infty, 1] \times [0, 1],$$
$$V_- = [0, 1] \times (-\infty, 1], \qquad V_+ = [0, 1] \times [0, \infty).$$

The two hallways are $L_- = H \cup V_-$ and $L_+ = H \cup V_+$; they are exchanged by ρ.

Definition 2.2. A passage of S through $L = H \cup V$ consists of continuous maps $\theta : [0, 1] \to \mathbb{R}$ and $c : [0, 1] \to \mathbb{R}^2$ such that

$$R_{\theta(0)}S + c(0) \subset H, \qquad R_{\theta(t)}S + c(t) \subset L \quad (0 \le t \le 1), \qquad R_{\theta(1)}S + c(1) \subset V.$$

Definition 2.3. An ambidextrous sofa is a nonempty compact connected set $S \subset \mathbb{R}^2$ with a passage (θ_-, c_-) through L_- and a passage (θ_+, c_+) through L_+ satisfying $R_{\theta_-(0)} = R_{\theta_+(0)}$.

There is no smoothness assumption on S or on either passage. The continuous angle lifts may differ initially by an integer multiple of 2π. Compactness makes area finite, and connectedness refers to the physical body, including its boundary. Initial translations need not agree: translating a placement farther along the incoming arm does not change its orientation.

2.2. **Caps and their roofs.** For a nonempty compact set K define its support function by

$$h_K(\theta) = \max_{p \in K} \langle p, u(\theta) \rangle.$$

It is unchanged by taking the convex hull. For $K \subset \Sigma$ let

$$K^{\downarrow} = \{(x, y) : 0 \le y \le y' \text{ for some } (x, y') \in K\}.$$

Definition 2.4. A downward cap is a nonempty compact convex set $U \subset \Sigma$ such that $(x, y) \in U$ and $0 \le y' \le y$ imply $(x, y') \in U$.

Its horizontal projection is an interval $I_U = [l_U, r_U]$, its width is $w_U = r_U - l_U$, and its roof is

$$f_U(x) = \max\{y : (x, y) \in U\} \quad (x \in I_U), \qquad f_U(x) = 0 \quad (x \notin I_U).$$

On I_U the roof is concave and takes values in $[0, 1]$. The middle half and exterior quarters are

$$J_U = [l_U + w_U/4, r_U - w_U/4], \qquad I_U \setminus J_U.$$

The cap domain permits width zero, height less than one, flat faces, and nonsmooth roofs. None of the later universal cap statements assumes symmetry.

2.3. **The physical barrier.** Place a copy of L_- in the body's frame with its outer walls on the supporting lines of U with normals $u(\theta)$ and $u^\perp(\theta)$. Its inner walls have heights

$$\begin{aligned} a_\theta(x) &= \frac{h_U(\theta) - 1 - x\cos\theta}{\sin\theta}, \\ b_\theta(x) &= \frac{h_U(\theta+\pi/2) - 1 + x\sin\theta}{\cos\theta} \qquad (0 < \theta < \pi/2). \end{aligned} \tag{2.3}$$

The forbidden open inner corner consists of points below both walls. Its vertical threshold is $c_\theta(x) = \min(a_\theta(x), b_\theta(x))$.

For $0 < \alpha \le \pi/2$ define the swept niche and full partial barrier by

$$n_\alpha(U)(x) = \max\left(0,\ \sup_{0<\theta<\alpha} c_\theta(x)\right), \tag{2.4}$$

$$N_\alpha(U)(x) = \max\big(n_\alpha(U)(x), a_\alpha(x)\big). \tag{2.5}$$

The first-wall formula in Equation (2.3) remains defined at $\alpha = \pi/2$. There $a_{\pi/2}(x) = h_U(\pi/2) - 1 \le 0$, so $N_{\pi/2}(U) = n_{\pi/2}(U)$. At a proper terminal angle the outgoing strip requires $y \ge a_\alpha(x)$ over the entire body, including abscissae where that wall lies above the two-ray niche. This is why the terminal wall appears separately in Equation (2.5).

Lemma 2.5 (Continuity of the barrier)**.** *For every downward cap U, $n_\alpha(U)$ and $N_\alpha(U)$ are continuous functions of x for $\pi/4 \le \alpha \le \pi/2$. They are bounded and integrable on the cap's projection interval.*

Proof. Set

$$\begin{aligned} m_U(x,t) &= \big(\min\{a_t(x), b_t(x)\}\big)_+ \quad (0 < t < \pi/2), \\ m_U(x,0) &= m_U(x,\pi/2) = 0. \end{aligned}$$

where $z_+ = \max(z, 0)$. On the open angular interval this is continuous. At $t = 0$, use

$$0 \le m_U(x,t) \le \left(\frac{h_U(t+\pi/2) - 1 + x\sin t}{\cos t}\right)_+.$$

The right side tends to zero, locally uniformly in x, because $h_U(\pi/2) \le 1$. At $t = \pi/2$ the analogous bound by $(a_t(x))_+$ tends to zero. Thus m_U is continuous on $\mathbb{R} \times [0, \pi/2]$. Continuity at a proper terminal angle also shows that

$$n_\alpha(U)(x) = \max_{0\le t\le\alpha} m_U(x,t).$$

A continuous function's maximum over a compact interval depends continuously on x: uniform continuity on any compact rectangle bounds the difference of the two maxima by the largest difference of the corresponding values. The terminal wall a_α is continuous, since $\sin\alpha > 0$, so taking its maximum with $n_\alpha(U)$ proves the claim for $N_\alpha(U)$. □

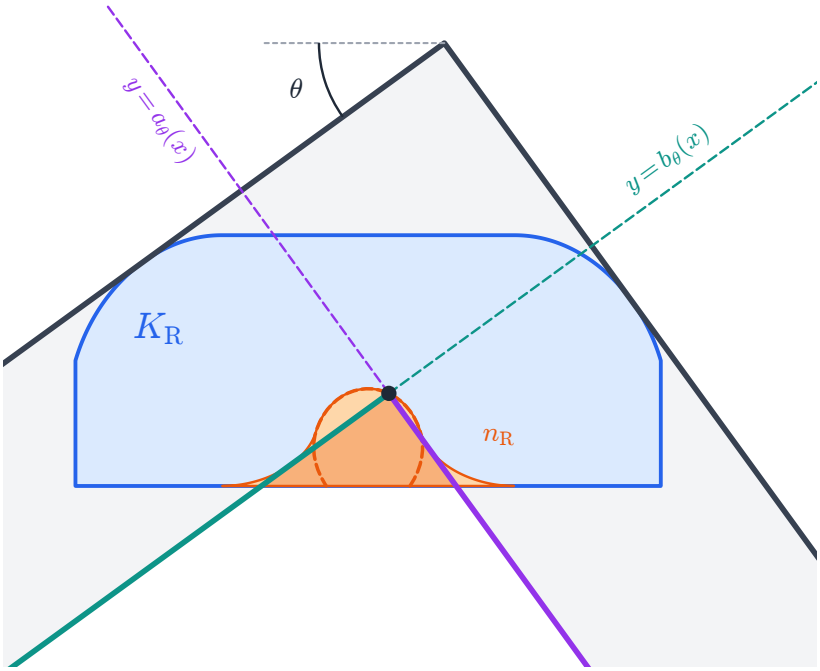


FIGURE 2. The supporting hallway at $\theta = \pi/5$ for Romik's cap. The inner walls meet at its moving corner. The dashed corner path sweeps the niche as the angle varies.

Definition 2.6. The score of a cap at angle α is

$$P_\alpha(U) = \int_{I_U \setminus J_U} f_U(x)\,\mathrm{d}x - \int_{J_U} N_\alpha(U)(x)\,\mathrm{d}x. \tag{2.6}$$

The middle roof contributes no positive term to the score; its support lines still affect the barrier. Both integration domains move when the cap's projection changes. The elementary bound

$$P_\alpha(U) \le w_U/2 \tag{2.7}$$

follows from $f_U \le 1$, $N_\alpha(U) \ge 0$, and $|I_U \setminus J_U| = w_U/2$.

3. From continuous passages to two cap scores

We first extract the angular information needed for the cap argument from the two physical passages. The motions may reverse their direction of rotation, and their terminal angles need not agree. Their initial rotation maps agree, as required in the definition of an ambidextrous sofa. Throughout this section the convex hull and both caps are formed from the original body after one rigid change of coordinates.

Incoming normalization. Write the two placements as

$$g_\pm(t)p = R_{\vartheta_\pm(t)}p + v_\pm(t), \qquad 0 \le t \le 1.$$

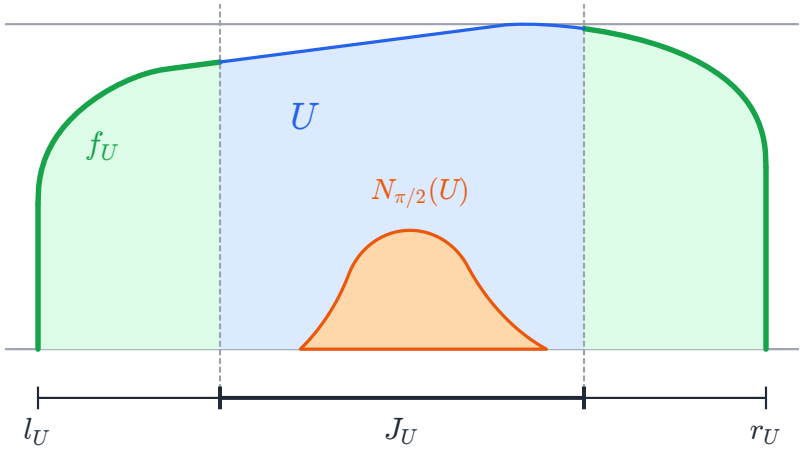


FIGURE 3. The score counts the roof over the two outer quarters (green) and subtracts the barrier over the middle half (orange). The cap drawn here is a generic cap, rather than Romik's.

Set $T = g_-(0)S$. Both motions can be rebased to act on T by composing with $g_-(0)^{-1}$. Their continuous angle lifts can then be chosen as $\vartheta_\pm(t) - \vartheta_\pm(0)$: equality of the initial rotation maps suffices, even if the original lifts differ by an integer multiple of 2π. Thus $T \subseteq \Sigma$, both rebased initial angles are zero, and $|T| = |S|$. The rebased translations are allowed to differ. Compactness and connectedness are preserved by this normalization.

Support tightening at a visited frame. For normals $u(\theta), u^\perp(\theta)$ and offsets A, B, write

$$\begin{aligned} H_\theta(A,B) = \{p : \langle p, u(\theta)\rangle \le A+1,\ \langle p, u^\perp(\theta)\rangle \le B+1, \\ \langle p, u(\theta)\rangle \ge A \text{ or } \langle p, u^\perp(\theta)\rangle \ge B\}. \end{aligned} \tag{3.1}$$

If this hallway contains T, then, for $K = \operatorname{conv} T$,

$$h_K(\theta) - 1 \le A, \qquad h_K(\theta + \pi/2) - 1 \le B.$$

Replace A, B by those support offsets. Both outer walls still contain K, and each inner alternative becomes weaker. Consequently

$$T \subseteq H_\theta(h_K(\theta) - 1, h_K(\theta + \pi/2) - 1). \tag{3.2}$$

This argument retains T; it makes no assertion that the nonconvex hallway contains its entire convex hull. Continuity of h_K makes these tightened placements continuous when the frame angle varies continuously.

Lemma 3.1 (The effective angle)**.** *Let $T \subseteq \Sigma$ be nonempty and compact, with $|T| > \sqrt{2}$. For any normalized passage through L_-, there is $\alpha \in (\pi/4, \pi/2]$ with the following properties: every angle in $[0, \alpha]$ occurs as a body-relative*

hallway frame; T satisfies the support-tightened hallway constraints at every interior angle; and

$$h_K(\alpha) - 1 \le \langle p, u(\alpha)\rangle \quad \textit{for every } p \in T. \tag{3.3}$$

Moreover, either the passage reaches a clockwise quarter turn and $\alpha = \pi/2$, or α is the clockwise magnitude of its actual terminal angle and $\alpha < \pi/2$.

Proof. Let ϑ be the rebased body angle and put $\theta(t) = -\vartheta(t)$, the hallway frame angle, so that $\theta(0) = 0$. First exclude the wrong frame $-\pi/4$. The two outer inequalities have normals $(1,-1)/\sqrt{2}$ and $(1,1)/\sqrt{2}$. For arbitrary offsets A, B, the horizontal section of (3.1) at ordinate y is

$$[m(y), m(y) + \sqrt{2}], \qquad m(y) = \min\{\sqrt{2}A + y, \sqrt{2}B - y\}.$$

Indeed the inner alternatives say that x is at least one of the two displayed affine expressions, and the two outer walls say that x is at most each of those expressions plus $\sqrt{2}$. Intersecting with $0 \le y \le 1$ gives area at most $\sqrt{2}$. The body therefore cannot visit this frame. Continuity also excludes every smaller frame angle: reaching one would first require visiting $-\pi/4$.

The actual outgoing arm at time 1 puts the whole body into a unit strip normal to $u(\theta(1))$. If $-\pi/4 < \theta(1) \le \pi/4$, its intersection with Σ has area $1/\cos\theta(1) \le \sqrt{2}$. To see this directly, a strip $A \le x\cos\theta + y\sin\theta \le A + 1$ has, at each fixed y, a horizontal section of length $1/\cos\theta$. Thus $\theta(1) > \pi/4$.

If the lift reaches $\pi/2$, take $\alpha = \pi/2$. Otherwise continuity forces $\theta(1) < \pi/2$, and we take $\alpha = \theta(1)$. The intermediate value theorem supplies every angle between zero and α in either case. Applying (3.2) at those times proves the interior hallway constraints, with no assumption that the original motion is monotone.

For a proper terminal angle the actual outgoing strip gives (3.3): its lower offset is at least $h_K(\alpha) - 1$. In the full-angle case it follows from the incoming strip alone, because $u(\pi/2) = (0,1)$, $h_K(\pi/2) \le 1$, and $y \ge 0$ on T.

□

Apply the lemma independently to the lower passage and to the reflection of the upper passage across $y = 1/2$. Reflection reverses the lifted body angle and changes L_+ into L_-; it preserves area. We obtain angles

$$\frac{\pi}{4} < \alpha \le \frac{\pi}{2}, \qquad \frac{\pi}{4} < \gamma \le \frac{\pi}{2}. \tag{3.4}$$

If an angle is proper, its terminal wall comes from its own passage. The two passages need not have the same visited times or translations.

The caps of the actual hull. Let $I = [l, r]$ be the horizontal projection of T, and set $w = r - l$. Connectedness makes this projection the whole interval I; taking the convex hull adds no abscissas. Write its vertical fibers as

$$K_x = [B(x), A(x)] \quad (x \in I),$$

and define the downward caps

$$U = K^{\downarrow}, \qquad V = (\rho K)^{\downarrow}. \tag{3.5}$$

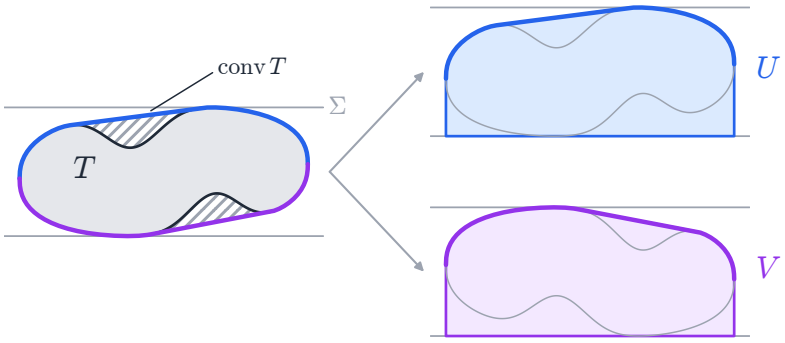


FIGURE 4. The original body T, its convex hull, and the two downward caps U and V. The upper hull boundary is the roof of U; reflecting the lower hull boundary gives the roof of V. The hatched part illustrates area added by convexification.

Their roofs satisfy $f_U = A$ and $f_V = 1 - B$, and both caps have projection I. In particular

$$K = U \cap \rho V. \tag{3.6}$$

The statements hold at the projection endpoints as well as in its interior. The caps are nonempty compact convex subsets of Σ.

Downward filling preserves support in every direction with nonnegative vertical component. Since both $u(\theta)$ and $u^{\perp}(\theta)$ point upwards for $0 < \theta < \pi/2$, the lower constraints of T use exactly the supports of U. The reflected upper constraints use those of V. Denote the resulting barriers by

$$N_U = N_\alpha(U), \qquad N_V = N_\gamma(V).$$

Each barrier includes the positive supremum of the two-wall minima over the entire open angular interval and the terminal first wall

$$a_\alpha(x) = \frac{h_U(\alpha) - 1 - x\cos\alpha}{\sin\alpha}$$

(with V, γ for the other passage). This last wall is redundant at a full quarter turn and is retained at every proper endpoint. Thus every $(x, y) \in T$ satisfies $N_U(x) \le y$ and $N_V(x) \le 1 - y$.

Physical fibers and the exact partition loss. Put $d_U = 1 - f_U$ and $d_V = 1 - f_V$. The actual four-wall envelope is

$$E = \{(x, y) : x \in I, \quad \max\{d_V(x), N_U(x)\} \le y \le 1 - \max\{d_U(x), N_V(x)\}\}. \tag{3.7}$$

We have $T \subseteq E$. Its signed fiber length is

$$\ell(x) = 1 - \max\{d_V(x), N_U(x)\} - \max\{d_U(x), N_V(x)\}. \tag{3.8}$$

For arbitrary wall data, the actual length of that fiber is $\ell(x)_+ = \max\{\ell(x), 0\}$. Here connectedness supplies a point of T at every $x \in I$, and containment

supplies a point of E in the same fiber. Consequently $\ell(x) \geq 0$ everywhere on I, and Cavalieri's formula gives

$$|E| = \int_I \ell(x)\,\mathrm{d}x \geq |T|. \tag{3.9}$$

Compactness gives measurable roofs. The barriers are continuous by Lemma 2.5, so all these functions are measurable and integrable on I.

We charge the inner walls on the middle half and the hull boundary on the exterior quarters. These two regions have the same horizontal length, so their constant unit-strip contributions will cancel. The price of using this division is a sum of positive parts, which records exactly where we charged the smaller of the two competing walls.

Let $J = [l + w/4, r - w/4] = J_U = J_V$ be the middle half of I. Define

$$\begin{aligned} D_{\mathrm{part}} = &\int_J \big[(d_V - N_U)_+ + (d_U - N_V)_+\big]\,\mathrm{d}x \\ &+ \int_{I\setminus J} \big[(N_U - d_V)_+ + (N_V - d_U)_+\big]\,\mathrm{d}x. \end{aligned} \tag{3.10}$$

It is nonnegative. The pointwise identity $\max\{a, b\} - a = (b - a)_+$ yields

$$\begin{cases} 1 - N_U - N_V - \ell = (d_V - N_U)_+ + (d_U - N_V)_+, & x \in J, \\ 1 - d_U - d_V - \ell = (N_U - d_V)_+ + (N_V - d_U)_+, & x \in I \setminus J. \end{cases}$$

Since $|J| = |I \setminus J| = w/2$, the constant terms cancel on integration. Recalling the score

$$P_\alpha(U) = \int_{I\setminus J} f_U\,\mathrm{d}x - \int_J N_U\,\mathrm{d}x,$$

we obtain the exact identity

$$P_\alpha(U) + P_\gamma(V) - |E| = D_{\mathrm{part}}. \tag{3.11}$$

Proposition 3.2 (Original-body score reduction)**.** *Every ambidextrous sofa S satisfies either $|S| \leq \sqrt{2}$, or, after an area-preserving rigid normalization, has the actual caps U, V and independent angles (3.4) with*

$$|S| \leq P_\alpha(U) + P_\gamma(V). \tag{3.12}$$

More precisely, for every real comparison constant c,

$$\begin{aligned} c - |S| = &\left(\frac{c}{2} - P_\alpha(U)\right) + \left(\frac{c}{2} - P_\gamma(V)\right) \\ &+ D_{\mathrm{part}} + |E \setminus T|. \end{aligned} \tag{3.13}$$

Proof. The small-area alternative is immediate. In the other case use the normalization, effective-angle lemma, and actual cap constructions above. The measurable bounded envelope contains T, so $|E \setminus T| = |E| - |T|$. Substitute this into (3.11), then use $|T| = |S|$. Dropping the two nonnegative geometric losses gives (3.12). □

The scalar score in Proposition 3.2 is defined on every downward cap, whether or not that cap arises from a feasible connected body. The signed algebra also holds on that larger domain. The conversion of $\int \ell$ into ordinary area used the actual body's connected projection; it is needed only for the reduction from physical sofas.

Remark 3.3. This distinction cannot be dropped for arbitrary convex hulls. Take $K = [-1,1] \times [0,1]$ and both full angles. At $x = 0$ the two inner walls for $\theta = \pi/4$ have height $2 - \sqrt{2}$. Thus $N_U(0), N_V(0) \geq 2 - \sqrt{2}$, and $\ell(0) \leq 2\sqrt{2} - 3 < 0$. The central envelope fiber is empty. The general area formula is $|E| = \int_I \ell_+ \, \mathrm{d}x$; (3.9) applies because the original connected body certifies that every fiber is nonempty.

4. Selecting polygons near a prescribed maximum

Put $L = \pi/2$. A *full maximum* is a cap maximizing P_L over all downward caps. A *joint maximum* is a pair (U, α) maximizing $P_\alpha(U)$ over all caps and all $\alpha \in [L/2, L]$. We shall construct polygons converging to any prescribed maximum. This quantifier will also be needed in the equality argument.

4.1. **Compactness and continuity.** A narrow cap has too little exterior area. A very wide cap pays too much for the diagonal tent: its reward grows linearly with width, while that tent's area grows quadratically. This confines every competitive cap to a common rectangle and makes compactness available.

Lemma 4.1 (Width bounds)**.** *If $P_\alpha(U) \geq M/2$ and $L/2 \leq \alpha \leq L$, then $M \leq w_U < 6$. After horizontal translation, U lies in $[-3,3] \times [0,1]$.*

Proof. Since $f_U \leq 1$ and $N_\alpha(U) \geq 0$, $P_\alpha(U) \leq w_U/2$. Center the projection and write $w = w_U$. The baseline endpoints give $\sqrt{2}h_U(L/2), \sqrt{2}h_U(3L/2) \geq w/2$. The two walls at $L/2$ therefore give

$$N_\alpha(U)(x) \geq (w/2 - \sqrt{2} - |x|)_+.$$

For $\alpha = L/2$ the terminal first wall dominates this minimum; for larger α it is one of the visited two-wall minima. If $w \geq 6$, the displayed tent is positive on $[-w/4, w/4]$, and

$$P_\alpha(U) \leq \frac{(1+\sqrt{2})w}{2} - \frac{3w^2}{16}.$$

This quadratic decreases for $w \geq 6$. Its value at six is $3\sqrt{2} - 15/4 < 3/4 < M/2$, a contradiction. Centering a cap of width less than six gives the asserted box. □

Lemma 4.2 (Continuity of the score)**.** *On the space of nonempty downward caps in a fixed bounded box, $P_\alpha(U)$ is jointly continuous in Hausdorff distance and in $\alpha \in [L/2, L]$.*

Proof. Take the box $[-B, B] \times [0, 1]$. Its support functions are uniformly bounded and uniformly Lipschitz in angle. For $x \in [-B, B]$,

$$(\min(a_t(x), b_t(x)))_+ \le 2B \min(\tan t, \cot t), \qquad (a_\alpha(x))_+ \le 2B \cot \alpha. \quad (4.1)$$

For example $h_U(t + L) \le B \sin t + \cos t$, so $b_t(x) \le (B + x) \tan t + 1 - \sec t \le 2B \tan t$. The first wall gives the other tail bound and the terminal bound.

If two cap supports differ by at most η, their walls differ by at most $\eta / \sin \delta$ on $[\delta, L - \delta]$. The omitted positive tails are at most $2B \tan \delta$. Taking maxima and minima preserves these estimates. Thus the full niche changes by at most

$$4B \tan \delta + \frac{\eta}{\sin \delta}.$$

For a varying endpoint below L, first restrict to a closed angular interval away from the axes. Uniform continuity there controls both the support change and the change of its upper limit. At $\alpha = L$ the second estimate in (4.1) makes the positive terminal wall tend uniformly to zero. Hence the complete barriers converge uniformly on the horizontal box.

Projection endpoints converge because they are horizontal supports. The roofs converge in L^1: Hausdorff convergence of bounded convex sets gives convergence of their indicators almost everywhere away from the limiting boundary, and integrating the vertical fibers gives the roof assertion. If the limit is degenerate, the approximating convex-set areas tend to zero and the same conclusion follows. The symmetric difference of the moving middle intervals has length tending to zero. Bounded roofs and barriers now give convergence of both integrals defining the score. □

Proposition 4.3 (Attainment)**.** *Full and joint maxima exist. A joint maximum can be chosen with its angle largest among all joint maximizing angles.*

Proof. The reference cap has $P_L(K_{\mathrm{R}}) = M/2$. By Lemma 4.1, every competitor at least this good can be centered in $[-3, 3] \times [0, 1]$. Blaschke's compactness theorem applies to the nonempty compact convex subsets of this box. Their downward closure is preserved under Hausdorff limits: approximate (x, y) by (x_n, y_n) and lower its ordinate to $\min(y', y_n)$ for $0 \le y' \le y$. The score continuity lemma therefore gives a maximum in the box, both with $\alpha = L$ fixed and with $\alpha \in [L/2, L]$ variable. Width bounds remove the box restriction from the conclusion. The joint maximizing level set is compact; maximize its angle coordinate to obtain the final assertion. □

4.2. **Canonicalization and used supports.** The middle roof affects the penalty but earns no area in the score. Lowering it to its endpoint chord therefore helps. Restoring height one raises the exterior roof and the middle barrier by equal amounts over intervals of equal length. These two observations explain the canonical form used in the variations.

Lemma 4.4 (Chord and height reduction)**.** *For any positive-width cap U and fixed $0 < \alpha \le L$, there is a cap $U^\sharp$ with the same projection, height one,*

and an affine roof on J_U, such that $P_\alpha(U^\sharp) \geq P_\alpha(U)$. Applied to a full or joint maximum, the construction preserves its score and its angle.

Proof. Let q join the roof values at the endpoints j_-, j_+ of J_U. Concavity gives $f_U \geq q$ on J_U and $f_U \leq q$ outside it. For the latter assertion, write j_- as a convex combination of $x < j_-$ and j_+ and solve the concavity inequality for $f_U(x)$; the right side follows by exchanging endpoints. In particular $q \geq 0$ throughout the projection. The cap $U^{\mathrm{c}} = U \cap \{y \leq q(x)\}$ therefore has the same projection and exterior roof. Inclusion decreases every support, hence every visited wall and the terminal first wall. Its barrier decreases, so its score increases.

If H is the height of U^{c}, put $\varepsilon = 1 - H$ and $U^\sharp = U^{\mathrm{c}} + [0, \varepsilon]e_y$. Its roof rises by ε. Supports in the two visited families rise by $\varepsilon \sin t$ and $\varepsilon \cos t$; both raw walls, including the terminal wall, rise by exactly ε. The maximum with the floor gives

$$0 \leq N_\alpha(U^\sharp) - N_\alpha(U^{\mathrm{c}}) \leq \varepsilon.$$

The exterior and middle intervals have equal length, so the exterior gain pays this middle penalty. The projection stays fixed, the middle roof stays affine, and the height becomes one. Maximality forces equality of scores when the starting cap is a maximum. □

We call a height-one cap with affine middle roof *canonical*. Write $f_U(x) = mx + c$ on its middle interval and let θ_c be the normal angle of this facet, so $\cot \theta_c = -m$.

Lemma 4.5 (Top localization)**.** *At every positive full or joint maximum the top face meets J_U. For a canonical maximum the higher middle endpoint has height one.*

Proof. Suppose the top face starts at $a > j_+$. Adjoin (j_+, H) and its downward segment, and take the convex hull. A changed upward support must have negative horizontal component, because the old top point (a, H) dominates the new point in all other directions. Its new support is $\langle (j_+, H), n \rangle$; for $x \in J_U$ and $y \geq 0$,

$$\langle (j_+, H) - (x, y), n \rangle - 1 = (j_+ - x)n_x + (H - y)n_y - 1 \leq 0.$$

Thus changed second walls contribute no positive niche on J_U; first walls and the terminal wall are unchanged. The barrier is unchanged, whereas the exterior roof strictly increases on (j_+, a). If the top lies left of J_U, insert (j_-, H). Changed supports then have positive horizontal component and the same inequality makes their first walls, including the terminal wall, nonpositive on J_U. Both cases contradict maximality. □

Lemma 4.6 (Saturation)**.** *For a canonical joint maximum* (U, α)*,*

$$\begin{aligned} U &= U^{\mathrm{sat}} \cap \{y \le mx + c\}, \\ U^{\mathrm{sat}} &= (I_U \times [0,1]) \cap \bigcap_{0 \le t \le \alpha} H_t, \\ H_t &= \{z : \langle z, u(t)\rangle \le h_U(t),\ \langle z, u^{\perp}(t)\rangle \le h_U(t+L)\}. \end{aligned} \tag{4.2}$$

On either unused open normal arc (α, L) *or* $(L+\alpha, \pi)$*, the upper curvature measure is zero except possibly at* θ_c*. Thus the support is harmonic on the unused arcs away from that normal.*

Proof. The saturated cap contains U, retains its baseline, and has the same support in each used direction: its defining inequalities give one bound and inclusion gives the opposite bound. In particular it has the same height and complete barrier. Maximality makes its exterior roof equal to f_U, since a strict increase at an interior exterior point would increase an area integral. Continuity gives equality at $j_{\pm}$; cutting to their chord now proves (4.2). Every boundary point of U^{sat} has an active constraint from the compact family of defining normals. At arclength almost every boundary point its outer normal is unique, so equals an active normal. There is consequently no surface-area measure on unused open arcs. The chord cut only adds its facet normal θ_c. The distributional identity $h'' + h = 0$ on the remaining arcs is the stated harmonicity. □

4.3. **A penalty that retains any chosen maximum.** Polygonal maxima can drift among different limiting caps. To recover equality for a particular body, we must keep the limit at its own cap. A small penalty for changing that cap's support values does this. Every early sample retains a positive weight, while the grid error tends to zero faster than the penalty coefficient.

Fix a full maximum U_*, or a joint maximum (U_*, α) and keep this angle fixed. Choose a horizontal box with unit slack beyond both projection endpoints. For $n = 2^k$ set $t_j = j\alpha/n$ and use the sampled barrier

$$N_{k,\alpha}(V) = \max\{0, \max_{0<j<n} \min(a_{t_j}, b_{t_j}), a_\alpha\}.$$

For $\alpha = L$ the last wall is redundant. Let $P_{k,\alpha}$ be the corresponding score. Sample outer normals on the two source grids and on an independent dyadic grid of the whole upper semicircle:

$$\Gamma_k = \{jL/n : 0 \le j \le 2n\} \cup \{t_j, L + t_j : 0 \le j \le n\}.$$

These sets are nested, contain the axes and top, and have dense union.

Lemma 4.7 (Uniform grid error and circumscription)**.** *On the fixed boxed cap domain,*

$$0 \le P_{k,\alpha}(V) - P_\alpha(V) \le e_k, \qquad e_k \longrightarrow 0.$$

Circumscription by the normals Γ_k *preserves every sampled support, projection, and sampled barrier, and increases the exterior roof. It therefore increases* $P_{k,\alpha}$*.*

Proof. On $[\delta, L-\delta]$ the wall functions have a common modulus of continuity in angle. A mesh of size $\alpha/2^k$ approximates their maximum uniformly there. Equation (4.1) makes the omitted endpoint contributions uniformly small. The terminal wall is already included exactly. First choose δ and then the mesh; this gives uniform barrier approximation. Multiplying its error by the bounded middle length proves the score estimate. The circumscribed intersection contains the original cap and retains each sampled support: an old attaining point remains in it, and the corresponding defining halfplane gives the upper bound. The axis supports preserve the projection, and all barrier supports are sampled. □

For $s \in \Gamma_m$ give its level-m sample weight $\nu_m = 2^{-m-1}/|\Gamma_m|$, and define the persistent penalty

$$D_k(V, U_*) = \sum_{m=0}^{k} \nu_m \sum_{s \in \Gamma_m} (h_V(s) - h_{U_*}(s))^2. \tag{4.3}$$

The total weight is at most one. A normal first sampled at an early level keeps that positive weight at every later level. In particular

$$\nu_m (h_V(s) - h_{U_*}(s))^2 \le D_k(V, U_*) \quad (s \in \Gamma_m,\ m \le k). \tag{4.4}$$

Theorem 4.8 (Selection at a prescribed maximum)**.** *There are levels $k_i \to \infty$, coefficients $\eta_i \downarrow 0$, and grid polygons U_i converging in Hausdorff distance to the specified cap U_*, such that U_i maximizes*

$$P_{k_i,\alpha}(V) - \eta_i D_{k_i}(V, U_*)$$

over every cap in the slack box. No middle affinity or regularity is required of the selected polygons.

Proof. Choose $\eta_i = 1/(i+1)$ and $k_i \ge i$ so that $e_{k_i} \le \eta_i^2$. Compactness and continuity give a boxed maximum of the penalized objective. Circumscription increases its score and preserves its penalty, so a maximizing cap can be taken to be a grid polygon. The circumscription of U_* has zero penalty and sampled score at least $P_\alpha(U_*)$. Hence

$$P_\alpha(U_*) \le P_{k_i,\alpha}(U_i) - \eta_i D_{k_i}(U_i, U_*) \le P_\alpha(U_*) + e_{k_i} - \eta_i D_{k_i}(U_i, U_*),$$

where the second inequality uses the maximality of U_* at its fixed angle. Thus $D_{k_i}(U_i, U_*) \le \eta_i \to 0$. Any Hausdorff cluster cap has, by (4.4), the same support as U_* at every sampled normal. Density gives equality on the upper semicircle. For a downward cap, downward supports are determined by its baseline and horizontal endpoints; the upper supports consequently determine the entire cap. Every cluster cap is U_*, which proves convergence of the whole sequence. Its artificial box sides are inactive for all sufficiently large i. □

If all support values have absolute value at most B, changing sampled supports by at most r changes the penalty by at most $4Br + r^2$. If only one normal changes, multiply this bound by its total persistent weight. These

bounds tend to zero after multiplication by η_i and will account for the error in the polygon variations.

5. Facet variations and limiting source laws

Fix a prescribed canonical full maximum, or a canonical joint maximum (U, α) with $\alpha < L$. Use the selections of Theorem 4.8. Write U_i for a selected polygon, h_i for its support, $A_i = f_{U_i}$, and q_i for its sampled complete barrier. Its projection and middle interval are I_i, J_i. All supports are bounded by B in absolute value, the mesh tends to zero, and the penalty coefficient η_i tends to zero. Moving a side outward adds an exterior strip and moves the corresponding inner wall into the cap. Maximality says that the first area cannot exceed the second. We take this variation for each polygon first, with i fixed; only then do we pass to the prescribed limiting cap.

5.1. Moving a floating side and an endpoint.

For a nonaxis, nontop sampled normal s, let $\ell_{i,s}$ be the outer facet length, $\ell^{\mathrm{w}}_{i,s}$ its part over $I_i \setminus J_i$, and $\ell^{\mathrm{m}}_{i,s}$ its part over J_i. Let $\tau_{i,s}$ be the length of the positive barrier graph over J_i carried by the inner line with this normal. The terminal first line is counted when $\alpha < L$; an unused normal has $\tau_{i,s} = 0$. Distinct source lines have distinct slopes, so their intersection points do not create ambiguity in these lengths.

Lemma 5.1 (Finite facet variation)**.** *With $w_{i,s}$ the total persistent weight at s, put $b_{i,s} = 4B\eta_i w_{i,s}$ and $\beta_i = \sum_s b_{i,s} \le 4B\eta_i$. Then*

$$\ell^{\mathrm{w}}_{i,s} \le \tau_{i,s} + b_{i,s}. \tag{5.1}$$

At a positive vertical end face the outward derivatives of the unpenalized score are $e_{i,R} - C_{i,R}$ and $e_{i,L} - C_{i,L}$, where

$$\begin{gathered} Q_{i,\pm} = A_i(j_{i,\pm}) + q_i(j_{i,\pm}), \\ C_{i,R} = (3Q_{i,+} - Q_{i,-})/4, \qquad C_{i,L} = (3Q_{i,-} - Q_{i,+})/4, \\ e_{i,R} = A_i(r_i), \qquad e_{i,L} = A_i(l_i). \end{gathered} \tag{5.2}$$

For all end faces, including zero ones,

$$e_{i,R} \le (C_{i,R} + b_{i,0})_+, \qquad e_{i,L} \le (C_{i,L} + b_{i,\pi})_+. \tag{5.3}$$

Proof. Move the supporting line at s outwards by $\varepsilon > 0$, keeping all other grid halfplanes. An old attaining point remains for each unchanged support, so those supports stay fixed. For a positive facet the moved support rises by ε for sufficiently small ε. The projection and J_i are unchanged. The gained outer strip has area $\varepsilon\ell^{\mathrm{w}}_{i,s} + O(\varepsilon^2)$. The gained forbidden area has area $\varepsilon\tau_{i,s} + O(\varepsilon^2)$: an exposed inner segment moves by its normal displacement, and finitely many line intersections, floor contacts, and window endpoints contribute quadratic errors. Thus the score derivative is $\ell^{\mathrm{w}}_{i,s} - \tau_{i,s}$. The penalty derivative has absolute value at most $b_{i,s}$. Maximality gives (5.1); at zero facet length it is automatic.

Move a positive right end face from r_i to $r_i+\varepsilon$. Nonaxis supports and $q_i(x)$ stay fixed, whereas the middle endpoints move by $\varepsilon/4, 3\varepsilon/4$. Differentiating the three integrals gives

$$e_{i,R} + \tfrac{1}{4}(A_i + q_i)(j_{i,-}) - \tfrac{3}{4}(A_i + q_i)(j_{i,+}) = e_{i,R} - C_{i,R}.$$

Moving the left face outwards moves the middle endpoints by $-3\varepsilon/4, -\varepsilon/4$ and gives $e_{i,L} - C_{i,L}$. If an end face has height zero, its retained last sloping halfplane and the floor already impose the endpoint. Relaxing the axis constraint then leaves the body unchanged. This proves precisely the positive-part bounds (5.3). □

5.2. **The four neighboring lines.** Only part of an inner wall remains visible after the neighboring hallway positions are imposed. Solving their four line inequalities bounds that visible part by a multiple of the angular spacing. Combined with the side variation, this will bound the outer curvature.

Lemma 5.2 (Neighbor exposure)**.** *At an interior first source t with source spacing δ, put $f = h_i(t)$, $g = h_i(t+L)$, $c = \cos\delta$, $s = \sin\delta$, and $T = \tan(\delta/2)$. Let*

$$\ell^{\mathrm{circ}} = \frac{h_i(t-\delta) + h_i(t+\delta) - 2cf}{s} \geq 0.$$

The full exposed first-wall length satisfies

$$\tau \leq \tan\delta(|q^+|+T)+(2T-\ell^{\mathrm{circ}})_+, \qquad q^+ = f+\frac{h_i(t+L+\delta) - cg}{s} - 1. \tag{5.4}$$

The corresponding second-source bound has the roles of the two normal families interchanged.

Proof. Parameterize the first inner ray by $z(v) = (f-1)u(t) + vu^{\perp}(t)$, $v \leq v_0 = g - 1$. To avoid both neighboring open quadrants, an exposed point must lie in

$$[\min(v_-, v_+), v_0] \;\cup\; [v_{\mathrm{lo}}, v_{\mathrm{hi}}],$$

where solving the four neighboring line inequalities gives

$$v_- = \frac{h_i(t+L-\delta) - 1 - (f-1)s}{c}, \quad v_+ = \frac{h_i(t+L+\delta) - 1 + (f-1)s}{c},$$
$$v_{\mathrm{lo}} = \frac{h_i(t+\delta) - 1 - (f-1)c}{s}, \qquad v_{\mathrm{hi}} = \frac{(f-1)c - h_i(t-\delta) + 1}{s}.$$

Indeed either a neighboring companion alternative holds, putting the point in the first interval, or both neighboring first alternatives must hold, putting it in the second. At zero the neighboring quadrant is below the floor; at a proper terminal endpoint it is dominated by the whole terminal first wall. These neighbors remain valid. Subtracting gives $v_{\mathrm{hi}} - v_{\mathrm{lo}} = 2T - \ell^{\mathrm{circ}}$. If $d_- = (gc - h_i(t+L-\delta))/s$ and $d_+ = (h_i(t+L+\delta) - gc)/s$, then $d_- \leq d_+$ and

$$v_0 - v_- = \tan\delta(f + d_- - 1 + T), \qquad v_0 - v_+ = \tan\delta(-q^+ + T).$$

The first interval therefore has length at most $\tan\delta(|q^+|+T)$. Floor clipping only shortens the exposed pieces, proving the bound. The second-source calculation is the same calculation in the interchanged orthonormal coordinates. □

In a full grid $\ell^{\mathrm{circ}} = \ell_{i,t}$; with additional outer normals $\ell_{i,t} \le \ell^{\mathrm{circ}}$. Uniform support and support-slope bounds imply, for a fixed constant K,

$$\tau_{i,t}, \tau_{i,t+L} \le K\delta, \qquad \ell_{i,s} \le K\delta + b_{i,s} + \ell^{\mathrm{m}}_{i,s} \tag{5.5}$$

at nonterminal source normals. Normals used only for circumscription satisfy $\ell^{\mathrm{w}}_{i,s} \le b_{i,s}$. The terminal first source instead satisfies $\tau_{i,\alpha} \le |J_i|/\sin\alpha$; this mass is retained. There is no terminal companion source, since its two-wall minimum is dominated by the terminal first wall.

Lemma 5.3 (Vanishing middle errors)**.** *On every compact normal arc separated from θ_c and L, $\sum_s \ell^{\mathrm{m}}_{i,s} \to 0$.*

Proof. Concavity and the bound $0 \le A_i \le 1$ give uniform roof-slope bounds near J_U, because this interval stays a positive distance from the projection endpoints. The roofs converge uniformly there. On $[j_- + \varepsilon, j_+ - \varepsilon]$, their one-sided slopes converge uniformly to the middle slope m: bound each derivative between its two secant slopes to points a fixed distance away and then pass to the uniform limit. Thus every middle facet there has normal arbitrarily close to θ_c. The two remaining boundary strips have total arclength at most a fixed constant times $\varepsilon + o(1)$. Let $i \to \infty$ and then $\varepsilon \downarrow 0$. □

Lemma 5.4 (The tilted middle junctions)**.** *If a canonical full or joint maximum has $m \neq 0$, then $A'_-(j_-) > m > A'_+(j_+)$. Its middle facet has projection exactly J_U.*

Proof. At its normal $n_c = u(\theta_c) = (-m, 1)/\sqrt{1+m^2}$ and $x \in J_U$,

$$h_U(\theta_c) - 1 - \langle (x,0), n_c\rangle = A(x)/\sqrt{1+m^2} - 1 \le -\delta_m,$$
$$\delta_m = 1 - (1+m^2)^{-1/2} > 0.$$

There is therefore an open normal arc Ω around n_c on which this deficit is at most $-\delta_m/2$, uniformly for $x \in J_U$. Its walls exclude every positive forbidden point above J_U. Changing only supports in Ω by less than $\delta_m/4$ leaves the complete middle barrier unchanged; this also covers the terminal wall if its normal belongs to Ω.

If, for example, $A'_+(j_+) = m$, concavity makes all upper normal cones at wing points $x \downarrow j_+$ tend to $\{n_c\}$. Choose such a point $p = (x, A(x))$ with $A(x) < 1$ and its normal cone inside Ω, and adjoin $p + \varepsilon e_y$ with its downward segment. Taking the convex hull raises the exterior roof near x. Outside Ω, the positive support gap of p has a positive minimum, so a sufficiently small ε leaves those supports unchanged. Inside Ω the barrier remains unchanged. The score strictly increases, a contradiction. A point below height one exists: for $m < 0$ the roof decreases to the right, and for $m > 0$ a matching right

slope forces $A(j_+) < 1$. The left junction uses the same argument, with the signs reversed. Concavity supplies the nonstrict inequalities initially, so both must be strict. A facet extending outside J_U would violate one of them. □

5.3. **Actual curvature densities and corner velocities.** The upper surface-area measure of a convex cap is $\sigma = h'' + h$ in the distributional sense; for a polygon it is $\sum_s \ell_s \delta_s$. Uniform support convergence gives weak convergence of these measures, since for a smooth test function φ their integrals are $\int h(\varphi'' + \varphi)$. Their total masses are bounded by the perimeter of the fixed box.

Theorem 5.5 (Regular curvature and shadow laws)**.** *For every prescribed canonical full maximum the upper curvature measure has a bounded density away from θ_c, L. For a prescribed proper joint maximum, remove also α; on unused arcs away from θ_c the density is zero. There is no singular continuous curvature, and the only possible interior atoms are at these removed normals.*

Put $F(t) = h_U(t)$, $G(t) = h_U(t + L)$ and, on their regular intervals,

$$p = F' - G + 1, \qquad q = G' + F - 1, \qquad u = F'' + F, \qquad v = G'' + G. \tag{5.6}$$

Then $F, G \in W^{2,\infty}_{\mathrm{loc}}$ there and

$$\begin{gathered} p' = u - 1 - q, \qquad q' = v - 1 + p, \qquad p \le 1, \quad q \ge -1, \\ 0 \le u \le \kappa(q), \qquad 0 \le v \le \kappa(p), \qquad \kappa(z) = \max\{|z|, (1 + |z|)/2\}. \end{gathered} \tag{5.7}$$

At almost every angle where both supports are regular,

$$\begin{aligned} p > 0, q > 0 &\implies u = 0, v \le 1/2, \\ p < 0, q < 0 &\implies v = 0, u \le 1/2. \end{aligned} \tag{5.8}$$

If the inner corner has negative height, $u = v = 0$; at height zero, $u, v < 1$ almost everywhere.

Proof. Sum (5.5) on a compact regular normal interval. The penalty sum tends to zero and the middle sum tends to zero by Lemma 5.3. The result bounds its limiting curvature by K times its angular length. Testing nonnegative continuous functions proves measure domination, hence a bounded density. The remaining finite set of normals can carry only atoms. Saturation supplies the assertion on unused arcs.

For the sharper bound, combine (5.4) with $\ell_{i,t} \le \tau_{i,t} + b_{i,t} + \ell^{\mathrm{m}}_{i,t}$. If $\ell_{i,t} \ge 2T$, the positive part vanishes; otherwise move its $-\ell_{i,t}$ term to the left. Since $\tan \delta = \delta + O(\delta^3)$ and $2T = \delta + O(\delta^3)$, the two cases give

$$\ell_{i,t} \le \delta \kappa(q^+_{i,t}) + O(\delta^2) + b_{i,t} + \ell^{\mathrm{m}}_{i,t}.$$

The constant is uniform in the boxed polygons. Support functions are uniformly semiconvex: $h_i'' \ge -h_i \ge -B$. Uniform convergence therefore makes nearby one-sided support secants converge to the limiting derivative at each differentiability point. Their common bound gives convergence in local L^1. Since κ is Lipschitz, summing the last inequality on an interval gives $\sigma_F \le \kappa(q)\,\mathrm{d}t$. Lebesgue differentiation yields $u \le \kappa(q)$ almost everywhere.

The interchanged source families give $v \le \kappa(p)$. The bounded-density equation $h'' = \rho - h$ gives the stated $W^{2,\infty}$ regularity. Differentiating (5.6) gives the two differential equations. The support points $Fu + F'u^{\perp}$ and $Gu^{\perp} - G'u$ lie in U; comparison with the companion supporting halfplanes gives $F' \le G$ and $-G' \le F$, hence $p \le 1$ and $q \ge -1$.

Here is the additional finite calculation for shadowing. Put $d^{+} = (h_i(t+\delta) - cf)/s$, $d^{-} = (cf - h_i(t-\delta))/s$, and use the corresponding secants $d_{\pm}$ at $t+L$. Let $p_{\pm} = d^{\pm} - g + 1$ and $q_{\pm} = f + d_{\pm} - 1$. In a full grid these are its one-sided velocities; additional outer normals may shorten the circumscribed facets. If $p_{+} > T, q_{+} > T$, its whole first inner ray lies inside the next quadrant: the two thresholds on that ray exceed its endpoint $g-1$ by $p_{+} - T$ and $\tan\delta(q_{+} - T)$. Its exposed first-source length is zero. If also $q_{-} + T \ge \tan\delta(p_{-} + T)$, the previous and next quadrants cover the companion ray outside $[-d_{-} - T, T - d_{+}]$. Its exposed length is at most $(2T - (d_{+} - d_{-}))_{+} \le (2T - \ell_{i,t+L})_{+}$. On a compact subset of $\{p > 0, q > 0\}$ all these inequalities hold eventually. The facet variation then gives

$$\ell_{i,t} \le b_{i,t} + \ell^{\mathrm{m}}_{i,t}, \qquad \ell_{i,t+L} \le T + b_{i,t+L} + \ell^{\mathrm{m}}_{i,t+L}.$$

Sum and pass to the limit as before, using $T/\delta \to 1/2$. Interchanging the local families and reversing the neighboring triple gives the negative-negative assertion. For proper angles this argument takes place inside the visited open source interval; on unused regular arcs the densities vanish already. It does not reflect the partial objective.

Finally set $c = (F-1)u(t) + (G-1)u^{\perp}(t)$, so $c' = pu + qu^{\perp}$. On a compact negative-height corner interval all sampled corners are eventually negative. Their quadrants have no positive exposed source; the facet and middle error bounds therefore force zero limiting curvature. On the level set $c_y = 0$, absolute continuity gives $c'_y = c''_y = 0$ almost everywhere. Put $z = c'_x$, $s = \sin t$, $k = \cos t$. Then

$$p = zk, \quad q = -zs, \quad su + kv = s + k - 2z.$$

The sharp bounds and $\kappa(a) \le |a| + 1/2$ imply $su + kv \le |z| + (s+k)/2$. This excludes $z \le 0$. Nonnegativity gives $z \le (s+k)/2$, hence $|p|, |q| \le (1+\sqrt{2})/4$ and $u, v \le (5+\sqrt{2})/8 < 1$. □

5.4. **Endpoint complementarity and source balance.** An endpoint variation also moves the middle window. The factors $1/4$ and $3/4$ below are its endpoint velocities, obtained by differentiating $j_{-} = l + (r-l)/4$ and $j_{+} = r - (r-l)/4$. Moving a positive end face in either direction balances its height against these window terms. A zero end face permits only the corresponding one-sided inequality.

Define $Q_{\pm} = A(j_{\pm}) + N_{\alpha}(U)(j_{\pm})$ and $C_R = (3Q_{+} - Q_{-})/4$, $C_L = (3Q_{-} - Q_{+})/4$. For a full maximum take $\alpha = L$.

Lemma 5.6 (Endpoint complementarity)**.** *For every prescribed maximum,* $e_R = (C_R)_{+}$ *and* $e_L = (C_L)_{+}$.

Proof. Trim the right end to $r_i - \varepsilon$. The retained roof is unchanged; all supports decrease, so the new barrier is at most the old barrier. Keeping the old barrier as an upper comparison on the new window gives

$$P_{k_i,\alpha}(U_i \cap \{x \le r_i - \varepsilon\}) - P_{k_i,\alpha}(U_i) \ge \varepsilon(C_{i,R} - e_{i,R}) + o(\varepsilon).$$

Concavity bounds the trimming distance by $\varepsilon\sqrt{1 + (w_i - \varepsilon)^{-2}}$: the possible vertical change is at most the horizontal change divided by $w_i - \varepsilon$, using the secant from the opposite endpoint. Eventually $w_i \ge 1$, so this is at most $\sqrt{5}\varepsilon$ for $\varepsilon \le 1/2$. The penalty change is at most $4\sqrt{5}B\eta_i\varepsilon + O(\eta_i\varepsilon^2)$. Maximality gives $e_{i,R} \ge C_{i,R} - 4\sqrt{5}B\eta_i$. The direct left trim gives the same left inequality. Together with (5.3) these give $e_{i,Q} = (C_{i,Q})_+ + o(1)$ for $Q = R, L$.

End heights require a separate convergence argument. Roof slopes over J_i are between -4 and 4 eventually. Facets in a sufficiently short punctured arc of either axis consequently have zero middle length. Equation (5.5) bounds their total length by $K(a + \delta_i) + \beta_i$ on an arc of length a. Weak convergence of the full surface-area measures therefore gives

$$\limsup e_{i,R} \le e_R \le \liminf e_{i,R} + Ka.$$

The first inequality tests the closed axis singleton; the second tests its open a-neighborhood. Let $a \downarrow 0$. The left argument is identical. Barriers and interior roof values converge uniformly, and the middle endpoints converge, so $C_{i,Q} \to C_Q$. Passing to the limit proves the lemma. □

Lemma 5.7 (Positive pressures)**.** *For every canonical full or joint maximum,*

$$A(j_-) > 1/2, \quad A(j_+) > 1/2, \qquad e_R = C_R > 0, \quad e_L = C_L > 0.$$

Proof. The higher middle endpoint has height one by Lemma 4.5. The exact one-angle estimate proved in Lemma D.1 gives $P_\alpha(U) < 4/5$ if the other endpoint is at most $1/2$, contradicting $P_\alpha(U) \ge M/2 > 4/5$. If $Q_+ \ge 1$ and $C_R \le 0$, then $Q_- \ge 3Q_+$; complementarity gives $e_L = C_L \ge 2Q_+ \ge 2$, impossible. If the remaining pressure is nonpositive, then $Q_+ \ge 3Q_-$ and $1 \ge e_R = C_R \ge 2Q_-$, forcing $A(j_-) \le 1/2$, again impossible. Exchanging sides proves the opposite tilt. Both pressures are positive, so complementarity removes their positive parts. □

Theorem 5.8 (The limiting source measures)**.** *Let ω be the actual normal-length measure of the upper boundary over the exterior quarters, omitting the vertical end faces and the horizontal top. The finite source measures $\nu_i = \sum_s \tau_{i,s}\delta_s$ have subsequences converging weakly to ω. For full maxima this measure has bounded densities on both open source quarters. For proper maxima it has bounded densities on $(0, \alpha)$ and $(L, L + \alpha)$ and at most one atom, at α. It has no atom at the companion endpoint $L + \alpha$, the axes, or the top, and no mass on unused open arcs.*

Proof. The mesh bounds and the retained terminal bound give weakly compact finite source measures. Equation (5.1) gives $\omega \le \nu$ for any joint weak limit of source and outer wing measures. To identify the outer limit, test an

angular function φ supported away from the axes and top. Its graph integral is

$$\int_{I_i\setminus J_i} \varphi\big(\arg(-A_i'(x),1)\big)\sqrt{1+A_i'(x)^2}\,\mathrm{d}x.$$

Concave roofs converge uniformly on interior intervals and their derivatives converge almost everywhere. The angular cutoff bounds the integrand even near steep slopes; dominated convergence and shrinking the endpoint strips give the actual wing measure. The coarse source bound excludes extra floating atoms at omitted normals. The terminal first normal is included, so its facet mass is retained in this identification.

For the reverse inequality use right horizontal erosion $E_\varepsilon^R = U_i \cap (U_i - \varepsilon e_x)$. Its roof is $\min(A_i(x), A_i(x+\varepsilon))$ and

$$h_{E_\varepsilon^R}(s) \le h_i(s) - \varepsilon(\cos s)_+.$$

The virtual supports on the right dominate the actual eroded barrier. Only first-source walls move in this comparison, including the whole terminal first wall. Their finite barrier-area derivative is $-\sum_{0<s<L}\tau_{i,s}\cos s$. The exterior roof derivative is $-\sum_{0<s<L}\ell^{\mathrm{w}}_{i,s}\cos s$. Include the end sliver and the moving-window terms to obtain

$$\Delta P_i \ge \varepsilon(D_{i,R} - e_{i,R} + C_{i,R}) + o(\varepsilon), \qquad D_{i,R} = \sum_{0<s<L}(\tau_{i,s} - \ell^{\mathrm{w}}_{i,s})\cos s.$$

For distance control, put $\lambda = \varepsilon/w_i$. For $p \in U_i$, both $(1-\lambda)p + \lambda(l_i, 0)$ and this point plus εe_x lie in U_i by convexity. Thus erosion moves points by at most $(\operatorname{diam} U_i/w_i)\varepsilon$, uniformly bounded by a fixed multiple of ε. Penalized maximality now gives

$$-\beta_i \le D_{i,R} \le e_{i,R} - C_{i,R} + O(\eta_i).$$

Direct left erosion gives the same bound with weight $-\cos s$ on the second quarter. Positive endpoint pressure and complementarity make both limiting defect moments zero. Their weights are strictly positive on their open quarters; at the first terminal atom the weight is $\cos\alpha > 0$. The source mesh estimates exclude atoms at the zero-weight top endpoint. The nonnegative defect $\nu - \omega$ therefore vanishes on both quarters, proving $\nu = \omega$. □

5.5. **The stationary length and projection budgets.** The outer boundary and the exposed inner walls now have the same normal-length measure. Integrating the support distance cancels their common support term, leaving one unit of length per source. Integrating the horizontal projection instead gives the length occupied by the positive barrier. These are the two budgets used below.

Let T_{w} be the length of the height-one top face lying outside J_U, and set $L_{\mathrm{w}} = \omega((0,\pi)) + T_{\mathrm{w}}$. Vertical end faces are excluded from this length.

Corollary 5.9 (Stationary wing identity)**.** *Every canonical full or joint maximum satisfies*

$$2P_\alpha(U) = L_{\mathrm{w}}, \qquad \max_{x\in J_U} N_\alpha(U)(x) \le (e_R + e_L)/2. \tag{5.9}$$

Proof. Put $O = \int_{I\setminus J} A$, $N = \int_J N_\alpha(U)$ and denote barrier endpoint values by $n_\pm$. On an outer graph segment, $h\,\mathrm{d}s = (A - xA')\,\mathrm{d}x$. Integration by parts on the two wings gives

$$2O = \int h\,\mathrm{d}\omega + T_{\mathrm{w}} + re_R - le_L + j_-A_- - j_+A_+.$$

On each finite positive inner graph segment the support distance is $h_i - 1$, so the same calculation gives

$$2\int_{J_i} q_i = \int (h_i - 1)\,\mathrm{d}\nu_i - j_{i,-}q_i(j_{i,-}) + j_{i,+}q_i(j_{i,+}).$$

Uniform convergence and weak source convergence give this identity in the limit. Subtract and use $\nu = \omega$:

$$2P_\alpha(U) = L_{\mathrm{w}} + re_R - le_L + j_-Q_- - j_+Q_+.$$

The last two terms equal $lC_L - rC_R$; pressure equality cancels all boundary terms.

Put $S = A_- + A_+$, $E = e_R + e_L$ and $Z = n_- + n_+$. Top localization makes the wings monotone towards their middle endpoints, so $\int |\cos s|\,\mathrm{d}\omega = S - E$. For each finite barrier, $\mathrm{TV}_{J_i}(q_i) = \int |\cos s|\,\mathrm{d}\nu_i$. Lower semicontinuity of total variation gives $\mathrm{TV}_J(N_\alpha(U)) \le S - E$. Continuity gives $2\max_J N_\alpha(U) - Z \le \mathrm{TV}_J(N_\alpha(U))$, while the endpoint equations give $E = (S + Z)/2$. Combining these proves the height bound. □

Corollary 5.10 (Positive-barrier projection)**.** *Every canonical full or joint maximum satisfies*

$$|\{x \in J_U : N_\alpha(U)(x) > 0\}| \le w_U/2 - T_{\mathrm{w}}. \tag{5.10}$$

Proof. The horizontal projection of a finite source segment of length $\tau_{i,s}$ is $\tau_{i,s}\sin s$. Their interiors are disjoint parts of the barrier graph, so

$$|\{x \in J_i : q_i(x) > 0\}| = \int \sin s\,\mathrm{d}\nu_i.$$

The outer projection budget similarly gives $\int \sin s\,\mathrm{d}\omega_i = w_i/2 - T_{i,\mathrm{w}}$. The limiting measure ω therefore has sine moment $w_U/2 - T_{\mathrm{w}}$, also obtained directly from its actual outer graph. Uniform barrier convergence gives lower semicontinuity of the positive-set length: each compact subset on which the limiting barrier is positive is eventually contained in the finite positive set. Apply inner regularity and $\nu_i \rightharpoonup \omega$. □

These identities use weak limits of finite source lengths. They do not require those lengths to converge to the ordinary arclength of the limiting niche: arbitrarily shallow positive pieces may persist in the source measure after their heights vanish.

6. The full-turn cap problem

We first determine the geometry of a maximizing cap. The support functional in the next section will then identify its value and its shape. Throughout this section $L = \pi/2$ and $P = P_L$.

6.1. **The two charged wings.** Let U be a prescribed canonical full maximum. Translate its projection to $[-2C, 2C]$, so $J_U = [-C, C]$. If the affine middle roof has positive slope, put

$$f_U(-C) = 1 - h, \qquad f_U(C) = 1, \qquad \operatorname{top}(U) = [C, C + T] \times \{1\}.$$

The higher endpoint is one by Lemma 4.5, and the middle facet ends at C by Lemma 5.4. The lower endpoint is greater than $1/2$ by Lemma 5.7. Thus $0 < h < 1/2$. The width and height estimates of Corollary G.4, followed by Corollary C.10, sharpen this to

$$1/2 < C < 37/50, \qquad 0 < h < 1/3. \tag{6.1}$$

The strict width cut applies with $a = 2C$: a cap with $C \geq 37/50$ would have $P < \underline{p}$, whereas a maximum has $P \geq M/2 > \underline{p}$.

We use the actual support of the high wing and the support of the low wing alone:

$$F(t) = h_U(t), \qquad G(t) = \max_{-2C \leq x \leq -C} \{-x \sin t + f_U(x) \cos t\}, \quad 0 \leq t \leq L. \tag{6.2}$$

In this section G omits the middle facet. Its endpoint values and derivatives are

$$\begin{aligned} F(0) = G(L) = 2C, \quad F(L) = 1, \quad G(0) = 1 - h, \\ F'(0) = e_R, \quad F'(L) = -C - T, \quad G'(0) = C, \quad G'(L) = -e_L, \end{aligned} \tag{6.3}$$

where $e_R = f_U(2C)$ and $e_L = f_U(-2C)$. Let

$$\begin{aligned} u = F'' + F, \quad v = G'' + G, \qquad p = F' - G + 1, \quad q = G' + F - 1, \\ z = (F - 1)u(t) + (G - 1)u^{\perp}(t), \\ B = z + pu^{\perp}(t), \qquad D = z - qu(t). \end{aligned} \tag{6.4}$$

Here u and v on the first line are scalar curvature densities; $u(t)$ and $u^{\perp}(t)$ on the second line are the normal vectors defined earlier. To distinguish the niche endpoint value from the corner vector, below we write $z_+ = N_L(U)(C)$. Differentiation gives, almost everywhere,

$$p' = u - 1 - q, \quad q' = v - 1 + p, \qquad B' = (u - 1)u^{\perp}(t), \quad D' = (1 - v)u(t). \tag{6.5}$$

The tangent endpoints are

$$\begin{aligned} B(0) = (2C - 1, e_R), \qquad B(L) = (C + T, 0), \\ D(0) = (-C, -h), \qquad D(L) = (1 - 2C, e_L). \end{aligned} \tag{6.6}$$

Lemma 6.1 (Removing the middle facet)**.** *The supports in Equation* (6.2) *belong to* $W^{2,\infty}(0,L)\cap C^1[0,L]$*. Their curvature densities are the charged wing densities of Theorem 5.5. The positive niche on* J_U *is unchanged if the actual companion support* $h_U(t+L)$ *is replaced by* $G(t)$*. For* $0<t<\delta:=\arctan(h/(2C))$ *one has* $u=v=0$*.*

Proof. Concavity of the roof and the pinned middle facet give

$$h_U(t+L)=\max\{G(t),-C\sin t+\cos t\}.$$

The added support term has inner-wall height $1-\sec t+(x-C)\tan t\le 0$ for $x\in J_U$. Distribute the two-wall minimum over this maximum and take the positive part; this proves the niche assertion angle by angle. The support G starts with the point $(-C,1-h)$ and contains no middle-facet atom. For $t\le\delta$ that point dominates every point of the low wing, by the supporting middle line. The actual companion is instead attained at $(C,1)$, whose wall is nonpositive on J_U. Thus no positive inner graph can carry either source at these angles. The exact source identity of Theorem 5.8 gives $u=v=0$ there. For $t>\delta$, the low wing gives the actual companion support. The regularity and density bounds of Theorem 5.5 therefore apply to G; matching the zero-density initial interval introduces no atom, because the middle-junction inequalities are strict. The traces are the coordinates of the extremal support points, giving Equation (6.3). □

6.2. **Excluding an inclined middle.** The essential estimate is that a first wing with curvature at most one cannot support an inclined maximum. The companion wing may have curvature greater than one and may have tangencies below the floor. The proof uses one joint source measure to keep track of the fraction of a corner assigned to each wall.

Proposition 6.2 (First-wing exclusion)**.** *A canonical full maximum with positive middle slope and* $1/2<C<4/5$*,* $0<h<1/2$ *cannot have* $u\le 1$ *almost everywhere.*

Proof. The contact geometry and the joint source decomposition are proved in Lemmas C.14 and C.15. The two clipped strip moments in Lemma C.16 give $T>0$ and $6CT<3T$ when the companion density is also at most one. If it exceeds one, Lemma C.17 pays for the extra fold and gives the same strict inequality. Since $C>1/2$, both conclusions contradict the required strip moment $6CT$. The common source occupations in that calculation account for measurable ties and later floor crossings. □

The first-wing exclusion reduces rigidity to controlling the initial corner. A dry endpoint gives a common width boundary: triangle payments exclude larger widths, while an energy test handles smaller ones. For small heights, the ordinary-sofa area bound supplies the same control.

Lemma 6.3 (Horizontal middle rigidity)**.** *Every canonical full maximum has middle slope zero.*

Proof. Suppose first that its middle slope is positive. The parameter range is Equation (6.1). For $C \le 2/3$, With the energy-balance height h_0 of (C.9), Lemma C.12 gives $u \le 1$, contradicting Proposition 6.2.

For larger widths, Lemma H.3 gives $N_L(U)(-C) = 0$. The initial radius test fixes a comparison width C_{rad}, defined in (C.10). The triangle payment of Lemma H.4 excludes $C \ge C_{\mathrm{rad}}$ for every allowed height. Below this width, With the energy-balance height h_0 of (C.9), Lemma C.12 gives $u \le 1$ whenever $h \ge h_0$.

For $0 < h \le h_0$, Lemma H.5 shows that failure of either initial-floor energy test contradicts the ordinary one-turn area bound. When both tests hold, Lemma C.11 again gives $u \le 1$. The first-wing exclusion therefore rules out every positive slope. Horizontal reflection preserves the full-angle cap score and changes the sign of the middle slope. Apply the positive-slope conclusion to the reflection to exclude negative slopes. □

The horizontal estimates of Lemma F.1 now place every canonical full maximum in the class with both wing curvatures at most one, top face exactly J_U, and ordered contact switches. We next evaluate the resulting support functional and recover every original equality cap.

7. The support functional and its maximum

Put $L = \pi/2$. This section concerns pairs of C^1 functions on $[0, L]$, without a convexity assumption. For such a pair $X = (F, G)$ let

$$p_X = F' - G + 1, \qquad q_X = G' + F - 1, \qquad p_- := \min(p, 0), \quad q_+ := \max(q, 0),$$

and define

$$\begin{aligned}\mathcal{A}(X) = \frac{1}{2}\int_0^L [F^2 + G^2 - F'^2 - G'^2 + (F-1)^2 + (G-1)^2 \\ + (F-1)G' - (G-1)F' - p_{X,-}^2 - q_{X,+}^2]\,\mathrm{d}t.\end{aligned} \tag{7.1}$$

The admissible traces at half-width a are

$$F(0) = G(L) = a, \qquad F(L) = G(0) = 1. \tag{7.2}$$

The support-area identity and the three niche contact arcs give this functional: their common corner terms cancel, leaving the two squared contact momenta as penalties. We can maximize it on a larger class of profiles than the convex supports. At fixed traces a stationary pair will be the unique maximum; varying the common side trace will then select the reference width.

7.1. **The fixed-trace remainder.** The positive-part squares already have nonnegative convex remainders. The remaining quadratic form has a mixed derivative term. Rotating the pair (ϕ, ψ) through angle $-t/2$ absorbs that term into a derivative square. A Dirichlet estimate then makes the remainder strictly positive for every nonzero variation with fixed traces.

For a variation $V = (\phi, \psi)$ write

$$B_0(V) = \frac{1}{2}\int_0^L [\phi'^2 + \psi'^2 - 2\phi^2 - 2\psi^2 - \phi\psi' + \psi\phi'] \, \mathrm{d}t.$$

The first variation of Equation (7.1) is

$$\begin{aligned} D\mathcal{A}(X)[V] = \int_0^L [&(2F-1)\phi + (2G-1)\psi - F'\phi' - G'\psi' \\ &+ \tfrac{1}{2}(\phi G' + (F-1)\psi' - \psi F' - (G-1)\phi') \\ &- p_{X,-}(\phi' - \psi) - q_{X,+}(\psi' + \phi)] \, \mathrm{d}t. \end{aligned} \tag{7.3}$$

These formulas make sense when the first derivatives are continuous; the stationary pairs constructed below have piecewise continuous second derivatives.

Lemma 7.1 (Dirichlet estimate)**.** *If $f \in C^1[0, L]$ and $f(0) = f(L) = 0$, then*

$$\int_0^L f^2 \le \frac{L^2}{6}\int_0^L f'^2$$

Proof. Cauchy–Schwarz applied on $[0, t]$ and $[t, L]$ gives

$$f(t)^2 \le t\int_0^t f'^2, \qquad f(t)^2 \le (L-t)\int_t^L f'^2.$$

Multiply the first inequality by $L - t$, the second by t, and add. The result is $Lf(t)^2 \le t(L-t)\int_0^L f'^2$. Integration gives the inequality, since $\int_0^L t(L-t)\,\mathrm{d}t = L^3/6$. □

Lemma 7.2 (Coercive remainder)**.** *For a variation V vanishing at all four traces, put $y(t) = e^{-it/2}(\phi(t) + i\psi(t))$. Then*

$$B_0(V) = \frac{1}{2}\int_0^L (|y'|^2 - \tfrac{9}{4}|y|^2)\,\mathrm{d}t \ge \frac{1}{2}\left(1 - \frac{3L^2}{8}\right)\int_0^L |y'|^2\,\mathrm{d}t. \tag{7.4}$$

Equality $B_0(V) = 0$ implies $V = 0$ on $[0, L]$.

Proof. For $z = \phi + i\psi$, $|y'|^2 = |z'|^2 + |z|^2/4 - \mathrm{Im}(\overline{z}z')$. This gives the identity in Equation (7.4). Apply Lemma 7.1 to the two real components of y. The coefficient is positive because $3L^2/8 = 3\pi^2/32 < 1$. If $B_0 = 0$, the derivative integral is zero, so y is constant. Its zero traces imply $y = 0$ and hence $V = 0$. □

Theorem 7.3 (Fixed-trace maximum)**.** *Suppose X satisfies Equation (7.2) and $D\mathcal{A}(X)[V] = 0$ for every zero-trace C^1 variation V. For every pair $\widetilde{X}$ with the same traces,*

$$\mathcal{A}(\widetilde{X}) \le \mathcal{A}(X) - B_0(\widetilde{X} - X). \tag{7.5}$$

Equality of the two functional values implies $\widetilde{X} = X$.

Proof. The function $s \mapsto s_-^2$ is convex and differentiable, with derivative $2s_-$. Its remainder

$$R_-(s,t) = t_-^2 - s_-^2 - 2s_-(t-s)$$

is nonnegative. Define $R_+(s,t)$ similarly with positive parts. Expanding the polynomial terms in Equation (7.1) gives the exact identity

$$\begin{aligned}\mathcal{A}(\widetilde{X}) = \mathcal{A}(X) + D\mathcal{A}(X)[\widetilde{X} - X] - B_0(\widetilde{X} - X) \\ - \tfrac{1}{2}\int_0^L (R_-(p_X, p_{\widetilde{X}}) + R_+(q_X, q_{\widetilde{X}}))\,\mathrm{d}t.\end{aligned}$$

Stationarity and nonnegativity give Equation (7.5). If the values agree, Lemma 7.2 makes the variation zero pointwise. □

The fluxes and forces associated with the first variation are

$$\begin{aligned} J_F &= -F' - (G-1)/2 - p_-, & K_F &= 2F - 1 + G'/2 - q_+, \\ J_G &= -G' + (F-1)/2 - q_+, & K_G &= 2G - 1 - F'/2 + p_-. \end{aligned}$$

Thus $D\mathcal{A}(X)[V] = \int (K_F\phi + K_G\psi + J_F\phi' + J_G\psi')$. If $J_F' = K_F$ and $J_G' = K_G$, integration by parts gives

$$D\mathcal{A}(X)[V] = [J_F\phi + J_G\psi]_0^L. \tag{7.6}$$

The equations can hold separately on finitely many subintervals, provided the fluxes are continuous at their endpoints. For a symmetric pair $G(t) = F(L-t)$ one has $J_G(t) = -J_F(L-t)$ and $K_G(t) = K_F(L-t)$. It therefore suffices to verify the first flux equation.

7.2. **The stationary family.** The stationary profiles come from the Euler equations on the three contact-sign regions. Substitution into $J_F' = K_F$, $J_G' = K_G$ gives

sign region	first equation	second equation
$p, q \geq 0$	$F'' + F = 0$	$G'' + G = 1/2$
$p \leq 0 \leq q$	$2F'' - G' + F = 0$	$2G'' + F' + G = 0$
$p, q \leq 0$	$F'' + F = 1/2$	$G'' + G = 0.$

Choose the first switch at b and impose reflection symmetry about $L/2$. The last switch is then $L - b$. Solving these linear equations and matching values and derivatives gives the coefficients below; the half-angle term belongs to the coupled middle equations.

For $b \in [0, \pi/4]$ put

$$\begin{aligned} \gamma &= b/2 + \pi/8, & k &= \cot\gamma, & a &= \frac{\sin b + 2k\cos b}{3}, \\ E &= \frac{1}{2} - \frac{\cos b}{2} + k\sin b, & R &= \frac{2}{3\sin\gamma}, & S &= \frac{1}{2} - \frac{\cos b}{6} + \frac{k\sin b}{3}. \end{aligned} \tag{7.7}$$

These letters denote functions of b in this subsection. Define

$$F_b(t) = \begin{cases} a\cos t + E\sin t, & 0 \le t \le b, \\ R\cos(t/2+\pi/8) + S\sin t, & b \le t \le L-b, \\ \frac{1}{2} + \frac{1}{2}\sin t + \frac{a}{2}\cos t, & L-b \le t \le L, \end{cases} \qquad G_b(t) = F_b(L-t). \tag{7.8}$$

Lemma 7.4 (Matching and stationarity)**.** *The pair $X_b = (F_b, G_b)$ is C^1, satisfies Equation (7.2) at width $a(b)$, and is stationary. Its contact signs are $p, q \ge 0$ on $[0, b]$, $p \le 0 \le q$ on $[b, L-b]$, and $p, q \le 0$ on $[L-b, L]$.*

Proof. The identities $R\sin\gamma = 2/3$ and $R\cos\gamma = 2k/3$ give value and derivative matching at b by substitution in Equation (7.8). Reflection gives matching at $L-b$. The axis values are $F_b(0) = a$ and $F_b(L) = 1$.

On the first phase, direct differentiation gives

$$p = \tfrac{1}{2} + (E - \tfrac{1}{2})\cos t - \tfrac{3}{2}a\sin t, \qquad q = -1 + \tfrac{3}{2}a\cos t + (E - \tfrac{1}{2})\sin t.$$

Substituting Equation (7.7), these become

$$p = \frac{1-\cos(b-t)}{2} + k\sin(b-t), \qquad q = -1 + k\cos(b-t) + \tfrac{1}{2}\sin(b-t).$$

They are nonnegative: $k \ge 1$, $0 \le b - t \le \pi/4$, and $\cos r + \frac{1}{2}\sin r \ge 1$ for $0 \le r \le \pi/4$. For the last inequality, square the nonnegative sides, or use $\sin r \le \cos r$ and $\sin r \le 1$ in $\cos^2 r + \sin^2 r = 1$. On the middle phase, put $\varphi = t/2 + \pi/8$. Then $p = 1 - 3R\sin\varphi/2$ and $q = 3R\cos\varphi/2 - 1$. Since $\gamma \le \varphi \le \pi/2 - \gamma$, both sine and cosine are at least $\sin\gamma$. This proves the middle signs. Reflection, namely $p(t) = -q(L-t)$, proves the last-phase signs.

The outer phases satisfy the uncoupled Euler equations by their sine and cosine form. In the middle, $G_b = R\sin\varphi + S\cos t$; the half-angle terms cancel in $2F_b'' - G_b' + F_b$, and the full-angle terms cancel separately. Reflection gives the other equation. The proved signs select these Euler equations, and C^1 matching makes the fluxes continuous at the switches. Thus Equation (7.6) gives stationarity. □

Lemma 7.5 (Value of the family)**.** *At the width $a(b)$,*

$$\mathcal{A}(X_b) - a(b) = H(b) := \frac{(2E-1)(a+\sin b) + 3a\cos b + b - 2}{2}. \tag{7.9}$$

Proof. Subtracting the first variation in the direction X from twice the functional cancels its quadratic terms. Write $A_X(t)$ for the density in Equation (7.1) and $D_X(t)$ for that first-variation density. By Equation (7.6),

$$\int_0^L D_X = 2\big(J_F(L)F(L) - J_F(0)F(0)\big).$$

On the first phase $2A_X - D_X$ has primitive

$$V_E(t) = t/2 + (E/2 + a/4 - 1/2)\sin t + (a + 1/4)\cos t.$$

On the last phase its primitive is

$$V_W(t) = t/2 - (E/2 + a/4 - 1/2)\cos t - (a + 1/4)\sin t.$$

On the middle phase $2A_X - D_X = (G' - F')/2$; this follows by substituting $p_- = p$ and $q_+ = q$ and cancelling the quadratic terms. Integrating the three primitives and adding the boundary term, using $F(0) = a$, $F(L) = 1$, gives

$$2\mathcal{A}(X_b) = (2E - 1)(a + \sin b) + 3a\cos b + b - 2 + 2a.$$

This is Equation (7.9). □

Lemma 7.6 (The unique scalar maximum)**.** *The width $a(b)$ is strictly decreasing from $a(0) = \frac{2}{3}\cot(\pi/8)$ to $a(\pi/4) = \sqrt{2}/2$. The function H has a unique maximum at $\beta = \arctan Y$, and $H(\beta) = M/2$.*

Proof. Differentiating k gives $k' = -(1 + k^2)/2$. Hence

$$a' = -\frac{k(k\cos b + 2\sin b)}{3} < 0, \qquad (2E - 1)' = k(2\cos b - k\sin b) > 0.$$

The second sign uses $k\sin b \le \cos b$, which follows from $\gamma \ge b$ and the sine difference formula. The derivative of the score follows from the boundary flux. Differentiate the three phase integrals defining $\mathcal{A}(X_b)$. Their moving-endpoint terms cancel because the profiles and their first derivatives match. The Euler equations remove the interior terms, leaving Equation (7.6). Only $F_b(0) = G_b(L) = a(b)$ vary, and $J_F(0) = -E$, $J_G(L) = E$. Consequently

$$\frac{\mathrm{d}}{\mathrm{d}b}\mathcal{A}(X_b) = 2E(b)a'(b), \qquad H'(b) = (2E(b) - 1)a'(b).$$

The optimal member therefore has endpoint height $E = 1/2$: its area derivative balances the width charge. This gives $2\cot\gamma\sin b = \cos b$. To identify its root, set $y = \tan b$. It gives $\tan\gamma = 2y$. Since $2\gamma = b + \pi/4$, the tangent double-angle formula gives

$$\frac{4y}{1 - 4y^2} = \frac{1 + y}{1 - y}, \qquad 4y^3 + 3y - 1 = 0.$$

The zero exists uniquely because $2E - 1$ is strictly increasing and has opposite signs at 0 and $\pi/4$; at that zero $\tan\gamma = 2y < 1$, so the denominators used above are positive. Thus it is $\beta = \arctan Y$. The signs of a' and $2E - 1$ make H strictly increase before β and strictly decrease after it. At β, the equation for k gives $a = 1/(3\sin\beta)$, and therefore

$$H(\beta) = \frac{\cot\beta + \beta - 2}{2} = \frac{1/Y + \beta - 2}{2} = M/2.$$

The last equality uses $1/Y = 4Y^2 + 3$. □

7.3. **The large-width tail.** Put $\gamma_0 = \pi/8$. For $r \sin\gamma_0 \geq 2/3$, define

$$F_r(t) = r\cos(t/2+\gamma_0) + (1 - r\sin\gamma_0)\sin t, \qquad G_r(t) = F_r(L-t).$$

Its half-width is $a = r\cos\gamma_0$.

Lemma 7.7 (Tail comparison)**.** *The tail pair is stationary at its traces, and*

$$\mathcal{A}(F_r, G_r) - a = 3r\cos\gamma_0 - 1 - 3r^2\sin\gamma_0\cos\gamma_0 < 2/3.$$

Proof. The momenta are $p = 1 - 3r\sin(t/2+\gamma_0)/2 \leq 0$ and $q = 3r\cos(t/2+\gamma_0)/2 - 1 \geq 0$. The middle-phase flux calculation of Lemma 7.4 applies on the whole interval and proves stationarity. Here $2A_X - D_X = (G' - F')/2$ throughout. Integrating this identity and the boundary flux gives $\mathcal{A} = 4r\cos\gamma_0 - 1 - 3r^2\sin\gamma_0\cos\gamma_0$. For $x = r\sin\gamma_0 \geq 2/3$, one has $x - x^2 \leq 2/9$, so the score is at most $\frac{2}{3}\cot\gamma_0 - 1$. Since $\cot(\pi/8) = 1 + \sqrt{2} < 5/2$, this is strictly less than $2/3$. □

Theorem 7.8 (Global calibration)**.** *If $X = (F, G) \in C^1[0, L]^2$ satisfies Equation (7.2) with $a \geq 4/5$, then*

$$\mathcal{A}(X) - a \leq M/2. \tag{7.10}$$

Equality holds precisely for $a = 1/(3\sin\beta)$ and $X = X_\beta$.

Proof. If $a \leq a(0)$, Lemma 7.6 and the intermediate value theorem give a unique $b \in [0, \pi/4]$ with $a(b) = a$, since $\sqrt{2}/2 < 4/5$. Apply Theorem 7.3 to the stationary pair X_b, followed by Lemma 7.6. Equality forces $b = \beta$ and then $X = X_\beta$ pointwise. If $a \geq a(0)$, choose $r = a/\cos\gamma_0$ and apply the same fixed-trace theorem to the tail. Its score is less than $2/3 < M/2$. The common boundary $a = a(0)$ is covered by either construction. □

Corollary 7.9 (Translation mode)**.** *Suppose $F(L) = G(0) = 1$ and $F(0) + G(L) \geq 8/5$. Then*

$$\mathcal{A}(F, G) - \frac{F(0) + G(L)}{2} \leq M/2.$$

Equality determines the reference support pair up to one horizontal translation.

Proof. The transformation $F_t(s) = F(s) + t\cos s$, $G_t(s) = G(s) - t\sin s$ preserves p and q. Expanding the density shows that its change is t times the derivative of

$$\frac{3(F(s)\sin s + G(s)\cos s) - \cos s - \sin s}{2}.$$

Hence $\mathcal{A}(F_t, G_t) = \mathcal{A}(F, G)$ when $F(L) = G(0)$. Choose $t = (G(L) - F(0))/2$; the new side traces both equal $a = (F(0) + G(L))/2$. Apply Theorem 7.8. The centered reference pair is exactly X_β, as substitution in its three support phases shows. Undoing the transformation gives the claimed translation mode. □

7.4. **Recovering every full maximum.**

Theorem 7.10 (Sharp full-angle cap bound)**.** *Every downward cap U satisfies*

$$P_L(U) \le M/2. \tag{7.11}$$

Equality holds if and only if U is a horizontal translate of the reference cap K_{R}.

Proof. By Proposition 4.3, a full maximum exists. Canonicalization in Lemma 4.4 preserves its maximizing value. Its middle is horizontal by Lemma 6.3; hence its roof is one on J_U. The horizontal estimates of Lemma F.1 give, after horizontal translation, support traces $F(0) = G(L) = a = w_U/2 > 1$, $F(L) = G(0) = 1$, both regular curvature densities at most one, top face exactly J_U, and ordered contact switches. The support-area and signed-niche identities of Corollary C.4 therefore give

$$P_L(U) = \mathcal{A}(F, G) - a \le M/2$$

by Theorem 7.8. Since the reference attains $M/2$, this is the global maximum and proves Equation (7.11) for every cap.

The same reasoning applies to any prescribed canonical maximum: Theorem 4.8 and its limiting laws target that cap itself. Equality in Theorem 7.8 forces $a = 1/(3\sin\beta)$ and $(F, G) = X_\beta$. By the reference support formulas, this is the centered reference cap. A downward cap is determined by its supports on the upper semicircle: the cap is the intersection of their supporting halfplanes with $y \ge 0$, and every lower support is already determined by its baseline. Thus every canonical maximum is a translate of K_{R}.

For an arbitrary equality cap, let V be its middle-chord cut and let $W = V + [0, \varepsilon]e_y$ be its height-one extrusion, as in Lemma 4.4. The proved bound makes all inequalities in

$$M/2 = P_L(U) \le P_L(V) \le P_L(W) \le M/2$$

equalities. The canonical classification gives $W = K_{\mathrm{R}} + (b, 0)$. The exact extrusion deficit of Lemma 7.11 then forces $\varepsilon = 0$, since the reference niche is continuous and zero at both endpoints of its middle window. Consequently $V = W$ and its roof is one throughout J_U. Inclusion $V \subset U \subset \mathbb{R} \times [0, 1]$ makes the original roof one there too. Outside J_U the chord cut preserved that roof. Hence $U = V = W$. Translation invariance and the reference value prove the converse. □

Lemma 7.11 (Extrusion deficit)**.** *Let V be a positive-width cap and suppose $W = V + [0, \varepsilon]e_y$ has height at most one. Then*

$$N_L(V) = (N_L(W) - \varepsilon)_+, \qquad P_L(W) - P_L(V) = \int_{J_V} (\varepsilon - N_L(W)(x))_+ \, \mathrm{d}x. \tag{7.12}$$

If $N_L(W)$ is continuous and vanishes at an endpoint of J_V, the last integral is positive whenever $\varepsilon > 0$.

Proof. Vertical Minkowski addition raises the roof by ε and every upward support by ε times its vertical normal coordinate. Both raw inner walls therefore rise by exactly ε. If r is their signed angular supremum before extrusion, then $N_L(V) = r_+$ and $N_L(W) = (r+\varepsilon)_+$. This proves the first identity, including points where r is negative. The exterior roof gain is $\varepsilon|I_V \setminus J_V|$; this equals $\varepsilon|J_V|$. Subtract the niche gain $N_L(W) - (N_L(W) - \varepsilon)_+$ to obtain the second identity. Continuity next to a zero endpoint gives an interval of positive length on which $N_L(W) < \varepsilon/2$, proving strictness. □

8. Partial turns and the terminal wall

Put $L = \pi/2$. In this section the whole terminal wall is retained:

$$N_\alpha(U)(x) = \max\{n_\alpha(U)(x), a_\alpha(x)\},$$
$$n_\alpha(U)(x) = \max\{0, \sup_{0<t<\alpha} \min(a_t(x), b_t(x))\}.$$

The two functions agree at $\alpha = L$, because $a_L \le 0$. Horizontal reflection exchanges the two wall families. At a proper angle it does not preserve the objective with its first terminal wall. All changes of orientation below concern a symmetric finite-angle comparison or the support equations, and keep this distinction.

The contradiction will compare two lengths. Moving the terminal facet inward shows that its strictly exposed middle interval has length at most the facet's horizontal length. Changing the angle bounds that facet length by a geometric reserve w. The companion moment will force strict exposure over an interval longer than w.

8.1. **The remaining geometry of a proper maximum.** Fix a joint maximizing pair (U, α), with $\alpha < L$. The canonical reductions of Lemma 4.4 keep its own angle and score. Center its projection and write

$$I_U = [-2C, 2C], \quad J_U = [-C, C],$$
$$f_U(-C) = 1 - h_L, \quad f_U(C) = 1 - h_R,$$
$$h_L, h_R \ge 0, \quad \min(h_L, h_R) = 0.$$

The saturated description in Lemma 4.6 is used throughout. In particular an unused normal interval has zero charged curvature, apart from a possible uncharged middle facet.

Lemma 8.1 (Angle and width cuts)**.** *For a canonical joint maximum at a proper angle,*

$$\alpha > \arctan(8/3), \qquad 1/2 < C < \frac{37}{50}, \qquad h_L + h_R < \frac{1}{3}.$$

Proof. The reference cap is a competitor, so the maximizing score is at least $M/2$. The initial-angle estimate in Proposition E.1 is strictly below $M/2$ for $\pi/4 \le \alpha \le \arctan(8/3)$. All three angles $\pi/8, \pi/4, 3\pi/8$ have therefore been visited. The finite-angle roof and triangle estimates in Corollary G.4 give $1/2 < C < 4/5$ and $h_L + h_R < 1/3$. They apply to the partial barrier

because it dominates the same three tents. If the high middle endpoint is on the left, reflect the cap and its actual barrier together in this comparison. The endpoint-pressure equations are merely relabeled; the outgoing wall is not changed to the wall of another variational problem. The revised floor-loss calculation in Proposition E.2 then excludes $C \geq 37/50$. □

Lemma 8.2 (Completion of a descending middle)**.** *Suppose $h_R = h > 0$ and put $\theta_0 = L - \arctan(h/(2C))$. If $\alpha \geq \theta_0$ or $h \geq 1 - \sin\alpha$, then $N_\alpha(U) \geq n_L(U)$ on J_U and $P_\alpha(U) \leq P_L(U)$. Otherwise $0 < h < 1 - \sin\alpha$ and $\alpha < \theta_0$.*

Proof. The negative affine middle has its top endpoint at $(-C, 1)$. Saturation gives, on the missing first-normal arc,

$$h_U(t) = \begin{cases} C\cos t + (1-h)\sin t, & \alpha < t < \theta_0, \\ -C\cos t + \sin t, & \theta_0 < t < L. \end{cases}$$

The first case is omitted when $\alpha \geq \theta_0$. No charged extension of the middle facet into an unused normal arc is possible. The second branch has first wall $(-C-x)\cot t + 1 - \csc t \leq 0$ on J_U. On the first branch put $z = C - x \in [0, 2C]$; its wall is

$$A_z(t) = 1 - h + \frac{z\cos t - 1}{\sin t}, \qquad A_z'(t) = \frac{\cos t - z}{\sin^2 t}.$$

For $z \geq \cos\alpha$ it decreases after α, and is at most the retained terminal wall. For $0 \leq z < \cos\alpha$ its maximum on $[\alpha, L]$ is $1 - h - \sqrt{1 - z^2} \leq 1 - h - \sin\alpha$. Under the asserted height condition this is nonpositive. Every missing two-wall minimum is consequently bounded by $\max(0, a_\alpha)$, proving the pointwise comparison and the score inequality. □

Discard the completed descending branch for the moment. In every other case set

$$c = \cos\alpha, \quad s = \sin\alpha, \quad H = 1 - h_R, \quad d = \frac{1 - Hs}{c}, \quad w = c - d. \tag{8.1}$$

Then $s < H \leq 1$, $0 < d < c$ and $0 < w < c/2$. For a horizontal or rising middle $H = 1$; for a descending one these inequalities follow from the residual alternative in Lemma 8.2. Let T_L, T_R be the height-one top overhangs outside J_U. They vanish on a strictly lower middle side. Immediately after α the first support is the point (b, H), where $b = C + T_R$; in the descending case this remains true up to θ_0. The companion support after α is the left endpoint $(-2C, e_L)$, and has no terminal atom. If τ is the horizontal length of the first terminal facet, its endpoints and shifted endpoints are

$$(b, H),\ (b + \tau, H - \tau c/s), \qquad B_+ = (b - c, H - s), \quad B_- = B_+ + (\tau, -\tau c/s). \tag{8.2}$$

In particular

$$a_\alpha(-C + z) = \frac{c}{s}(2C + T_R - d - z), \qquad r_0 = b - d \tag{8.3}$$

is the terminal wall and its zero.

Lemma 8.3 (Top projection)**.** *For the residual geometry above, $T_L+2T_R \le d$ and $T_R < d$.*

Proof. By Corollary 5.10,

$$2C - T_L - T_R \ge |\{x \in J_U : N_\alpha(U)(x) > 0\}| \ge |J_U \cap (-\infty, r_0)|.$$

If $T_R \ge d$, the last interval has length $2C$, forcing $T_L = T_R = 0$, contrary to $T_R \ge d > 0$. Thus $T_R < d$. Also $r_0 > C - c > 0$, and the last interval has length $2C + T_R - d$. Rearrangement proves the result. □

8.2. **One-sided angle variations and the terminal facet.**

Lemma 8.4 (The angle derivative with ties)**.** *Let $E = \{x \in J_U : a_\alpha(x) > n_\alpha(U)(x)\}$ and $Z = \{x \in J_U : a_\alpha(x) = n_\alpha(U)(x)\}$. At a proper angle the right derivative, with the cap fixed, is*

$$\partial_+ P_\alpha(U) = -\frac{1}{s^2}\left[\int_E (x - (B_+)_x)\,\mathrm{d}x + \int_Z (x - (B_+)_x)_+\,\mathrm{d}x\right].$$

At a joint maximum the bracket is nonnegative.

Proof. The right support derivative chooses the upper/left endpoint of the supporting face. Uniformly on J_U, $a_{\alpha+\varepsilon}(x) = a_\alpha(x) + \varepsilon(x - (B_+)_x)/s^2 + o(\varepsilon)$. If $a_\alpha > n_\alpha$, the terminal companion wall is strictly lower, since otherwise its limit from earlier angles would already give $n_\alpha \ge a_\alpha$. Newly visited corners therefore cannot alter the first-order terminal derivative. If $a_\alpha < n_\alpha$, the old history supplies a strict gap, giving derivative zero. On a tie,

$$\max(n_\alpha, a_{\alpha+\varepsilon}) \le N_{\alpha+\varepsilon} \le \max(n_\alpha, \sup_{0\le t\le\varepsilon} a_{\alpha+t});$$

both bounds have derivative $((x - (B_+)_x)/s^2)_+$. The supports have locally bounded one-sided derivatives, and the angle denominators are bounded away from zero. The quotients are uniformly bounded on J_U, so dominated convergence gives the integrated formula. Its sign follows from maximality. □

Lemma 8.5 (A short facet for each prescribed maximum)**.** *Every residual canonical proper-angle joint maximum satisfies $\tau \le w$. No largest-angle choice is required.*

Proof. The terminal corner z_α satisfies

$$(z_\alpha)_x - (B_+)_x = s[1 - (b + 2C)s + (H - e_L)c] < 0.$$

Indeed $b \ge C > 1/2$, $H \le 1$, $e_L \ge 0$, and $s > 12/13, c < 5/13$, so $3Cs - c > 1$. Suppose $\tau > w$. Then $(B_-)_x > r_0$. For $(z_\alpha)_x < x < (B_-)_x$, the companion terminal wall exceeds the first one, while $\partial_- a_\alpha(x) = (x - (B_-)_x)/s^2 < 0$. The companion has no terminal atom. Sufficiently close earlier angles have both walls above $a_\alpha(x)$, and hence $n_\alpha(U)(x) > a_\alpha(x)$ on this interval. For $x \ge (B_-)_x$ the first wall is negative. Thus strict terminal exposure and relevant ties lie to the left of $(z_\alpha)_x < (B_+)_x$. Lemma 8.4 forces $|E| = 0$. Continuity then gives $N_\alpha = n_\alpha$ on J_U.

Choose

$$\max(-C,(z_\alpha)_x) < \zeta < (B_+)_x < r_0 < r_* < \min(C,(B_-)_x).$$

Immediately after α the same cap has first support (b,H), so $\partial_t a_t(x) = (x - b + \cos t)/\sin^2 t$. For $x \le \zeta$ its first walls decrease for a small angle increment. On $[\zeta, r_*]$ the old history has a uniformly positive gap above a_α; uniform convergence covers the new walls there. The new zero $b - (1 - H\sin t)/\cos t$ remains below r_*, so the new first walls are nonpositive to its right. Choose the increment below $\theta_0 - \alpha$ when the middle descends. For some $\beta > \alpha$, all new first walls are consequently below n_α on J_U. Their two-wall minima are below the same bound. The old history remains present, and $N_\beta = N_\alpha$, so (U,β) is another joint maximum.

The cap and its window have remained fixed. Its saturation persists: $U \subseteq U_\beta^{\mathrm{sat}} \cap \{y \le \ell_U\} \subseteq U_\alpha^{\mathrm{sat}} \cap \{y \le \ell_U\} = U$. The original terminal facet has positive charged length and normal $\alpha \in (0,\beta)$, away from the middle normal. Theorem 5.5 for this prescribed new maximizing pair excludes such an interior charged atom, a contradiction. □

Lemma 8.6 (Strict terminal exposure)**.** *For a canonical proper-angle maximum, $|E| \le \tau$.*

Proof. Move its terminal supporting line inward by ε. The endpoint baseline points remain present, so the projection and the window do not change. The exterior area lost to first order is $\varepsilon\tau/s$: the charged facet has length τ/s. At every strict terminal contact the barrier drops to first order by ε/s. Every other used support can only decrease, which can only improve the score. Restrict first to a compact subset of E with a positive contact gap and then exhaust E. The inward derivative of the score is at least $(|E| - \tau)/s$. Maximality makes it nonpositive, proving the inequality. At a zero-length facet the exterior loss is $o(\varepsilon)$ and gives the same conclusion. □

8.3. **A weighted companion comparison.** For a rising middle use the left charged-wing support G alone. The omitted right-wing points have $X \ge C, Y \le 1$, and their companion walls on J_U are at most $1 - \sec t + (x - C)\tan t \le 0$. The middle segment is controlled by its endpoints. This replacement therefore preserves positive niche contacts. For the other signs use the actual companion support. In all cases $G(0) = 1 - h_L$, $G'(0+) = C + T_L$ and $v = G'' + G \ge 0$. Put

$$\begin{aligned} A_0 = C - T_L, \quad \theta = \arcsin A_0, \quad \lambda = \tan\theta, \\ \mathcal{E}_\theta = \int_0^\theta (v(t) - 1)_+ (\lambda\cos t - \sin t)\,\mathrm{d}t. \end{aligned} \tag{8.4}$$

Lemma 8.7 (Localized companion moment)**.** *For $0 \le z \le w$,*

$$n_\alpha(U)(-C + z) \le \sqrt{1 - (C - T_L)^2} - \sqrt{1 - (C + z)^2} + \mathcal{E}_\theta.$$

Proof. Write the shifted companion support point as $D = (G-1)u^{\perp} - G'u$. Then

$$D'_x = (1-v)\cos t, \quad D'_y = (1-v)\sin t, \quad b'_t(x) = \frac{x - D_x(t)}{\cos^2 t}.$$

Its outer abscissa is $D_x - \sin t \geq -2C$. Also $0 < A_0 < C + w < C + c/2 < s$; the final inequality follows from $C < 37/50, c < 3/\sqrt{73}$ by squaring $37/50 + 3/(2\sqrt{73}) < 8/\sqrt{73}$. A positive maximum of the companion-wall family on $[0, \alpha]$ is interior: its value at zero is $-h_L$, and its derivative is negative once $\sin t > C + z$. At a maximizing angle $D_x = -C + z$, $b_t = D_y$ and $t \leq \arcsin(C + z)$. Absolute continuity of the regular support gives

$$D_y(t) + h_L - \lambda(T_L + z) = \int_0^t (1 - v(r))(\sin r - \lambda \cos r)\,\mathrm{d}r.$$

Before θ its integrand is bounded above by the weighted excess. After θ it is at most $\sin r - \lambda \cos r$ because $v \geq 0$. For $t \geq \theta$, integration bounds $D_y + h_L$ by

$$\sqrt{1 - A_0^2} - \cos t + \lambda(C + z - \sin t) + \mathcal{E}_\theta.$$

The part preceding the excess increases with $t \geq \theta$, and its maximum at $t = \arcsin(C + z)$ is the asserted circle difference. For $t < \theta$, use instead $D_y + h_L \leq \lambda(T_L + z) + \mathcal{E}_\theta$ and

$$\lambda(T_L + z) \leq \int_{A_0}^{C+z} \frac{x}{\sqrt{1 - x^2}}\,\mathrm{d}x.$$

Finally the niche is bounded by the positive part of the largest companion wall. The asserted right side is nonnegative, and discarding $-h_L$ proves the result even when there is no positive wall maximum. □

Lemma 8.8 (The terminal contradiction)**.** *No canonical joint maximum has proper angle and residual tail geometry.*

Proof. The circle comparison and reflected-excess estimate in Propositions E.3 and E.6 give, throughout the reserve interval,

$$a_\alpha(-C + z) - \left[\sqrt{1 - (C - T_L)^2} - \sqrt{1 - (C + z)^2}\right] > \mathcal{E}_\theta \qquad (0 \leq z \leq w).$$

Thus the terminal wall stays above the circle envelope even after paying for the companion's curvature excess. Together with Lemma 8.7, this makes the entire interval $[-C, -C + w]$ strictly terminal-exposed. Its right endpoint is interior to J_U, so continuity extends exposure a positive distance farther right. Thus $|E| > w \geq \tau$, contradicting Lemma 8.6. □

8.4. **The sharp theorem for every cap and its own angle.**

Theorem 8.9 (Sharp cap theorem)**.** *Every downward cap U and every $\alpha \in [\pi/4, L]$ satisfy*

$$P_\alpha(U) \leq M/2.$$

Equality holds if and only if $\alpha = L$ and $U = K_{\mathrm{R}} + (t, 0)$ for some $t \in \mathbb{R}$.

Proof. Choose a joint maximum with largest terminal angle, whose existence is given by compactness. Canonicalization retains that angle. If it is proper, Lemma 8.1 applies. In a completed descending branch, Lemma 8.2 gives $P_\alpha(U) \le P_L(U) \le M/2$ by Theorem 7.10. Since the reference is a competitor, both inequalities are equalities and the same cap at L is a joint maximum, contrary to the largest-angle choice. Every other proper branch is excluded by Lemma 8.8. Thus the joint value is the full-angle value $M/2$, proving the universal bound.

Now let the *prescribed* original pair (U, α) have score $M/2$. The bound makes this pair a joint maximum. Its canonical replacement keeps its own angle and score. If its angle is proper, the completed descending branch would make that replacement a full maximum. Full-angle affine rigidity forces its middle slope to vanish, contradicting descent. Every residual branch is excluded by the same terminal contradiction; Lemma 8.5 applies to this individual pair, without any largest-angle selection. Hence its angle is L. The original full-angle equality classification in Theorem 7.10 then identifies U itself as a horizontal reference translate. The converse follows from the reference full score and horizontal translation invariance. □

9. Optimality and the equality caps

The cap theorem now applies separately to the two caps of the original body. Its equality statement also forces both effective angles to be full quarter turns. The following section uses these caps to recover the hull and the body itself, including every boundary point.

Theorem 9.1 (Optimality)**.** *Every ambidextrous sofa has area at most M. The value M is attained by S_{R}.*

Proof. The small-area branch of Proposition 3.2 is strictly below M by Lemma 2.1. In the remaining branch, the reduction gives

$$|S| \le P_\alpha(U) + P_\gamma(V).$$

Both caps satisfy Theorem 8.9, so each score is at most $M/2$. The reference construction in Theorem B.4 attains their sum. □

Proposition 9.2 (The two original equality caps)**.** *If an ambidextrous sofa has area M, then in the normalization of Section 3 its two caps satisfy*

$$\alpha = \gamma = \frac{\pi}{2}, \qquad U = K_{\mathrm{R}} + (a, 0), \qquad V = K_{\mathrm{R}} + (b, 0) \quad \text{for some } a, b \in \mathbb{R}.$$

All four terms in the deficit identity (3.13) vanish for $c = M$.

Proof. Suppose $|S| = M$. In the normalization of Section 3, $|T| = M > \sqrt{2}$. Apply the exact deficit (3.13) with $c = M$. Every term on its right-hand side is nonnegative: two by the scalar cap theorem, one by the positive-part formula (3.10), and one because it is the area of a set. Hence all four vanish:

$$P_\alpha(U) = P_\gamma(V) = \frac{M}{2}, \qquad D_{\mathrm{part}} = 0, \qquad |E \setminus T| = 0. \tag{9.1}$$

In particular, Theorem 8.9 gives

$$\alpha = \gamma = \frac{\pi}{2}, \qquad U = K_{\mathrm{R}} + (a, 0), \qquad V = K_{\mathrm{R}} + (b, 0) \quad \text{for some } a, b \in \mathbb{R}. \tag{9.2}$$

These are the caps defined from $\operatorname{conv} T$ in (3.5).

The zero partition defect also gives, almost everywhere,

$$\begin{aligned} d_V \le N_\alpha(U), \quad d_U \le N_\gamma(V), \quad x \in J, \\ N_\alpha(U) \le d_V, \quad N_\gamma(V) \le d_U, \quad x \in I \setminus J. \end{aligned}$$

The vanishing filling defect says that the physical envelope differs from T by a null set. Exact recovery requires the additional topological argument below.

□

10. Recovering the body and its quarter turns

The caps in Proposition 9.2 determine the convex hull. The constraints of the two original passages then put the original body inside Romik's sofa. We finally use closedness to recover its boundary as well as its area.

Lemma 10.1 (Closed subsets of a regular closed set)**.** *Let $F \subseteq \mathbb{R}^2$ be compact and regular closed, meaning $F = \operatorname{cl}(\operatorname{int} F)$. If a closed set T satisfies $T \subseteq F$ and $|T| = |F|$, then $T = F$.*

Proof. Finite equal area gives $|F \setminus T| = 0$. The set $\operatorname{int} F \setminus T$ is open and has area zero, so it is empty: every nonempty open subset of the plane contains a disk of positive area. Thus $\operatorname{int} F \subseteq T$. Closedness of T gives $F = \operatorname{cl}(\operatorname{int} F) \subseteq T$. □

Theorem 10.2 (Exact uniqueness)**.** *For an ambidextrous sofa S,*

$$|S| = M \quad \Longleftrightarrow \quad S = R_\theta S_{\mathrm{R}} + v \quad \textit{for some } \theta \in \mathbb{R},\ v \in \mathbb{R}^2.$$

The equalities are equalities of compact sets.

Proof. Assume $|S| = M$ and use (9.2). Both actual caps have the same horizontal projection I. Comparing the right endpoints of their translated reference projections gives $r_{K_{\mathrm{R}}} + a = r_{K_{\mathrm{R}}} + b$, hence $a = b$. Reflection commutes with horizontal translation. The hull identity (3.6) therefore gives

$$K = U \cap \rho V = (K_{\mathrm{R}} \cap \rho K_{\mathrm{R}}) + (a, 0) = \hat{K}_{\mathrm{R}} + (a, 0). \tag{10.1}$$

The final equality is the reference hull construction in Theorem B.4.

Translate horizontally by $-a$. Every point of $T - (a, 0)$ is in $\hat{K}_{\mathrm{R}}$. The full-angle passage constraints also impose

$$n_{\mathrm{R}}(x) \le y \le 1 - n_{\mathrm{R}}(x).$$

Indeed the full-angle barriers of the translated caps are the translated reference niche. By the definition of S_{R} this proves

$$T \subseteq S_{\mathrm{R}} + (a, 0). \tag{10.2}$$

Since T is compact, it is closed. The reference body is regular closed by Theorem B.4, and both sets in (10.2) have area M. Thus Lemma 10.1 gives $T = S_{\mathrm{R}} + (a, 0)$. If the normalization was $T = R_\eta S + v_0$, inversion gives the explicit equality

$$S = R_{-\eta} S_{\mathrm{R}} + R_{-\eta}\big((a, 0) - v_0\big).$$

Conversely, every rotated and translated copy of S_{R} is ambidextrous: compose its two reference passages with the inverse rigid map. Rigid maps preserve area, so every such copy has area M. □

Theorem 10.3 (Quarter turns in every witnessing pair)**.** *Let S be compact and connected, and let (ϑ_-, v_-) and (ϑ_+, v_+) be any pair of passages through L_- and L_+ with the same initial rotation map. If $|S| = M$, then there are $t_-, t_+ \in [0, 1]$ such that*

$$\vartheta_-(t_-) = \vartheta_-(0) - \frac{\pi}{2}, \qquad \vartheta_+(t_+) = \vartheta_+(0) + \frac{\pi}{2}.$$

Proof. Apply the effective-angle construction to this particular pair of passages, rebasing each lift at its own initial angle. Equality of area forces both scalar scores to equal $M/2$, so Theorem 8.9 forces $\alpha = \gamma = \pi/2$. For each motion the proper-terminal alternative in Lemma 3.1 has angle strictly below $\pi/2$ and is therefore excluded. Each original passage reaches its correctly handed quarter turn. □

The theorem concerns angles reached during a passage. A witnessing motion may continue rotating afterwards, reverse direction, or pause; its terminal angle and its time parametrization are unrestricted by this conclusion. Together, Theorems 9.1, 10.2 and 10.3 prove Theorem 1.1.

11. Further questions

The theorem determines the maximizing body and the angular reach of its passages. It leaves several questions about the proof and nearby geometric problems.

A quantitative equality theorem. If an ambidextrous sofa has area at least $M - \varepsilon$, how close must it be, after a rigid motion, to S_{R}? The exact partition deficits and the coercive support-function remainder are possible starting points. A Hausdorff estimate would also have to control the geometry of nearly maximizing caps before they satisfy the exact maximizer source laws. We do not claim such an estimate here.

The ordinary-sofa input. Two strict exclusions use the ordinary-sofa bound 2.2199 from Baek's theorem. A direct ambidextrous estimate strong enough to replace those uses could shorten the dependency of the proof. Removing that input requires estimates for the same genuinely feasible auxiliary bodies; replacing them by arbitrary envelopes would not justify the ordinary-sofa theorem.

Passage paths. The quarter-turn theorem does not classify all motions of the optimal body. A classification would need to account for waiting, time reparametrizations, translations in straight arms, and any additional angular excursions. The explicit paths in Equation (B.7) prove existence but do not provide that classification.

Other corridors. The common incoming orientation, equal widths, and right angles are part of the problem proved here. Unequal hallway widths and other corner angles change both the supporting-wall formulas and the terminal constraints. The cap method supplies a framework for those questions, but the sharp calibration and equality shape would need to be determined again.

For a hallway width $d > 0$, apply $p \mapsto p/d$ to the body and both passages. The normalized hallways have unit width and area is divided by d^2. The largest ambidextrous area is therefore $d^2 M$, attained exactly by the scaled rigid images of S_{R}.

Variation methods with moving contact sets. The polygon approximation, curvature estimates, and graph-area identities used here may apply to other extremal problems in which contact regions change with the shape. A general formulation would need to retain the moving integration boundaries and the shared occupation of the two inner walls at a corner.

Appendix A. The ordinary-sofa bound used in the proof

An ordinary moving sofa is a nonempty compact connected planar set with a passage through L_- as in Definition 2.2. Baek's optimality theorem for Gerver's sofa supplies the following bound.

Theorem A.1 (Baek [1])**.** *Every ordinary moving sofa A satisfies*

$$|A| \le |S_{\mathrm{G}}| \le 2.2199,$$

where S_{G} is Gerver's sofa.

We use this theorem as an area obstruction. Each application first constructs a connected survivor with an explicit ordinary passage, then shows that its area would exceed the bound. The exact value of $|S_{\mathrm{G}}|$ is immaterial here. We use the rational relaxation $G_0 = 2.2199$ in two strict estimates for full-turn caps. No classification of ordinary equality cases enters the argument.

A.1. **Why a cap survivor is an ordinary sofa.** For a height-one downward cap U, let

$$A_U = \{(x, y) \in U : n_{\pi/2}(U)(x) \le y\}.$$

An arbitrary cap need not leave a connected survivor: its niche can reach above its roof. The proof therefore establishes the following geometric condition before applying the ordinary bound.

Lemma A.2. *Suppose U is a downward cap of height one and*

$$n_{\pi/2}(U)(x) \le f_U(x) \quad (x \in I_U).$$

Then A_U is a nonempty compact connected ordinary moving sofa. Its area is

$$|A_U| = |U| - \int_{I_U} n_{\pi/2}(U)(x)\,\mathrm{d}x.$$

Proof. The niche is continuous by Lemma 2.5, so the survivor is closed in U and compact. Each of its fibers is the nonempty interval $[n_{\pi/2}(U)(x), f_U(x)]$. The concave roof is continuous on the interior of I_U, and is continuous at its endpoints as well: concavity gives the lower limit there, while compactness gives the upper limit. Its graph is consequently connected. Every point of A_U joins this graph by a vertical segment in A_U, proving connectedness. The area formula follows by integrating these fiber lengths.

Translate U horizontally so that its right endpoint is 1. Write $F(t) = h_U(t)$ and $G(t) = h_U(t + \pi/2)$, and use

$$p \longmapsto R_{-t}p + (1 - F(t), 1 - G(t)), \qquad 0 \le t \le \pi/2.$$

The outer walls contain U by definition of the support function. At every interior angle, the inequality $y \ge n_{\pi/2}(U)(x)$ places every survivor point on or above at least one of the two inner walls. Thus the placement lies in L_-. At $t = 0$, $F(0) = G(0) = 1$, so the initial placement is the identity and lies in H. At $t = \pi/2$, its first coordinate is y, in $[0, 1]$, and its second is $1 - h_U(\pi) - x \le 1$. Hence it lies in V_-. The support function is continuous, so this is a continuous passage. □

The full-turn estimates verify the displayed roof inequality for the caps at which G_0 is used. They retain the complete niche, including any part outside the middle half. This distinguishes the area in Lemma A.2 from the spatial score $P_{\pi/2}$, which charges only the exterior roof and the middle barrier. The proof converts between them by explicit exterior and niche terms before applying Theorem A.1.

Appendix B. Romik's sofa

We describe Romik's shape [10, Theorem 5] with its initial contact point at $(1, 1/2)$. This fixes the coordinates of the cap K_{R}, hull $\hat{K}_{\mathrm{R}}$, and body S_{R} used throughout the proof.

B.1. **The cubic and support coordinates.** The function $4y^3 + 3y$ is strictly increasing, is zero at $y = 0$, and exceeds one at $y = 1$. Thus it has a unique positive root Y; substitution gives $2/7 < Y < 3/10$. Set

$$\begin{aligned} \beta &= \arctan Y, & L &= \pi/2, \\ A &= \frac{1}{4\sin\beta}, & k &= 1 - \frac{4}{3}A, \\ T &= \frac{3}{2}(\pi/4 - \beta), & R &= \frac{\cos\beta}{\cos T}. \end{aligned} \tag{B.1}$$

In particular $0 < \beta < \pi/4$ and $\cos T > 0$. On $[0, L]$ define the two support coordinates

$$F(t) = \begin{cases} \cos t + \frac{1}{2}\sin t, & 0 \le t \le \beta, \\ R\cos(t/2 + \pi/8) + k\cos t + \frac{1}{2}\sin t, & \beta \le t \le L - \beta, \\ (1 - \frac{2}{3}A)\cos t + \frac{1}{2}\sin t + \frac{1}{2}, & L - \beta \le t \le L, \end{cases} \tag{B.2}$$

and

$$G(t) = \begin{cases} (2A - 1)\sin t + \frac{1}{2}\cos t + \frac{1}{2}, & 0 \le t \le \beta, \\ R\sin(t/2 + \pi/8) - k\sin t + \frac{1}{2}\cos t, & \beta \le t \le L - \beta, \\ (\frac{8}{3}A - 1)\sin t + \frac{1}{2}\cos t, & L - \beta \le t \le L. \end{cases} \tag{B.3}$$

Lemma B.1. *The formulas for F, G agree in value and first derivative at both switches. They satisfy*

$$F(0) = G(0) = F(L) = 1, \qquad G(L) = \frac{8}{3}A - 1,$$

and

$$F(L - t) = G(t) + 2k\sin t, \qquad G(L - t) = F(t) - 2k\cos t.$$

Proof. Write $s = \sin\beta$ and $c = \cos\beta$. The matching identities at the first switch are

$$R\cos(\beta/2 + \pi/8) = \frac{4}{3}A\cos\beta, \qquad R\sin(\beta/2 + \pi/8) = \frac{2}{3}, \qquad 4A\sin\beta = 1.$$

The cubic is precisely the angle-matching equation. Put $\gamma = \beta/2 + \pi/8$. Since $0 < Y < 1/3$, all the following angles lie in the first tangent branch. The cubic gives

$$\frac{4Y}{1 - 4Y^2} = \frac{1 + Y}{1 - Y},$$

so the double-angle and addition formulas yield $2\arctan(2Y) = \beta + \pi/4$, or $\tan\gamma = 2Y$. Also $T = \pi/2 - (\beta + \gamma)$, and therefore

$$\cos T = \sin\beta\cos\gamma + \cos\beta\sin\gamma = \tfrac{3}{2}\cos\beta\sin\gamma.$$

It follows that $R\sin\gamma = 2/3$ and $R\cos\gamma = 1/(3Y) = (4A/3)\cos\beta$. Substitution into the first two phases proves agreement in value and first derivative at β. The two complement identities follow phase by phase from Equations (B.2) and (B.3) and transfer matching to $L - \beta$. The endpoint values follow directly. □

Define the half-plane pairs

$$\mathcal{H}_t = \{p : \langle p, u(t)\rangle \le F(t),\ \langle p, u^{\perp}(t)\rangle \le G(t)\} \quad (0 \le t \le L),$$

and then the hull and cap:

$$\hat{K}_{\mathrm{R}} = \bigcap_{t\in[0,L]} (\mathcal{H}_t \cap \rho^{-1}\mathcal{H}_t), \qquad K_{\mathrm{R}} = \hat{K}_{\mathrm{R}}^{\downarrow}. \tag{B.4}$$

This definition makes convexity and reflection symmetry immediate. At the axis normals it confines the hull to $[1-8A/3,1]\times[0,1]$. The point $(k,1/2)$ satisfies the half-plane inequalities, so the hull is nonempty and compact.

Lemma B.2 (Realization of the support coordinates)**.** *The hull $\hat{K}_{\mathrm{R}}$ is nonempty, compact, and convex. Its support values at t and $t+L$ are $F(t)$ and $G(t)$, respectively, for $0\le t\le L$. Its upper boundary consists of the two arcs*

$$B_F(t)=F(t)u(t)+F'(t)u^{\perp}(t),\qquad B_G(t)=G(t)u^{\perp}(t)-G'(t)u(t)$$

and the segment joining them at the top normal.

Proof. Define an upper support h on $[0,\pi]$ by $h(t)=F(t)$ and $h(t+L)=G(t)$, and extend it to a periodic function by $h(-t)=h(t)-\sin t$. The latter identity expresses reflection in $y=1/2$. On each open phase its curvature radius is

	$(0,\beta)$	$(\beta,L-\beta)$	$(L-\beta,L)$
$F''+F$	0	$\frac{3}{4}R\cos(t/2+\pi/8)$	$\frac{1}{2}$
$G''+G$	$\frac{1}{2}$	$\frac{3}{4}R\sin(t/2+\pi/8)$	0

All entries are nonnegative. Their middle-phase maximum is $A\cos\beta=1/(4Y)<7/8$, by the matching identities and $Y>2/7$. The one-sided derivatives of h have an upward jump $4A/3$ at L and at $-L$, and no other jumps. Hence its curvature measure $h\,\mathrm{d}t+\mathrm{d}h'$ is nonnegative.

Here is the support criterion directly in these coordinates. For a smooth phase, the projection of its support point on a fixed normal is $h(t)\cos(\phi-t)+h'(t)\sin(\phi-t)$, whose derivative is

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(h(t)\cos(\phi-t)+h'(t)\sin(\phi-t)\big)=(h(t)+h''(t))\sin(\phi-t).$$

As t runs from $\phi-\pi$ to ϕ, this projection increases, and from ϕ to $\phi+\pi$ it decreases. At a derivative jump, join the one-sided support points by the segment in the tangent direction; its projected increments have the same sign. Every point of the resulting closed curve therefore has projection at most $h(\phi)$, and the support point at ϕ attains $h(\phi)$. Taking its convex hull realizes all the half-planes defining $\hat{K}_{\mathrm{R}}$. In particular $B_F(0)=(1,1/2)$ belongs to the hull, so it is nonempty. The axis inequalities put it in the bounded rectangle $[1-8A/3,1]\times[0,1]$; the half-plane intersection is closed, and hence compact. The displayed arcs and the top segment give its upper boundary. □

It follows that the projection and middle half are

$$I_{K_{\mathrm{R}}}=[1-8A/3,1],\qquad J_{K_{\mathrm{R}}}=[1-2A,1-2A/3],\qquad w_{K_{\mathrm{R}}}=8A/3. \tag{B.5}$$

Indeed $B_F(L)=(1-2A/3,1)$ and $B_G(0)=(1-2A,1)$, since $F'(L)=2A/3-1$ and $G'(0)=2A-1$. Thus the top segment is precisely $J_{K_{\mathrm{R}}}\times\{1\}$. The roof is one throughout $J_{K_{\mathrm{R}}}$, and reflection gives the bottom face over that interval. Convexity therefore gives $J_{K_{\mathrm{R}}}\times[0,1]\subset\hat{K}_{\mathrm{R}}$, so the hull has nonempty interior.

B.2. **The niche and the two passages.** The supporting outer walls contain the hull at every angle. Removing the points below both inner walls therefore gives a shape that passes the corner. Removing the reflected niche gives the second passage. We check that these two removals leave a connected band, so they describe an admissible body. Write $n_{\mathrm{R}} = n_L(\hat{K}_{\mathrm{R}})$, using the two-wall formula of Equation (2.4). Downward filling preserves upward support values, so this is also $n_L(K_{\mathrm{R}})$. Define

$$S_{\mathrm{R}} = \{(x, y) \in \hat{K}_{\mathrm{R}} : n_{\mathrm{R}}(x) \le y \le 1 - n_{\mathrm{R}}(x)\}. \tag{B.6}$$

Lemma B.3 (The reference niche)**.** *The niche n_{R} is continuous, vanishes outside $J_{K_{\mathrm{R}}}$, and satisfies $0 \le n_{\mathrm{R}}(x) < 1/2$ for every $x \in \mathbb{R}$.*

Proof. Continuity follows from Lemma 2.5. For confinement, put $r_0 = 1 - 2A/3$ and $D(t) = 1 + r_0 \cos t - F(t)$. At L, $D(L) = D'(L) = 0$, and $D'' + D = 1 - (F'' + F) \ge 0$. Solving this terminal-value equation gives

$$D(t) = \int_t^L [1 - F(s) - F''(s)] \sin(s - t)\, \mathrm{d}s \ge 0.$$

Consequently $F(t) - 1 \le r_0 \cos t$, so $a_t(x) \le 0$ whenever $x \ge r_0$. The complement identities exchange the two walls under $x \mapsto 2k - x$, so $n_{\mathrm{R}}(2k - x) = n_{\mathrm{R}}(x)$. This gives the same conclusion for $x \le 1 - 2A$.

The height of the inner corner is

$$q_*(t) = (F(t) - 1) \sin t + (G(t) - 1) \cos t.$$

Every two-wall minimum is at most this height: multiply its two inequalities by $\sin^2 t$ and $\cos^2 t$ and add. In the first phase, with $s = \sin \beta$,

$$q_*(t) - \tfrac{1}{2} = \cos t \left(\frac{\sin t}{2s} - \frac{1}{2} \right) - \sin t < 0.$$

The expression is negative at $t = 0$ as well. In the middle phase, with $z = t - \pi/4$,

$$q_*(t) - \tfrac{1}{2} = R \cos(3z/2) - \sqrt{2} \cos z.$$

Here $|z| \le \pi/4 - \beta < \pi/6$, so $\cos(3z/2) \le \cos z$ and $\cos z > 0$. Also $R = \cos \beta / \cos T < \sqrt{2}$, since $0 < T < \pi/4$. Thus this expression is negative. The last phase is the reflection of the first. The continuous corner height therefore has a compact maximum strictly below $1/2$, which bounds the entire niche. □

Theorem B.4. *The set S_{R} is an ambidextrous sofa of area M and* $\operatorname{conv} S_{\mathrm{R}} = \hat{K}_{\mathrm{R}}$. *It is invariant under reflection in $y = 1/2$ and is regular closed:* $\operatorname{cl}(\operatorname{int} S_{\mathrm{R}}) = S_{\mathrm{R}}$.

Proof. The set is closed in the compact hull. Every vertical fiber can be joined within S_{R} to the midline $y = 1/2$, and the entire midline segment over $I_{K_{\mathrm{R}}}$ is present. This proves connectedness. For the lower passage put $\phi = Lt$ and use the explicit placement

$$p \longmapsto R_{-\phi} p + (1 - F(\phi), 1 - G(\phi)), \qquad 0 \le t \le 1. \tag{B.7}$$

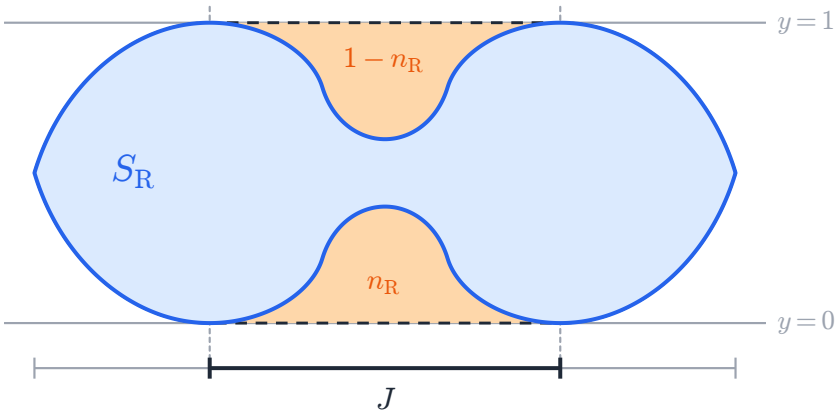


FIGURE 5. Romik's sofa is the hull (dashed) with the two open niches removed. Both niches lie over the middle half of the projection, and leave a positive band about $y = 1/2$.

The outer supporting half-planes give the two outer wall constraints. The niche inequality gives the inner-corner alternative at every interior angle. At $t = 0$ the placement is the identity and the sofa lies in H; at $t = 1$ it is $(x, y) \mapsto (y, 2 - 8A/3 - x)$, which lies in V_- by the projection and strip bounds. Reflection gives the upper passage with angle ϕ and the same initial placement.

The boundary support points of $\hat{K}_{\mathrm{R}}$ survive the niche trimming. The upper boundary arcs have abscissae outside the interior of the top interval, where the niche is zero; their reflected arcs give the remaining supports. Thus S_{R} has the same support function as $\hat{K}_{\mathrm{R}}$, which proves the convex-hull identity. Finally the continuous strict niche band and the interior of the hull approximate every boundary point from inside S_{R}. At points on either niche boundary, a small displacement toward the midline gives the approximation; the strict $1/2$ margin preserves the other constraint. This proves regular closedness.

The area calculation follows below. □

B.3. **Exact value.** The value is obtained from the support functional $\mathcal{A}$ of Section 7, before any optimality theorem is used. For the reference pair, integration by parts in its Euler equations evaluates the functional without integrating its algebraic boundary. Put $p = F' - G + 1$ and $q = G' + F - 1$. On the first phase,

$$p = \tfrac{1}{2} - 2A \sin t, \qquad q = 2A \cos t - 1;$$

on the middle phase, with $\varphi = t/2 + \pi/8$,

$$p = 1 - \tfrac{3}{2} R \sin \varphi, \qquad q = \tfrac{3}{2} R \cos \varphi - 1.$$

The last phase is obtained by $p(L-t) = -q(t)$. Thus $p \geq 0$ on $[0, \beta]$ and $p \leq 0$ on $[\beta, L]$; $q \geq 0$ on $[0, L-\beta]$ and $q \leq 0$ afterwards. At the switches $p(\beta) = q(L-\beta) = 0$.

The Euler equations of Equation (7.1) on the three phases are, respectively,

$$\begin{aligned} F'' + F = 0, \quad G'' + G = \tfrac{1}{2}; \\ 2F'' - G' + F = 0, \quad 2G'' + F' + G = 0; \\ F'' + F = \tfrac{1}{2}, \quad G'' + G = 0. \end{aligned}$$

The displayed formulas for F, G satisfy them by differentiation. The two derivative fluxes are

$$J_F = -F' - (G-1)/2 - p_-, \qquad J_G = -G' + (F-1)/2 - q_+.$$

They are continuous across both switches because the corresponding contact momentum vanishes there. Their endpoint values are

$$J_F(0) = -\tfrac{1}{2}, \quad J_G(0) = 2 - 4A, \qquad J_F(L) = 1, \quad J_G(L) = \tfrac{1}{2}.$$

Integration by parts therefore gives

$$D\mathcal{A}(F,G)[F,G] = [J_F F + J_G G]_0^L = \frac{16}{3}A - 1.$$

If A_X denotes the functional density and D_X its self-variation density, expansion gives

$$2A_X - D_X = 2 - F - G + (F' - G')/2 - p_- + q_+.$$

Primitives of this remainder on the three phases are

$$\begin{aligned} V_1(t) &= t/2 + (2A - 1/4)\cos t + (A - 3/4)\sin t, \\ V_2(t) &= (G(t) - F(t))/2, \\ V_3(t) &= t/2 - (1/4 - A/3)\cos t - (3/4 + 2A/3)\sin t. \end{aligned}$$

Differentiating each primitive, with the corresponding contact signs, verifies the remainder. Summing their increments and using the matching values gives

$$2\mathcal{A}(F,G) - D\mathcal{A}(F,G)[F,G] = 4A\cos\beta - \frac{8}{3}A - 1 + \beta.$$

Consequently

$$\begin{aligned} \mathcal{A}(F,G) &= \frac{4}{3}A + 2A\cos\beta - 1 + \frac{\beta}{2}, \\ P_L(K_{\mathrm{R}}) &= \mathcal{A}(F,G) - \frac{w_{K_{\mathrm{R}}}}{2} = 2A\cos\beta - 1 + \frac{\beta}{2}. \end{aligned} \tag{B.8}$$

For the second equality, Lemma C.1 expresses the cap area by its support energy. The curvature bounds, contact signs, and top face established above verify all the hypotheses of Corollary C.4; its signed graph calculation identifies the niche integral. This uses only the reference geometry, independently of any optimality conclusion.

Since $4A\cos\beta = \cot\beta = 1/Y$ and the cubic gives $1/Y = 4Y^2 + 3$, Equation (B.8) yields $P_L(K_{\mathrm{R}}) = M/2$. Cavalieri's principle now gives the

body's area. Outside $J_{K_{\mathrm{R}}}$ the niche is zero; on $J_{K_{\mathrm{R}}}$ the hull fills the full unit-height fiber. Hence

$$|S_{\mathrm{R}}| = 2|K_{\mathrm{R}}| - w_{K_{\mathrm{R}}} - 2\int_{J_{K_{\mathrm{R}}}} n_{\mathrm{R}} = 2P_L(K_{\mathrm{R}}) = M.$$

For comparison with Romik's closed form, put $a = \sqrt[3]{\sqrt{2}+1}$ and $b = \sqrt[3]{\sqrt{2}-1}$, so $ab = 1$. Expansion shows that $Y = (a-b)/2$ solves the cubic and is positive. Also $1 + 4Y^2 = a^2 + b^2 - 1$. Therefore

$$M = \sqrt[3]{3+2\sqrt{2}} + \sqrt[3]{3-2\sqrt{2}} - 1 + \arctan\left(\frac{\sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}}{2}\right). \quad \text{(B.9)}$$

Appendix C. Full-turn integral and contact estimates

C.1. **Cap area and the signed niche integral.** Green's formula expresses both cap area and niche area through their supports. The cap calculation follows the outer boundary. The niche calculation joins a companion tangent, a reversed corner arc, and a first tangent. Keeping these arcs oriented makes the formulas valid even when a parametrized graph folds. Their difference will produce the support functional of Section 7.

For vectors $z, z' \in \mathbb{R}^2$, write $\det(z, z')$ for their determinant. All oriented integrals below retain their sign when endpoints are reversed.

Lemma C.1 (Cap support area)**.** *Let U be a downward cap of height one, with quarter supports $F(t) = h_U(t)$ and $G(t) = h_U(t + L)$. Suppose $F, G \in C^1[0, L]$ and their first derivatives are absolutely continuous. Then*

$$2|U| = \int_0^L (F^2 - F'^2 + G^2 - G'^2)\,\mathrm{d}t. \quad \text{(C.1)}$$

Proof. The upper boundary, oriented from right to left, has support parametrization $z(\theta) = h(\theta)u(\theta) + h'(\theta)u^{\perp}(\theta)$. On regular arcs $\det(z, z') = h(h + h'')$ almost everywhere. At the top normal L, the parametrization jumps along the top segment. Its determinant integral is its horizontal length, because its height is one. Put its endpoints at $x = x_{\mathrm{tl}}, x_{\mathrm{tr}}$. The one-sided support derivatives are $h'(L-) = -x_{\mathrm{tr}}$ and $h'(L+) = -x_{\mathrm{tl}}$. Integration by parts on the two quarter arcs therefore subtracts exactly $x_{\mathrm{tr}} - x_{\mathrm{tl}}$, cancelling the top segment's contribution.

At the right axis normal the derivative is the right end height e_R; at the left normal it is $-e_L$. The remaining boundary terms are $l_U e_L - r_U e_R$. The two vertical end faces contribute $r_U e_R - l_U e_L$, and the baseline contributes zero. Green's area formula now gives Equation (C.1). The absolutely continuous integration by parts is valid without pointwise second derivatives. Approximation by polygonal boundary arcs, or by smooth convex caps, justifies the same Green formula when a curvature density vanishes on an interval. □

For two supports with top traces $F(L) = G(0) = 1$, define

$$z = (F-1)u + (G-1)u^{\perp},$$
$$B = (F-1)u + F'u^{\perp} = z + pu^{\perp},$$
$$D = (G-1)u^{\perp} - G'u = z - qu.$$

Then

$$\det(z, z') = (F-1)^2 + (G-1)^2 + (F-1)G' - (G-1)F'. \tag{C.2}$$

Lemma C.2 (Global contact from unit curvature)**.** *Suppose the actual support densities satisfy $0 \le F'' + F \le 1$ and $0 \le G'' + G \le 1$ almost everywhere. At every interior angle s, the following points lie on the graph of the complete positive niche:*

$$B(s) \quad \text{if } p(s) \le 0, \qquad D(s) \quad \text{if } q(s) \ge 0, \qquad z(s) \quad \text{if } p(s) \le 0 \le q(s).$$

Proof. The complementary profile $k = 1 - F$ has $k'' + k \ge 0$. Variation of constants, or two integrations by parts, gives

$$k(t) - k(s)\cos(t-s) - k'(s)\sin(t-s) = \int_s^t (k'' + k)(r)\sin(t-r)\,\mathrm{d}r \ge 0$$

for $s, t \in [0, L]$, with the integral interpreted as oriented. Thus $F(t) - 1 \le \langle B(s), u(t)\rangle$ for every t. The analogous inequality is $G(t) - 1 \le \langle D(s), u^{\perp}(t)\rangle$. In particular $B_y, D_y \ge 0$, by using $t = L$ and $t = 0$.

At $B(s)$ the first-wall equality holds; the second-wall height is at least B_y precisely when $p(s) \le 0$. At all other angles the first-wall height is at most B_y. Taking the supremum of the two-wall minima proves its niche contact. The proof for $D(s)$ is the same. For a corner $z(s)$ use the identities

$$\langle z(s), u(t)\rangle = \langle B(s), u(t)\rangle - p(s)\sin(t-s),$$
$$\langle z(s), u^{\perp}(t)\rangle = \langle D(s), u^{\perp}(t)\rangle + q(s)\sin(s-t).$$

For $t \ge s$ the first gives the required wall alternative, and for $t \le s$ the second does. Both equalities hold at s. Also $z_y = B_y - p\cos s \ge 0$. These facts prove global contact with the whole angular supremum, not only with neighboring angles. □

Lemma C.3 (Signed-roof integral)**.** *Suppose $F, G \in C^1[0, L]$ have absolutely continuous first derivatives, $F(L) = G(0) = 1$, and $0 \le b \le d \le L$ satisfy $p(b) = q(d) = 0$. Let N be continuous, and suppose*

$$N(D_x(t)) = D_y(t) \quad (0 < t < d),$$
$$N(z_x(t)) = z_y(t) \quad (b < t < d),$$
$$N(B_x(t)) = B_y(t) \quad (b < t < L).$$

Then

$$2\int_{-G'(0)}^{-F'(L)} N(x)\,\mathrm{d}x = -\int_0^L \det(z, z')\,\mathrm{d}t + \int_b^L p^2\,\mathrm{d}t + \int_0^d q^2\,\mathrm{d}t. \tag{C.3}$$

In particular, if $p \geq 0$ before b, $p \leq 0$ after b, $q \geq 0$ before d, and $q \leq 0$ after d, the last two integrals are $\int_0^L p_-^2$ and $\int_0^L q_+^2$.

Proof. For an absolutely continuous graph parametrization $Z = (X, Y)$ with $Y = N(X)$, the chain rule applied to $\int^X N$ gives $\int_X^{X'} N = \int Y X'$. Also

$$2YX' = (XY)' - \det(Z, Z').$$

This calculation needs no monotonicity of X. Join the three oriented graph arcs $D : 0 \to d$, $z : d \to b$, and $B : b \to L$. The joins agree because $p(b) = q(d) = 0$. Their outer endpoints have height zero and abscissae $-G'(0)$ and $-F'(L)$. The XY terms cancel, giving

$$2\int N = -\int_0^d \det(D, D') + \int_b^d \det(z, z') - \int_b^L \det(B, B').$$

Direct differentiation gives

$$\det(B, B') = \det(z, z') - p^2 + [(F-1)p]',$$
$$\det(D, D') = \det(z, z') - q^2 + [(G-1)q]'.$$

Their endpoint terms vanish by the contact and top equations. Combining the remaining corner integrals proves Equation (C.3). The sign assumptions give the final assertion. □

Corollary C.4 (Actual horizontal score)**.** *Suppose a height-one horizontal canonical cap has unit curvature on both quarters, its top face is precisely J_U, and there are ordered contact switches b, d with the signs in Lemma C.3. Then*

$$P_L(U) = \mathcal{A}(F, G) - w_U/2.$$

Proof. By Lemma C.2, all three graph identities hold for $N = n_L(U)$. The top face gives $-G'(0) = l_U + w_U/4$ and $-F'(L) = r_U - w_U/4$. Subtract Equation (C.3) from Equation (C.1) and use Equation (C.2). Since $f_U = 1$ on J_U, the score is $|U| - w_U/2 - \int_{J_U} N$. □

C.2. **The three-tent estimate.** The diagonal hallway bounds the exterior roof and supplies one middle tent. The two adjacent hallways add more niche area unless their supports move inward; moving those supports inward cuts area from the exterior roof. We combine these two payments by completing weighted squares. The estimate uses only three visited angles and the endpoint laws. Put

$$r = \sqrt{2}, \quad k = r - 1, \quad c = \cos(\pi/8), \quad s = \sin(\pi/8),$$
$$A = 1 - k, \quad b = 2 - k, \quad d = 3r - 1, \quad \gamma = 1 - r/2.$$

These letters are local to this subsection. In particular c, s do not refer to a terminal angle, and A is not the reference parameter. Useful identities are

$$\frac{s}{c} = k, \quad k^2 = 1 - 2k, \quad \frac{1}{s} + \frac{1}{c} = 4c, \quad \frac{r}{4c^2} = k, \quad 2d + b^2 = 9. \tag{C.4}$$

Let a cap have projection $[-a,a]$, middle half $[-a/2,a/2]$, and affine middle roof rising from $1-h$ to one. Lift the left wing by h and put the middle roof at one; retain the right wing. Concavity is preserved, because the left slopes are at least h/a and the right slopes are nonpositive. Denote the lifted cap by $\widehat{U}$, and define $u = rh_{\widehat{U}}(\pi/4) - 1$ and $v = rh_{\widehat{U}}(3\pi/4) - 1$. Then $a/2 \le u, v \le a$. Put $m = (u+v)/2$ and define

$$\begin{aligned} F_h(a,u,v) &= a - \frac{(a-u)^2 + (a-v)^2}{2} - \frac{ah}{2} - (m - h/2 - k)^2 \\ &\qquad + \frac{(u-k-a/2)_+^2 + (v-h-k-a/2)_+^2}{2}, \qquad \text{(C.5)} \\ S(a,m,h) &= a/k + (k-2)m + 3/2 + 2k - 4c + h/4, \\ G_h(a,u,v) &= F_h(a,u,v) - S(a,m,h)_+^2/d. \end{aligned}$$

Lemma C.5 (Full three-tent score)**.** *Suppose* $37/25 \le a \le 8/5$, $0 \le h \le 1/3$, *and the actual endpoint laws are*

$$e_R = \tfrac{1}{2} + \tfrac{h}{4} + (3n_+ - n_-)/4, \qquad e_L = \tfrac{1}{2} - \tfrac{3h}{4} + (3n_- - n_+)/4,$$

where $n_\pm = n_L(U)(\pm a/2)$. *With* $t = 2c - r$ *and* $d_0 = 3/5 - t$,

$$P_L(U) \le G_h(a,u,v) + L(h), \qquad L(h) = \gamma[(d_0 - h/4)^2 + (d_0 + 3h/4)^2]. \quad \text{(C.6)}$$

Proof. The first diagonal support of U is unchanged by lifting, because $(a/2, 1)$ dominates all lifted left-wing points. The second diagonal parameter is $v - h$. Thus the actual diagonal niche is $[\min(u - k - x, v - h - k + x)]_+$. Its apex lies in the window and is positive. Its exact integral is

$$(m - h/2 - k)^2 - \tfrac{1}{2}[(u - k - a/2)_+^2 + (v - h - k - a/2)_+^2].$$

The lifted roof lies below $f_0(x) = \min(1, 1 + u - x, 1 + v + x)$. Let D be its exterior deficit from this roof. Integrating f_0 and undoing the left lift gives $P_L(U) \le F_h - D - (\int_J n_L - \int_J n_{45})$.

The box-wall bound gives $n_\pm \le 1 - \sqrt{1 - (a/2)^2} \le 2/5$. The endpoint laws then imply

$$e_R \ge 2/5 + h/4, \quad e_L \ge 2/5 - 3h/4, \quad e_R + e_L \ge 1 - h/2.$$

Writing $w_R = a - u$ and $w_L = a - v + h$, support containment gives

$$e_R \le 1 - w_R, \quad e_L \le 1 - w_L, \quad w_R \le 3/5 - h/4, \quad w_L \le 3/5 + 3h/4.$$

Let $\delta_L = c + sv - h_{\widehat{U}}(5\pi/8)$ and $\delta_R = c + su - h_{\widehat{U}}(3\pi/8)$. Both are nonnegative. On the two flat pieces of f_0, the near vertical supporting lines cut disjoint triangles of total area $D_0 = r(\delta_L^2 + \delta_R^2) \le D$. They fit before the middle window since $\delta_L \le s(v - a/2)$ and $\delta_R \le s(u - a/2)$.

The niche at $\pi/8$ dominates the tent

$$\begin{aligned} f_R(x) &= \min(b_R + kx, B_R - x/k), \\ b_R &= 1 - 1/c + kv - h - \delta_L/c, \qquad B_R = e_R + a/k - 1/s. \end{aligned}$$

Against the descending diagonal branch its signed peak is $Z_R = (B_R + b_R)/2 - u + k$, with crossover endpoints

$$x_L = (u - k - b_R)/r, \qquad x_R = (B_R - u + k)/r.$$

The reflected construction at $3\pi/8$ gives

$$b_L = 1 - 1/c + ku - \delta_R/c, \quad B_L = e_L + a/k - 1/s,$$
$$Z_L = (B_L + b_L)/2 - v + h + k.$$

The two triangles lie on opposite sides of the diagonal apex: their crossover displacements from it are $k(m \pm h/2) + 2s - 1 + \delta/(rc) > 0$. Indeed $m - h/2 \geq 3/4$, $k > 2/5$, and $2s > 3/4$. Their outer endpoints satisfy $x_R \leq 1 + a - 2c < a/2$, since $a/2 \leq 4/5$ and $c > 9/10$; the reflected endpoint satisfies the same inequality. Thus both complete signed gain triangles fit inside J whenever their peaks are positive.

For a signed triangle of height $Z > 0$ and slopes $r, -r$, whose baseline zero exceeds its right endpoint by Δ, the positive floor loses at most $\gamma\Delta_+^2$ from the signed area Z^2/r. To check this, use the coordinate $q = x_R - x$. If $0 < \Delta \leq rZ$, the loss is

$$\int_0^{k\Delta} rq \, \mathrm{d}q + \int_{k\Delta}^{\Delta} (\Delta - q) \, \mathrm{d}q = \tfrac{1}{2}[rk^2 + (1-k)^2]\Delta^2 = \gamma\Delta^2.$$

If $\Delta > rZ$, the remaining positive tent has area $r[(1+1/r)Z - \Delta]_+^2$. When its bracket is nonnegative, subtracting $Z^2/r - \gamma\Delta^2$ leaves $(1 + r/2)(\Delta - rZ)^2 \geq 0$. When the bracket is negative the latter bound is already nonpositive. The case $Z \leq 0$ needs no positive interval. For the two triangles the losses satisfy $\Delta_R \leq (w_R - t)_+$ and $\Delta_L \leq (w_L - t)_+$, so their total is at most $L(h)$.

Write $Z_R = Z_R^0 - \delta_L/(2c)$ and $Z_L = Z_L^0 - \delta_R/(2c)$. The endpoint-height sum gives $Z_R^0 + Z_L^0 \geq S(a, m, h)$. If $H_0 = ((Z_R)_+^2 + (Z_L)_+^2)/r$, Cauchy–Schwarz gives

$$S_+ \leq \sqrt{2rH_0} + \sqrt{rD_0/(4c^2)}, \qquad S_+^2 \leq (2r + r/(4c^2))(H_0 + D_0) = d(H_0 + D_0).$$

Combining the exterior deficit, diagonal niche, additional two triangles, and floor correction proves Equation (C.6). Each exterior deficit is charged once, and the two niche gain regions are disjoint. □

Lemma C.6 (Stationarity and the width derivative)**.** *Fix $h \geq 0$ and a comparison width a_0. Put*

$$\sigma = s_0(a_0) - Ah/4, \quad s_0(a) = da/2 - (4c - 2), \quad m_* = (a_0 + k)/2 + b\sigma/9 + h/4.$$

Suppose

$$\sigma \geq 0, \qquad m_* \leq k + a_0/2, \qquad r - a_0 - 2d\sigma/9 \leq 0.$$

Then for $a \geq a_0$ and diagonal data with nonnegative apex,

$$G_h(a, u, v) \leq B(a_0) - kh/2 - h^2/8 - 2\sigma^2/9, \qquad B(a) = a - (a - k)^2/2.$$

Proof. Use coordinates (a, m, Δ), with $\Delta = (u-v)/2$. Before the two clipping terms the Hessian of F_h is

$$\begin{pmatrix} -2 & 2 & 0 \\ 2 & -4 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

Each active clip adds the outer product of $(-1/2, 1, 1)$ or $(-1/2, 1, -1)$. With both present the Hessian is

$$\begin{pmatrix} -3/2 & 1 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

which is negative semidefinite. Any other clipping pattern subtracts a positive semidefinite matrix. The gradients match at the interfaces, and subtracting S_+^2/d preserves joint concavity.

At $u = v = m$ in the unclipped region put $m' = m - h/4$ and $m_0(a) = (a + k)/2$. Completion of squares gives

$$G_h = B(a) - kh/2 - h^2/8 - 2(m' - m_0(a))^2 - [s_0(a) - b(m' - m_0(a)) - Ah/4]^2/d.$$

Since $2d + b^2 = 9$, both support derivatives vanish at $(a, u, v) = (a_0, m_*, m_*)$. The clipping hypothesis makes the right clip zero; the left clip is smaller by h. Also $S = 2d\sigma/9 \geq 0$. The value there is the asserted bound, and its width derivative is $r - a_0 - 2d\sigma/9$. The tangent plane of a concave function is an upper bound everywhere. Its support derivatives vanish and its width derivative is nonpositive, so it bounds every larger width by this value. □

Lemma C.7 (Three-tent supporting plane)**.** *Fix* $0 \leq h \leq 1/3$ *and put*

$$a_0 = 37/25, \qquad B(a) = a - (a - k)^2/2,$$
$$s_0(a) = da/2 - (4c - 2), \qquad m_0(a) = (a + k)/2.$$

On $a \geq a_0$, $a/2 \leq u, v \leq a$, *the function* G_h *is jointly concave and*

$$G_h(a, u, v) \leq K_0(h) := B(a_0) - kh/2 - h^2/8 - \tfrac{2}{9}[s_0(a_0) - Ah/4]^2. \quad \text{(C.7)}$$

Proof. Apply Lemma C.6. At $a_0 = 37/25$ its stationary support coordinate is

$$u_* = v_* = (a_0 + k)/2 + \frac{b}{9}[s_0(a_0) - Ah/4] + h/4.$$

To check clipping, keep the angle parameters linked by $k^2 + 2k = 1$ and $2c^2(1 - k) = 1$. With $\mathcal{N}_q(x) = (x + q/x)/2$, define

$$k_+ = \mathcal{N}_2(\mathcal{N}_2(3/2)) - 1, \qquad k_- = 2/(1 + k_+) - 1,$$
$$Q_\pm = 2 - 2k_\pm, \qquad c_- = [\mathcal{N}_{Q_-}(\mathcal{N}_{Q_-}(1))]^{-1}, \qquad c_+ = \mathcal{N}_{Q_+}(\mathcal{N}_{Q_+}(1))/Q_+. \quad \text{(C.8)}$$

Squaring the Newton inequality gives $k_- < k < k_+$ and $c_- < c < c_+$. Write $s_+ = (2 + 3k_+)a_0/2 - (4c_- - 2)$. Substitution in these formulas gives

$2/3 < s_0(a_0) < s_+ < 3/4$ and $s_0(a_0) - Ah/4 > 0$. Since $bA = 3 - 5k$, the larger clip obeys

$$u_* - k - a_0/2 < -k_-/2 + (2 - k_-)s_+/9 + (6 + 5k_+)(1/3)/36 < 0.$$

The last inequality follows by inserting the Newton formulas; no separately chosen bound on the support coordinate is needed. The other clip is smaller by h. Also $a_0/2 < u_* < a_0$, and $S > 0$ there. The width derivative is negative because

$$r - a_0 - \frac{2d}{9}[s_0(a_0) - Ah/4] < r - a_0 < 0.$$

These checks verify the hypotheses of Lemma C.6, which gives the bound $K_0(h)$ for all $a \geq a_0$. □

The floor corrections in the full, proper, and dry-endpoint comparisons have the same form. The following payment separates the geometric loss from the final arithmetic check.

Lemma C.8 (Tilt pays for the floor)**.** *Let $\sigma = s_0(a_0)$ and put*

$$\Psi(h) = B(a_0) - kh/2 - h^2/8 - \tfrac{2}{9}(\sigma - Ah/4)^2 + \gamma[(p + \lambda h)^2 + (q + \mu h)^2].$$

If $k \geq 2/5$, $k \leq 1$, $\sigma \leq 3/4$, $\lambda p + \mu q \leq 1/6$, $\lambda^2 + \mu^2 \leq 5/8$, and $0 \leq h_0 \leq h$ with $h + h_0 \leq 1/2$, then $\Psi(h) \leq \Psi(h_0)$.

Proof. Subtract at h_0 and factor out $h - h_0$. The remaining factor is

$$-k/2 + A\sigma/9 + A(\lambda p + \mu q) + (h + h_0)[-1/8 - A^2/72 + A(\lambda^2 + \mu^2)/2].$$

Since $0 \leq A \leq 3/5$, it is at most

$$-1/5 + 1/20 + 1/10 + 1/32 < 0.$$

The first term is the direct tilt cost; the next two pay for moving the stationary point and the floor cuts. The last pays for their quadratic remainder. □

For the three applications, $t = 2c - (1 + k)$ and the data are

	p	q	λ	μ	h_0
full	$3/5 - t$	$3/5 - t$	$-1/4$	$3/4$	0
proper	0	$2/3 - t$	0	$2/3$	0
dry endpoint	$1/2 - t$	$7/12 - t$	$-1/4$	$3/4$	0

The Newton enclosures give $5/12 < t < 1/2$. Thus $\lambda p + \mu q \leq 1/6$ in every row, and the largest $\lambda^2 + \mu^2$ is $5/8$, from the endpoint-law slopes $-1/4, 3/4$. The allowed heights satisfy $h + h_0 < 1/2$. One tilt payment therefore covers all three exclusions.

The width fixed by the initial radius. A dry low endpoint makes the initial momentum $B = b(h) + 3n_+/4$, where $b(h) = -1/2 + 5h/4$. The radius test allows span $R(h) = \sqrt{5 - b(h)^2}$. The box gives $n_+ \le 1/3$, so the largest momentum at height h is $b(h)+1/4$. Choose the height where its floor-energy test meets four at that allowed radius:

$$R(h_0)^2+[b(h_0)+1/4]^2+[b(h_0)+1/4](1-h_0)-R(h_0)\sqrt{h_0(2-h_0)}=4. \quad \text{(C.9)}$$

This determines a unique $h_0 \in (0, 1/3)$. To check existence and obtain bounds, put

$$\begin{aligned} V(h) &= R(h)^2 + [b(h) + 1/4]^2 + [b(h) + 1/4](1 - h) - 4, \\ F(h) &= V(h)^2 - R(h)^2 h(2 - h). \end{aligned}$$

Here V is positive; F decreases on $[0, 1/3]$ and changes sign between its endpoints. The squared equation therefore gives the same height. Starting with $[l, u] = [0, 1/3]$, three simultaneous updates

$$(l, u) \longmapsto \left(l - \frac{F(l)}{F'(l)},\, u - \frac{F(u)}{F'(l)}\right)$$

give bounds $h_- \le h_0 \le h_+$. The divided difference of F lies above $F'(l)$ on each bracket, so these are enclosing updates, rather than fitted approximations. The shared radius width is now

$$\begin{gathered} b_0 = b(h_0), \quad R_0 = R(h_0), \quad C_{\mathrm{rad}} = R_0/3, \\ R_+ = \mathcal{N}^{\circ 2}_{R(h_+)^2}(2), \qquad R_- = R(h_-)^2/R_+, \qquad a_{\mathrm{rad}} = 2R_-/3. \end{gathered} \quad \text{(C.10)}$$

The span lower bound $d \ge 2$ supplies the Newton seed. The brackets give $R_- < R_0 < R_+$. The triangle comparison below excludes every larger width, even when $h < h_0$, and can be evaluated at zero tilt like the other two rows. Both boundaries are therefore fixed by the two energy tests.

Lemma C.9 (A common endpoint comparison)**.** *Use k_+, c_+ from (C.8) and define*

$$\begin{aligned} E &= (a + 1 - h/2)k + a - (a^2 + 1)/2 - h^2/8 \\ &\quad - 2z^2/9 + \tfrac{A}{2}(wD_R^2 + D_L^2), \\ z &= (2 + 3k)a/2 - (4c - 2) - Ah/4, \\ D_i &= b_i - (2c - 1 - k). \end{aligned}$$

For each of the following geometric endpoint data, $E(a, h, k, c; b_R, b_L, w) \le E(a, h, k_+, c_+; b_R, b_L, w) < \underline{p}$:

	a	h	b_R	b_L	w
full	37/25	0	3/5	3/5	1
proper	37/25	0	2/3	2/3	0
dry endpoint	a_{rad}	0	1/2	7/12	1

Here $w = 0$ records the proper comparison's single joint floor square.

Proof. The identity $k^2 + 2k = 1$ rewrites $B(a)$ as the linear term in E. To bound the whole expression, first increase c to c_+, then increase k to k_+. On these segments, the Newton enclosures give $3/5 \le z \le 3/4$ and $0 \le D_i \le 1/4$. The two changes both raise E:

$$\begin{aligned}\partial_c E &= 16z/9 - 2A(wD_R + D_L) > 0,\\ \partial_k E &= a + 1 - h/2 - \tfrac{4}{9}(3a/2 + h/4)z\\ &\quad - \tfrac{1}{2}(wD_R^2 + D_L^2) + A(wD_R + D_L)\\ &\ge a/2 + 15/16 - 7h/12 > 0.\end{aligned}$$

Here $0 \le w \le 1$, and all three rows have $h = 0$. The first comparison retains the improvement in the stationary square alongside the loss in floor payment; the second retains the linear area reward. Both can equally be obtained by subtracting the two squares. Inserting the same Newton formulas in each row gives $E < \underline{p}$ using (2.2). □

Corollary C.10 (Full width cut)**.** *Every cap satisfying the hypotheses of Lemma C.5 has $P_L(U) < \underline{p}$.*

Proof. By Lemma C.7, $P_L(U) \le K_0(h) + L(h)$. The full row of Lemma C.8 bounds this at $h = 0$. Its value is the full row of Lemma C.9, which is below $\underline{p}$. □

C.3. **The first-wing energy test.** The following estimates use the tilted notation of Section 6. In particular $I = [-2C, 2C]$, $J = [-C, C]$, the middle roof rises from $1 - h$ to 1, and F, G are the high-wing and low-wing supports. Write

$$d = 3C, \qquad B = e_R - 1 + h, \qquad s_h = \sqrt{h(2-h)}, \qquad H(C) = 1 - \sqrt{1 - C^2}.$$

The target $u \le 1$ follows if $q \le 1$ whenever the first source can be exposed. At its first p-zero this means $q + 1 \le 2$. Before the corner reaches the floor, the support trajectory is a free rotation. The two tests below bound its energy depending on which event comes first; the constants four and five are the corresponding squared radii.

Lemma C.11 (Initial-floor test)**.** *Suppose $1/2 < C < 4/5$ and $0 < h < 1/3$. At a canonical full maximum, the two inequalities*

$$d^2 + B^2 \le 5, \qquad d^2 + B^2 + B(1-h) - ds_h \le 4 \tag{C.11}$$

imply $u \le 1$ almost everywhere on $(0, L)$.

Proof. On the initial component where $p, q > 0$ and $D_y < 0$, both source densities vanish. The first vanishes by same-sign shadowing. A positive second-source contact with companion slack would have to be the stationary second-wall contact D, whose height is negative. A corner is shadowed because both its walls increase with the angle. The source identity therefore makes the second density zero as well. This reasoning also excludes a floor contact: a nonstationary one with slack is raised by a nearby angle, and a

same-sign floor corner is raised on both walls. Compact exhaustion in angle justifies the statement at the level of the limiting source measures.

Consequently, until this component meets the floor or $p = 0$,

$$\begin{aligned} F &= 2C\cos t + e_R \sin t, \qquad G = C\sin t + (1-h)\cos t, \\ D_y &= 1 - h - \cos t, \\ p - 1 &= B\cos t - d\sin t, \qquad q + 1 = d\cos t + B\sin t. \end{aligned} \tag{C.12}$$

Put $\gamma_0 = \arccos(1-h)$ and let α be the first p-zero on the initial positive-q component. Stop at $\min(\alpha, \gamma_0)$. The rotating vector in Equation (C.12) has squared norm $R^2 = d^2 + B^2$. Thus

$$\begin{aligned} E_{++} := (p-1/2)^2 + (q+1)^2 &= R^2 + p - 3/4 \\ &\le R^2 + 1/4 + \max\{B(1-h) - ds_h, -1\} \end{aligned}$$

at the stopping time. The -1 alternative is the value when $p = 0$; the other alternative is the value at the floor. Both tests in Equation (C.11) say that this upper bound is at most $17/4$.

If the floor is reached first, the same-sign bounds $u = 0$, $v \le 1/2$ give

$$E'_{++} = 2(p-1/2)(-1-q) + 2(q+1)(v-1+p) = 2(q+1)(v-1/2) \le 0.$$

At the first p-zero, $1/4 + (q+1)^2 \le 17/4$, hence $q \le 1$. While $p < 0$ and $q > 0$, the arm estimate gives $q' \le -1/2$: for $-1 \le p \le 0$ use $v \le (1-p)/2$, and for $p < -1$ use $v \le -p$. The positive-q component therefore has $q \le 1$ after this zero.

Any later positive-q component starts at $q = 0$ with $0 < p \le 1$; entry with $p \le 0$ is excluded by the preceding derivative bound. While $p, q > 0$, $u = 0$, $p' \le -1$ and $q' \le p - 1/2$. Its maximum q is at most $\int_{1/2}^{1}(x - 1/2)\,\mathrm{d}x = 1/8$. Once p is negative, q decreases again. If the initial positive-q component ends before the stopping time, only these later components need be considered. Finally $q \ge -1$ everywhere, and $u \le \kappa(q)$, with $u = 0$ in the positive-positive sector. The proved bounds $-1 \le q \le 1$ on all remaining sectors give $u \le 1$. □

Lemma C.12 (Short widths and the radius width)**.** *The first density of a canonical full maximum is at most one if $C \le 2/3$. If its low middle endpoint is dry, the same conclusion holds for $C \le C_{\mathrm{rad}}$ and $h_0 \le h < 1/3$, with C_{rad}, h_0 from (C.10).*

Proof. For $C \le 2/3$, write $e = e_R \in [0,1]$. Then $d^2 \le 4$, $|B| = |e - 1 + h| \le 1$ and $s_h \ge h$. The first test follows at once. The second follows from

$$B^2 + B(1-h) = e(e-1+h), \qquad e(e-1+h) - ds_h \le h(e-d) \le 0,$$

since $d > 3/2 \ge e$.

Now assume $2/3 < C \le C_{\mathrm{rad}}$, $h \ge h_0$, and $n_- = 0$. Since $9C_{\mathrm{rad}}^2 = 5 - b_0^2 < 5$, the box height obeys $H(C) \le 1/3$. The dry endpoint pressure therefore gives

$$-1/2 + 5h/4 \le B \le -1/4 + 5h/4.$$

On $[h_0, 1/3]$ both endpoints have absolute value at most $-b_0$, and hence

$$d^2 + B^2 \le 9C_{\mathrm{rad}}^2 + b_0^2 = 5.$$

This is exactly why the radius width was chosen.

For the floor test put $Q(d, B, h) = d^2 + B^2 + B(1-h) - d\sqrt{h(2-h)}$. It increases with B on this interval, since $2B + 1 - h \ge 3h/2 \ge 0$, and with $d \ge 2$. Thus it suffices to use $d = R_0$ and $B = -1/4 + 5h/4$. The resulting function is convex in h: its polynomial part has quadratic coefficient $5/16$, and the negative semicircle is convex. Only the two endpoints $h = h_0, 1/3$ remain. At h_0 the test equals four by its defining equation. At $1/3$, use $R_0^2 \le 5$, $R_0 \ge 2$ and $\sqrt{(1/3)(2-1/3)} \ge 2/3$. These give

$$Q(R_0, 1/6, 1/3) \le 5 + (1/6)^2 + (1/6)(2/3) - 2(2/3) < 4.$$

Convexity supplies the whole interval. Apply Lemma C.11. □

C.4. **Ordered contact geometry.** Here $F, G, u, v, p, q, z, B, D, C, h, T$ have the tilted meanings in Equations (6.2) and (6.4). In particular $G(0) = 1 - h$; it is the low-wing support, not the support containing the middle facet.

Lemma C.13 (Contact signs)**.** *If $1/2 < C < 4/5$, $0 < h < 1/2$ and $u \le 1$ almost everywhere, there are $0 < \alpha < \eta < L$ such that*

t	$(0, \alpha)$	(α, η)	(η, L)
p	> 0	< 0	< 0
q	> 0	> 0	< 0.

Moreover $p(\alpha) = q(\eta) = 0$.

Proof. Initially $p(0) = e_R + h > 0$, $q(0) = 3C - 1 > 0$, whereas $p(L) = 1 - 3C - T < 0$ and $q(L) = -e_L < 0$. Before $\delta = \arctan(h/(2C))$, the zero-density solution gives $q' \ge -1$ while p is positive. Since $\delta < h < 1/2 < q(0)$, q cannot reach zero there. After δ, the arm bound gives $p \le 1$. On the positive-positive sector, $u = 0$, and

$$[(p-1)^2 + (q+1)^2]' = 2(q+1)v \ge 0.$$

Its initial value is at least $9C^2 > 2$. A first exit through $q = 0$ would instead have $0 \le p \le 1$ and energy at most two. Thus the first exit is $p = 0$, at a time α, with $q > 0$.

For $q \ge 0$, the arm bound gives $p' = u - 1 - q \le \kappa(q) - 1 - q \le -1/2$. For $p \le 0$, the companion bound similarly gives $q' \le -1/2$. Consequently p stays negative while $q > 0$, and q reaches zero at some $\eta > \alpha$ with $p(\eta) < 0$. In the ensuing negative-negative sector $v = 0$, $u \le 1/2$ and

$$[(p-1)^2 + (q+1/2)^2]' = 2(p-1)(u-1/2) \ge 0.$$

At η this energy is strictly greater than $5/4$. A return through $p = 0$ would have $-1 \le q \le 0$ and energy at most $5/4$, since the lower arm bound is $q \ge -1$. Such a return is impossible. As long as $p < 0$, the preceding $q' \le -1/2$ bound also prohibits a return through $q = 0$. This proves all the signs, including strictness at the ends of the open sectors. The argument uses

first crossing times of continuous functions and integrated almost-everywhere derivative bounds, so it covers zero-density intervals. □

Lemma C.14 (First-unit contact geometry)**.** *Under the hypotheses of Lemma C.13, B_x is nondecreasing and B_y is nonincreasing. Let β be its first crossing of C (use $\beta = L$ when $T = 0$). Then*

$$\alpha < \beta, \qquad \beta \geq \arccos C, \qquad \sin\beta \geq 3/5, \qquad \cot\beta \leq 4/3. \tag{C.13}$$

The positive niche is precisely $(x_*, C+T)$ for one $x_* < 1-2C$, and

$$T = (C + x_*)_+. \tag{C.14}$$

If the companion density exceeds one, its first bad entry τ satisfies

$$\alpha < \beta < \tau < \eta, \quad p(\tau) = -1, \quad 0 < q(\tau) \leq 1/4, \qquad v \leq 1 \text{ a.e. on } (0,\tau). \tag{C.15}$$

In particular $T = 0$ implies $v \leq 1$ almost everywhere.

Proof. The monotonicity follows from $B' = (u-1)u^\perp$ and $u \leq 1$. If $B_x \geq C$, its associated outer point $(X, Y) = B + u(t)$ lies in the cap, so $\cos t \leq C$ and $X \geq C + \cos t$, $Y \leq 1$. The low endpoint gives $G \geq C\sin t + (1-h)\cos t$, whence

$$p \leq 1 - (2C + \cos t)\sin t + h\cos t < 0.$$

For the last sign put $k = \cos t \in [0, C]$. The upper bound $1 - (2C + k)\sqrt{1-k^2} + k/2$ is convex in k. At zero it is $1 - 2C < 0$; at C it is negative because

$$9C^2(1-C^2) - (1 + C/2)^2 > 0 \quad (1/2 \leq C \leq 4/5).$$

This polynomial has derivative $(2-3C)(24C^2+16C-1)/2$, hence its minimum is at an endpoint; its endpoint values are $1/8$ and $71/625$. Thus $B_x(\alpha) < C$, proving the first inequality in Equation (C.13); the outer point at the crossing gives the remaining three.

For the positive set write

$$R_t(x) = \frac{F(t) - 1 - x\cos t}{\sin t}, \qquad S_t(x) = \frac{G(t) - 1 + x\sin t}{\cos t},$$
$$L_0(t) = \frac{1 - G(t)}{\sin t}.$$

On $x < 1 - 2C$, the envelope $\sup_t S_t(x)$ is finite, convex, and nondecreasing in x. Its endpoint limits are $-h$ at zero and $-\infty$ at L. It is at most $-h$ for $x \leq -2C$, by the low-wing box $[-2C, -C] \times [0, 1-h]$; at $x = 1-2C$ its L-limit is $e_L > 0$. There is exactly one zero $x_* \in (-2C, 1-2C)$. Indeed a zero is attained at an interior angle and that wall has positive slope $\tan t$, so all points to its right have positive envelope. All points to its left have negative envelope. Equivalently $x_* = \inf_t L_0(t)$.

The derivative of $(F-1)/\cos t$ is $B_y/\cos^2 t \geq 0$. Its intercept therefore rises from $2C-1$ to $C+T$. Every first wall is positive for $x \leq 1-2C < 2C-1$, and every first wall is nonpositive for $x \geq C + T$. For $1 - 2C < x < C + T$, angles approaching L give $R_t(x) = (C + T - x)(L - t) + o(L - t) > 0$ and

$S_t(x) \to +\infty$. This proves the positive interval. The additional actual high-point wall is nonpositive on J and does not affect it there.

The finite positive graph has horizontal length equal to the sine/cosine moment of its two source measures. Source convergence and Equation (6.3) give its limiting length as

$$\int_0^L (u \sin t + v \cos t)\, \mathrm{d}t = 2C - T.$$

Uniform niche convergence gives the lower bound from the positive interval. To get the reverse bound, take a compact interval strictly left of x_*. All surrogate second walls there are strictly negative when their angles lie in $[\varepsilon, L-\varepsilon]$. The additional actual high-point wall is also strictly negative there. The finite positive graph on this compact interval must therefore use endpoint-angle sources. Their total mass is $O(\varepsilon) + o(1)$ by Equation (5.5). Exhaust the zero interval and then let $\varepsilon \downarrow 0$. Thus the finite positive lengths converge to $C - \max\{-C, x_*\}$. Equating the two lengths proves Equation (C.14).

Let τ be the first entry into $\{p < -1, q > 0\}$. Before it, $v \le 1$: use $v \le \kappa(p)$ when $p \ge -1$, and $v = 0$ in the negative-negative sector when $p < -1$. On a level set $q = 0$, its derivative is zero almost everywhere, whereas $p < -1$ would give $q' \le -1$; no exceptional mass remains. Entry through $q = 0$ is impossible because $q' \le -1/2$ for $p \le 0$. Thus $p(\tau) = -1$, $q(\tau) > 0$.

For $\alpha < t < \tau$, past D_x is nondecreasing, whereas future B_x is globally nondecreasing. Hence the positive core corner is the unique active angle: earlier second walls and later first walls are strictly below it. It lies in J, since $-C < D_x + q\cos t = z_x = B_x + p \sin t < C$. It contributes first-source density q. Before β, the disjoint visible B tangency contributes another $1-u$. The source identity gives $u \ge (1+q)/2$ before β, and $u \ge q$ after it. Plateaus contribute zero tangent length and do not change these bounds. If $\tau \le \beta$, the bounds $u \le 1$ and $u \le \kappa(q)$ make $q \le 1$, $u = (1+q)/2$ on the core up to τ. There $E = (p-1)^2 + (q+1)^2$ is nonincreasing and $E(\alpha) \le 5$; but $E(\tau) = 4 + (q(\tau)+1)^2 > 5$, a contradiction. The same strict comparison at the crossing excludes equality $\tau = \beta$.

On (β, τ), put $K = q + (p-1)^2/4$. Since $u \ge q$ and $-1 < p < 0$, $v \le (1-p)/2$ gives $K' \le v + (p-1)/2 \le 0$. At β, $E \le 5$ yields

$$K(\beta) \le 1 + q(\beta)/2 - q(\beta)^2/4 \le 5/4.$$

As $K(\tau) = 1 + q(\tau)$, this proves Equation (C.15). If $T = 0$, $\beta = L$ and the first bad entry is impossible; the arm and same-sign bounds then give $v \le 1$ everywhere. □

C.5. **The joint occupation of a corner.** A visible corner belongs to both wall families. Resolving its tangent vector into those two wall directions fixes the ratio of their source lengths. Thus both families must use the same occupied fraction of a corner. This shared fraction is what lets the energy argument pay for folds without counting a contact twice.

Lemma C.15 (Common source occupation)**.** *Under the first-unit hypotheses, suppose the companion density exceeds one on a set of positive measure, and let the ordered contacts and β be as above. There are measurable $d, \chi \in [0,1]$ such that, almost everywhere,*

$$\begin{array}{c|cc} & u & v \\ \hline (0,\alpha) & 0 & (1-v)d \\ (\alpha,\eta) & (1-u)b+q\chi & (1-v)d-p\chi \\ (\eta,L) & 0 & 0, \end{array} \qquad b = \mathbf{1}_{\{t<\beta\}}. \tag{C.16}$$

The second tangent occupation satisfies $d = 0$ where $v > 1$ or $D_y < 0$. Before a first bad entry τ, one has $\chi = 1$ on (α, τ).

Proof. Keep the source angle and graph abscissa together in the finite polygons. Assign to every positive first-source segment its arclength measure at (t, x); call this $\mu_{i,F}$, and define $\mu_{i,G}$ similarly. Their exact graph-vector identities on the window interior are

$$\begin{aligned} \pi_{x\#}(\sin t\, \mu_{i,F} + \cos t\, \mu_{i,G}) &= \mathbf{1}_{\{N_i>0\}\cap J_i}\, \mathrm{d}x, \\ \pi_{x\#}(-\cos t\, \mu_{i,F} + \sin t\, \mu_{i,G}) &= \mathrm{d}(N_i|_{J_i}). \end{aligned}$$

The second is a distributional identity. Bounded mass gives a joint weak subsequence. The no-ghost argument in Lemma C.14 passes the first identity to Lebesgue measure on (x_*, C). Uniform graph convergence passes the second to $N'(x)\,\mathrm{d}x$ on this positive interval. The angular marginals are exactly $u\,\mathrm{d}t, v\,\mathrm{d}t$ by Theorem 5.8. There are no endpoint-angle atoms. The graph is locally Lipschitz on its positive interval: on a compact positive part, the endpoint-angle walls tend to zero or below, so all active angles remain in a compact subinterval of $(0, L)$. These identities also exclude extra mass at a spatial singleton or at a floor point.

At a regular positive active source with first-wall slack, angular stationarity gives $x = B_x(t)$ and height $B_y(t)$; this needs $p < 0$. The analogous second-wall contact is D, with $q > 0$. A simultaneous contact is z. A corner with two positive momenta is shadowed by a slightly larger angle, and one with two negative momenta by a slightly smaller angle. Thus only core corners remain. At a second tangent

$$\partial_t^2 S_t(D_x(t)) = -(1 - v(t))/\cos t.$$

When $v > 1$, it is an angular local minimum with strict companion slack, and cannot be an active full-envelope maximum. The same conclusion follows at density points from the integrated second-derivative inequality. Second-tangent occupation vanishes there and when $D_y < 0$.

The first-unit bound makes B the global first-wall envelope. On (α, β) its companion has strict slack. Its graph has negative slope and lies to the right of every core corner, because z_x strictly decreases from $B_x(\alpha)$. An active D tangent has positive slope and cannot coincide on a set of positive graph measure. Thus the entire first-tangent contribution is $(1-u)\mathbf{1}_{(\alpha,\beta)}\,\mathrm{d}t$. Plateaus have zero spatial projection and hence zero extra mass.

On the measurable corner contact set $N(z_x(t)) = z_y(t)$, the chain rule and the derivative-zero property on level sets give

$$N'(z_x(t)) = \frac{p\sin t + q\cos t}{p\cos t - q\sin t} \quad \text{a.e.}$$

On a compact core interval z_x is bi-Lipschitz, so null sets in angle and abscissa cause no difficulty. Disintegrate the two vector identities at $x = z_x(t)$. Any competing D contact has the same tangent $(1, N'(x))$; subtract its parallel flux. For the remaining first and second arclength densities A_F, A_G the vertical identity becomes

$$(-\cos t - N'\sin t)A_F + (\sin t - N'\cos t)A_G = 0.$$

Substitution gives $A_G = (-p/q)A_F$. Their nonnegative horizontal share is at most one. Since the increasing-abscissa corner displacement is $-z'(t)\,\mathrm{d}t$, one common number $\chi \in [0,1]$ therefore assigns angular densities $q\chi$ and $-p\chi$. Exhaust compact core intervals to obtain a measurable χ on the whole core. Strict unique corner visibility before τ makes $\chi = 1$ there.

For a visible regular second tangent, $D'_x = (1-v)\cos t$. The area formula pulls its horizontal share back to angle as $(1-v)d\,\mathrm{d}t$, for a measurable $d \in [0,1]$. At almost every graph abscissa there is at most one visible second tangent, since its slope $\tan t$ determines its angle. Set $d = 0$ on absent or forbidden contacts. All positive second tangents and core corners lie in J: their abscissa exceeds $L_0(t) \geq x_* > -C$, whereas $D_x \leq 1 - C < C$ and $z_x \leq B_x(\alpha) < C$.

These three classes exhaust the limiting source measure. There is no nonnegative remainder, since the weighted spatial projection already equals the whole positive graph projection. Equating angular marginals gives Equation (C.16). In the final sector the first tangencies are outside J, as $\beta < \eta$ whenever this occupation is used in the nonunit branch; the second density already vanishes by same-sign shadowing. □

C.6. **The two clipped strip moments.** The top overhang T appears as a horizontal moment at the low-wing floor crossing and as a vertical moment at the high-wing window crossing. Integrating the contact energy compares these moments. Endpoint stationarity cancels the end heights, leaving the required term $6CT$ and convolution remainders that the two unit-curvature bounds can control.

Lemma C.16 (Two unit wings)**.** *A canonical full maximum with positive middle slope,* $1/2 < C < 4/5$ *and* $0 < h < 1/2$ *cannot have both* $u, v \leq 1$ *almost everywhere.*

Proof. Both B and D have nondecreasing abscissae; B_y decreases to zero and D_y increases from $-h$ to $e_L > 0$. Moreover $B_x - D_x = q\cos t - p\sin t \geq 4C - 2 > 0$. Let γ be a floor crossing of D, and β the right-window crossing

of B. Ordered signs and the exact wall derivatives

$$\partial_t R_t(x) = \frac{x - B_x(t)}{\sin^2 t}, \qquad \partial_t S_t(x) = \frac{x - D_x(t)}{\cos^2 t}$$

show that the positive graph in the window consists of $D[\gamma, \eta]$, the reversed core $z[\alpha, \eta]$, and $B[\alpha, \beta]$. Indeed at a tangent its sole active wall is the global angular maximum with companion slack; at a core corner every earlier second wall and later first wall is strictly below it. The source-box estimate gives $\alpha < \beta$, and positivity of $z_y(\eta)$ gives $\gamma < \eta$. No relative order of β and γ is needed. Flat tangent parameter intervals have zero projection.

The no-ghost identity now says $T = C + D_x(\gamma) > 0$. Strict positivity also follows directly from $h = \int_0^\gamma (1 - v) \sin t > 0$. Local finite graph-vector flux gives source lower bounds

$$\begin{aligned} u \, \mathrm{d}t &\geq [(1 - u)\mathbf{1}_{(\alpha,\beta)} + q\mathbf{1}_{(\alpha,\eta)}] \, \mathrm{d}t, \\ v \, \mathrm{d}t &\geq [(1 - v)\mathbf{1}_{(\gamma,\eta)} - p\mathbf{1}_{(\alpha,\eta)}] \, \mathrm{d}t. \end{aligned}$$

For a compact strict corner arc, resolve its increasing-abscissa displacement $-z'$ in the two wall directions: it assigns first length q and second length $-p$. Strict contact localization passes these local bounds from the polygons. The sine/cosine weighted sum of their right sides is the actual positive graph projection, namely $2C - T$, which is also the weighted source total. The nonnegative defects therefore have zero positive weighted moment and vanish. This proves equality in both bounds.

Put $z_+ = N_L(U)(C)$. Since $N_L(U)(-C) = 0$, the endpoint pressures and tangent moments are

$$\begin{aligned} e_R = \tfrac{1}{2} + \tfrac{h}{4} + \tfrac{3z_+}{4}, \qquad e_L = \tfrac{1}{2} - \tfrac{3h}{4} - \tfrac{z_+}{4}, \\ T = \int_0^\gamma (1 - v) \cos t \, \mathrm{d}t, \quad h = \int_0^\gamma (1 - v) \sin t \, \mathrm{d}t, \\ T = \int_\beta^L (1 - u) \sin t \, \mathrm{d}t, \quad z_+ = \int_\beta^L (1 - u) \cos t \, \mathrm{d}t. \end{aligned} \tag{C.17}$$

Use the continuous contact energy

$$\mathcal{H}(p, q) = \begin{cases} (p - 1/2)^2 + (q + 1)^2 + 3/4, & p, q \geq 0, \\ (p - 1)^2 + (q + 1)^2, & p \leq 0 \leq q, \\ (p - 1)^2 + (q + 1/2)^2 + 3/4, & p, q \leq 0. \end{cases} \tag{C.18}$$

The formulas agree on the two switching axes. Substitution of the exact source equations gives

$$\mathcal{H}' = -(q + 1)(1 - v)\mathbf{1}_{(0,\gamma)} + (1 - p)(1 - u)\mathbf{1}_{(\beta,L)}.$$

For example in an unclipped core the equations are $u = (1+q)/2$, $v = (1-p)/2$ and the derivative is zero; removing floor tangencies replaces the second equation by $v = -p$ and adds the first displayed term; removing window tangencies replaces the first by $u = q$ and adds the second. The same

substitution in the outer sign sectors gives the same terms. Thus every crossing order is included.

Let

$$I_D = \int_0^{\gamma} (q+1)(1-v), \qquad I_B = \int_{\beta}^{L} (1-p)(1-u),$$
$$R_D = \int_0^{\gamma} (1-v(t)) \int_0^{t} u(s)\sin(t-s)\,\mathrm{d}s\,\mathrm{d}t, \tag{C.19}$$
$$R_B = \int_{\beta}^{L} (1-u(t)) \int_t^{L} v(s)\sin(s-t)\,\mathrm{d}s\,\mathrm{d}t.$$

Integration variables on the first line are t. Endpoint substitution gives

$$\mathcal{H}(L) - \mathcal{H}(0) = 6CT + T^2 - (h+z_+)(h+z_+/2).$$

Variation of constants in the momentum equations gives

$$\begin{aligned} q(t)+1 &= 3C\cos t + (e_R+h)\sin t \\ &\quad - \int_0^t (1-v(s))\cos(t-s)\,\mathrm{d}s + \int_0^t u(s)\sin(t-s)\,\mathrm{d}s, \\ 1-p(t) &= (3C+T)\sin t + e_L\cos t \\ &\quad - \int_t^L (1-u(s))\cos(s-t)\,\mathrm{d}s + \int_t^L v(s)\sin(s-t)\,\mathrm{d}s. \end{aligned}$$

For any integrable function a, symmetry of the cosine kernel gives

$$\int_\xi^\zeta a(t) \int_\xi^t a(s)\cos(t-s)\,\mathrm{d}s\,\mathrm{d}t = \tfrac{1}{2}\left[\left(\int_\xi^\zeta a\cos\right)^2 + \left(\int_\xi^\zeta a\sin\right)^2 \right].$$

Applying it to $1-v$ and $1-u$ yields

$$I_D = 3CT - T^2/2 + e_R h + h^2/2 + R_D,$$
$$I_B = 3CT + T^2/2 + e_L z_+ - z_+^2/2 + R_B.$$

Since the energy change is $I_B - I_D$, the pressures simplify these identities to

$$6CT = (z_+ - h)(2 - h - z_+)/4 + R_B - R_D. \tag{C.20}$$

Here $R_D \geq 0$, $h + z_+ < 2$, $z_+ \leq \cot\beta\, T \leq 4T/3$, and

$$R_B \leq \int_\beta^L (1-u)(1-\sin t)\,\mathrm{d}t \leq \frac{1-\sin\beta}{\sin\beta} T \leq 2T/3.$$

The first term in Equation (C.20) is at most $z_+/2$. Hence $6CT \leq 4T/3$, contradicting $C > 1/2$, $T > 0$. □

C.7. **Paying for a folded companion.** When the companion curvature exceeds one, its tangent can fold back and the two-unit calculation acquires an error. The common source occupations bound that error by the unused first-wing curvature. Its sine moment is already T, so the fold can be paid from the same strip budget.

Lemma C.17 (Folded-wing energy)**.** *Under the first-unit hypotheses, a companion density exceeding one also contradicts maximality.*

Proof. Let $\alpha < \beta < \tau < \eta$ be the contacts in Lemma C.14. That lemma forces $T > 0$, $x_* = -C + T$ and $N_L(U)(-C) = 0$. Use the common occupations d, χ of Lemma C.15. On the core they give

$$u = \frac{b + q\chi}{1 + b}, \qquad v = \frac{d - p\chi}{1 + d}, \qquad b = \mathbf{1}_{\{t<\beta\}}. \tag{C.21}$$

Let γ be the first zero of D_y. We claim $\gamma < \tau$. There is a zero before η, since $D_y(0) = -h$ and $D_y(\eta) = z_y(\eta) > 0$. Suppose D_y has stayed nonpositive until τ. Then $L_0' = D_y/\sin^2 t \le 0$ on this initial interval. At a positive core corner, every earlier second wall is strictly lower, because

$$S_r(z_x(t)) = \tan r\,[z_x(t) - L_0(r)] < \tan t\,[z_x(t) - L_0(t)] = z_y(t) \quad (r < t).$$

If the earlier bracket is negative the inequality is immediate; otherwise both its factors are no greater and the tangent is strictly smaller. Every later first wall is strictly lower by global B_x monotonicity. Thus this corner is uniquely exposed and $\chi = 1$ as long as D_y remains nonpositive. After τ, while this condition persists, $b = d = 0$ and Equation (C.21) gives $u = q$, $v = -p$, $p' = q' = -1$. Therefore $D_y' = (1 + p)\sin t < 0$. It cannot leave the nonpositive region before q reaches zero, contradicting $D_y(\eta) > 0$.

This reasoning includes $D_y(\tau) = 0$ by the following continuation argument. At τ the corner is positive and inside the window, and its two local angular derivatives have opposite strict signs. The remote-angle gaps are strict and uniform on compact remote angle sets. Both properties persist on a neighborhood, giving $\chi = 1$ immediately to the right. Then $p' = -1$; for $p < -1$, the fractional formula gives $v = (d - p)/(1 + d) > 1$, forcing $d = 0$. Hence $D_y' < 0$ there as well. This excludes an immediate positive departure from zero. The claim follows, including a possible zero plateau.

Put $S = C + D_x(\gamma)$. At this floor crossing $D_x(\gamma) = L_0(\gamma) \ge x_*$. The low-wing outer point has abscissa at most $-C$, so $D_x \le 1 - C$. Together with $x_* < 1 - 2C$, these facts give

$$\begin{aligned} 0 < T \le S \le 1, \qquad & T \le C, \\ S = \int_0^\gamma (1 - v)\cos t, \qquad & h = \int_0^\gamma (1 - v)\sin t, \\ T = \int_\beta^L (1 - u)\sin t, \qquad & z_+ = \int_\beta^L (1 - u)\cos t. \end{aligned} \tag{C.22}$$

All four integrals use $\mathrm{d}t$. The quantity S records the horizontal displacement up to the first floor crossing; its excess over T will have a favorable sign in the energy comparison.

On (τ, η), $b = 0$ and $p' = q\chi - 1 - q \le -1$. Thus $p \le -1$ and $0 < q \le 1/4$ there. For $w = q - p$, substitution in the fractional equations gives the exact identity

$$w' = (1 - \chi)\left[p + q + \frac{d}{1 + d}\right] + \frac{\chi d(1 + p)}{1 + d} \le 0.$$

Both terms are nonpositive since $p+q \le -3/4$ and $d/(1+d) \le 1/2$. As $w(\tau) \le 5/4$, $v \le -p \le 5/4 - q$ there. Before τ, $v \le 1$; after η, $v = 0$. Hence $v \le 5/4$ globally. Where $v > 1$, $d = 0$, $v = -p\chi$, $u = q\chi$ and

$$(v-1)_+ \le 1/4 - q \le (1-u)/4. \tag{C.23}$$

Use the continuous energy Equation (C.18). Before β, the source equations and $\chi = 1$ give $\mathcal{H}' = -(q+1)(1-v)(1-d)$. On the core after β, direct differentiation gives

$$\mathcal{H}' - (1-p)(1-u) + (q+1)(1-v)(1-d) = (1-\chi)(p+q). \tag{C.24}$$

The residual is zero before τ and nonpositive afterwards. On the final sector $\mathcal{H}' = 1-p$. On $[0,\gamma]$, $d = 0$ and $v \le 1$, so its exact contribution is $-I_D$. All later pieces with $v \le 1$ are favorable. The remaining fold defect is paid by Equation (C.23), since $q+1 \le 5/4$ there:

$$\int_{\{v>1\}} (q+1)(v-1)(1-d) \le \tfrac{5}{16} \int_\beta^L (1-u) \le \frac{5T}{16\sin\beta}.$$

It follows that

$$\mathcal{H}(L) - \mathcal{H}(0) \le I_B - I_D + 5T/(16\sin\beta). \tag{C.25}$$

The endpoint pressures are those in Equation (C.17). The harmonic reconstructions used in the two-unit proof remain valid, but the first horizontal moment is now S rather than T. Thus

$$\begin{aligned} \mathcal{H}(L) - \mathcal{H}(0) &= 6CT + T^2 - (h+z_+)(h+z_+/2), \\ I_D &= 3CS - S^2/2 + e_R h + h^2/2 + R_D, \\ I_B &= 3CT + T^2/2 + e_L z_+ - z_+^2/2 + R_B. \end{aligned}$$

The remainders have exactly the definitions in Equation (C.19); $R_D \ge 0$ because $v \le 1$ before γ. Substitute these expressions into Equation (C.25) and use the endpoint pressures. The result is

$$\begin{aligned} 6CT \le{}& (z_+ - h)(2 - h - z_+)/4 + R_B - R_D \\ &+ (S-T)[(S+T)/2 - 3C] + 5T/(16\sin\beta). \end{aligned} \tag{C.26}$$

The displacement term is nonpositive: $S \ge T$, $(S+T)/2 \le (1+C)/2 < 3C$. Using $v \le 5/4$ in the backward sine convolution gives

$$R_B \le \tfrac{5}{4} \frac{1-\sin\beta}{\sin\beta} T, \qquad z_+ \le \cot\beta\, T.$$

Since $h + z_+ < 2$, the first term is at most $z_+/2$. After dropping $-R_D$ and the displacement term, the three remaining costs are the endpoint niche, the flux remainder, and the fold defect. Together they bound the right side of Equation (C.26) by

$$f(\beta)T, \qquad f(\beta) = \frac{1}{2}\cot\beta + \frac{5}{4}\frac{1-\sin\beta}{\sin\beta} + \frac{5}{16\sin\beta}.$$

f decreases on $(0, \pi/2]$. Since $\beta > \arctan(3/4)$, the 3–4–5 sine and cosine values give $f(\beta) < f(\arctan(3/4)) < 3$. But $6CT > 3T$ for $C > 1/2$ and $T > 0$, the required contradiction. □

Appendix D. The low middle-height estimate

The low endpoint limits the area in one exterior quarter. As the cap gets wider, the diagonal tent removes a quadratic amount of middle area. We compare these two quantities, splitting only when a tent tail reaches a boundary of the middle interval.

Lemma D.1. *Let U be a height-one downward cap with affine middle roof. If its higher middle endpoint has height one and its lower middle endpoint has height at most $1/2$, then, for every $\alpha \in [\pi/4, \pi/2]$,*

$$P_\alpha(U) < \frac{4}{5} < M/2.$$

Proof. The barrier dominates the positive two-wall tent at $\pi/4$. The resulting one-angle upper comparison is invariant under horizontal reflection, since reflection exchanges its two diagonal supports. We may therefore center the projection and put

$$I = [-2C, 2C], \quad J = [-C, C], \quad A(C) = 1, \quad A(-C) = a \le 1/2.$$

For $x = -2C + z$, $0 \le z \le C$, concavity below the middle chord gives

$$A(x) \le (3a - 1)/2 + (1 - a)z/(2C) \le 1/4 + z/(4C).$$

The exterior reward is consequently at most $11C/8$. For $C \le 1/2$ this is at most $11/16$, proving the claim in that range. Suppose $C \ge 1/2$, and write

$$\max_U (x + y) = 2C + u, \qquad \max_U (-x + y) = 2C + v.$$

The baseline endpoints, $(C, 1)$, and the preceding roof bound give

$$\max(0, 1 - C) \le u \le 1, \qquad 0 \le v \le 1/4.$$

For the last upper bound, on the left wing $-x + A(x) \le 2C + 1/4 - (1 - 1/(4C))z$. On the middle chord it is at most $C + 1/2 \le 2C + 1/4$, and the right wing gives an even smaller bound.

On the right wing set $z = 2C - x$; then $A(x) \le \min(1, u + z)$. Since $1 - u \le C$, its integral is at most $C - (1 - u)^2/2$. On the left wing compare $v + z$ with $1/4 + z/(4C)$. Their crossing is $z_0 = (1/4 - v)/(1 - 1/(4C)) \le C$. The missing triangle gives

$$R \le \frac{11C}{8} - \frac{(1 - u)^2}{2} - \frac{(1/4 - v)^2}{2}.$$

The actual diagonal tent is $(h - |x - d|)_+$, where

$$h = 2C + (u + v)/2 - \sqrt{2}, \qquad d = (u - v)/2.$$

Its apex lies in J. Put $C_0 = \sqrt{2} - 1/4$. If $1/2 \le C \le C_0$, its left tail cannot reach the left endpoint, since $h - C - d = C + v - \sqrt{2} \le C + 1/4 - \sqrt{2} \le 0$.

Thus C_0 is the width at which that tail can first reach the left endpoint; no additional width tolerance is needed. Writing $z = C + u - \sqrt{2}$, exact integration therefore gives the upper comparison

$$P_\alpha(U) \le F(C,u,v) := \frac{11C}{8} - \frac{(1-u)^2}{2} - \frac{(1/4-v)^2}{2} - h_+^2 + \frac{z_+^2}{2}. \quad \text{(D.1)}$$

If $z \le 0$, put $x_0 = 1-u$, $y_0 = 1/4 - v$ and $H = 2C + 5/8 - \sqrt{2} > 0$. With $w = (x_0+y_0)/2$,

$$\frac{x_0^2+y_0^2}{2} + (H-w)_+^2 \ge w^2 + (H-w)_+^2 \ge H^2/2.$$

The second inequality follows by completing the square for $w \le H$, and by $w^2 \ge H^2$ for $w \ge H$. Completing the square in C now yields

$$F \le \frac{11\sqrt{2}}{16} - \frac{99}{512} < \frac{11}{16}\frac{10}{7} - \frac{99}{512} < 4/5,$$

using $\sqrt{2} < 10/7$.

For $z > 0$, the allowed interval in u starts at $u_0 = \sqrt{2} - C$. Here $h > 0$, and direct differentiation gives

$$F_u = 1 - C - (u+v)/2, \qquad F_{uu} = -1/2.$$

Its maximum is either at u_0, already covered, or at $u_* = 2 - 2C - v$. The upper endpoint introduces no extra case, since $C \ge 1/2$ implies $u_* \le 1$. An interior stationary point in the clipped range satisfies $C + v \le 2 - \sqrt{2}$. Hence $1/4 - v \ge \sqrt{2} - 5/4 > 0$. Discard the other nonnegative losses in (D.1) to get

$$F(C,u_*,v) \le \frac{11(2-\sqrt{2})}{8} - \frac{1}{2}(\sqrt{2}-5/4)^2 = \frac{31}{32} - \frac{\sqrt{2}}{8} < \frac{31}{32} - \frac{1}{8}\frac{7}{5} < 4/5,$$

using $\sqrt{2} > 7/5$.

Finally, for $C \ge C_0$ the baseline diagonal tent alone gives

$$P_\alpha(U) \le \begin{cases} 11C/8 - (2C - \sqrt{2})^2, & C_0 \le C \le \sqrt{2}, \\ (11/8 + 2\sqrt{2})C - 3C^2, & C \ge \sqrt{2}. \end{cases}$$

Both expressions decrease on their stated ranges and agree at $C = \sqrt{2}$. Their largest value is therefore at the geometric transition $C_0 = \sqrt{2} - 1/4$. At this point it is

$$\frac{11C_0}{8} - (2C_0 - \sqrt{2})^2 = \frac{19\sqrt{2}}{8} - \frac{83}{32} < \frac{19}{8}\frac{10}{7} - \frac{83}{32} < \frac{4}{5}.$$

This bounds the remaining widths and completes the proof. □

Appendix E. Estimates for the terminal tail

E.1. **The initial-angle exclusion.** At a small terminal angle, the outgoing wall alone removes enough area. For the remaining angles we retain that wall together with the visited diagonal pair. Their outer supports bound the reward, and their inner walls bound the penalty. Integrating fixed pieces

of these walls gives supporting planes; two square completions bound the relaxation.

Proposition E.1. *For every downward cap U and $\pi/4 \le \alpha \le \arctan(8/3)$,*

$$P_\alpha(U) < \frac{M}{2}.$$

Proof. First put $\kappa = \cot\alpha$ and center the projection as $I = [-2C, 2C]$. Write $A = f_U$ and $H = h_U(\alpha)/\sin\alpha$. The roof satisfies $A(x) \le H - \kappa x$ and the whole terminal wall is $H - \csc\alpha - \kappa x$. The left exterior roof is at most one; integrate the support bound on $[C, 2C]$ and the terminal lower bound on $[-C, 0]$, even where that affine bound is negative. This gives

$$P_\alpha \le (1 + \sqrt{1+\kappa^2})C - 2\kappa C^2 \le \frac{(1+\sqrt{1+\kappa^2})^2}{8\kappa}.$$

On $4/5 \le \kappa \le 1$ the last expression decreases: its derivative has the sign of $(\sqrt{1+\kappa^2} - 2)(\sqrt{1+\kappa^2} + 1)$. Its largest value is $(33 + 5\sqrt{41})/80 < \underline{p}$. This handles $\alpha \le \arctan(5/4)$.

The support data forced by the passage. Lemma 4.4 and Proposition 4.3 supplies height extrusion, middle-chord reduction, coercivity, attainment, and top insertion. The attainment proof also applies with α fixed: if a cap has score greater than $\underline{p}$, the fixed-angle maximum is attained inside the same compact width interval. Indeed the width-six calculation in Lemma 4.1 is below $3/4$, so every cap with score above $\underline{p}$ has width below six. A maximizing sequence at this fixed angle can therefore be centered in the same compact box; score continuity and Blaschke compactness give its maximum. Its canonical representative has height one, an affine middle roof, and a top face meeting the middle window. Thus, a hypothetical cap with score at least $M/2$ yields such a canonical fixed-angle maximizer with at least the same score. It suffices to rule out that maximizer.

Center the projection and introduce the parameters

$$\begin{gathered} I = [-2C, 2C], \quad J = [-C, C], \\ K = \sqrt{2} - 1, \quad \kappa = \cot\alpha, \quad d = \frac{\kappa}{1 + \sqrt{1+\kappa^2}}. \end{gathered} \tag{E.1}$$

The outgoing-wall estimate above handles $\alpha \le \arctan(5/4)$. For the new interval it suffices to take

$$\begin{gathered} \frac{3}{8} \le \kappa \le \frac{4}{5}, \qquad \kappa d = \sqrt{1+\kappa^2} - 1, \\ \kappa = \frac{2d}{1-d^2}, \qquad 0 < d < K. \end{gathered} \tag{E.2}$$

Use the actual supports to define

$$u = \max_U (x+y) - 1, \quad v = \max_U (-x+y) - 1, \quad b = \max_U \left(x + \frac{y-1}{\kappa} \right). \tag{E.3}$$

The three outer support lines are $1+u-x$, $1+v+x$, and $1-\kappa(x-b)$. The actual outgoing inner wall is $\kappa(b-d-x)$. Consequently

$$\begin{aligned} A(x) &\le \min\{1, 1+u-x, 1+v+x, 1-\kappa(x-b)\}, \\ N_\alpha(U)(x) &\ge \max\{0, \min(u-K-x, v-K+x), \kappa(b-d-x)\}. \end{aligned} \tag{E.4}$$

The 45-degree angle is genuinely visited throughout the new interval. Only the displayed lower bound on the true niche is used.

There is an essential compatibility condition between the two first supports:

$$\boxed{b \le u \le 2(1-\kappa)C + \kappa b, \qquad u, v, b \le 2C.} \tag{E.5}$$

Indeed, $y \le 1$ and $\kappa < 1$ give $x + (y-1)/\kappa \le x + y - 1$, hence $b \le u$. At a point attaining u,

$$\kappa b \ge \kappa x + y - 1 = u - (1-\kappa)x \ge u - 2(1-\kappa)C.$$

The remaining upper bounds follow directly from the containing box.

Since the middle is affine and its top meets J, either $A(-C) = 1$ or $A(C) = 1$. The first alternative gives $v \ge C$ and $u, b \ge -C$; the second gives $u, b \ge C$ and $v \ge -C$. These alternatives will be handled directly.

The natural diagonal width. Put

$$g_C(z) = \int_C^{2C} (x - z)_+ \, dx = \frac{(2C-z)_+^2 - (C-z)_+^2}{2}.$$

For the diagonal comparison only, reflect the cap if necessary so that $v \in [C, 2C]$. Dropping the terminal wall gives

$$P_\alpha \le F_{45}(u, v) := 2C - g_C(u) - g_C(v) - \int_{-C}^{C} [\min(u - K - x, v - K + x)]_+ \, dx. \tag{E.6}$$

The diagonal tent and the two exterior losses give

$$F_{45} \le \max\{2/3, \ 2C - 2(C - K/2)^2\}. \tag{E.7}$$

Here is a direct way to check the clipping. A nonpositive apex, or an apex to the left of the window, gives the first bound from the exterior losses alone. Otherwise, if $u \ge C$, integrate the positive tent and use the square supporting inequality $z_+^2 \ge 2qz - q^2$, $q \ge 0$, at the balanced supports $u = v = C + K/2$. If $u < C$, the first wall comparison below, with $\kappa = 0$, gives the same second bound. If $2C < K$, use $P_\alpha \le 2C < K < 2/3$ instead.

For $C \le K$, the quadratic in (E.7) increases with C, and its endpoint value is

$$2K - K^2/2 = 3K - 1/2 < 3/4 < \underline{p}.$$

Thus the angle exclusion needs only $C > K$. For later use, derive the width needed to separate the terminal corner. Its horizontal displacement is at

least $3C$, and its height increase is at most one, so the required inequality is $3C\sin\alpha > 1+\cos\alpha$. Put

$$\alpha_0 = \arctan(8/3), \qquad \kappa_0 = \cot\alpha_0 = 3/8,$$
$$C_{\mathrm{corner}} = \frac{1+\cos\alpha_0}{3\sin\alpha_0} = \frac{\sqrt{1+\kappa_0^2}+\kappa_0}{3}. \tag{E.8}$$

The diagonal bound already excludes $C \le C_{\mathrm{corner}}$. Indeed one Newton step from 9 bounds $\sqrt{73}$, while the two steps defining k_+ bound K. With $C_+ = (\mathcal{N}_{73}(9)+3)/24$, direct substitution gives

$$2C_{\mathrm{corner}} - 2(C_{\mathrm{corner}} - K/2)^2 \le 2C_+ - 2(C_+ - k_+/2)^2 < \underline{p}.$$

Both comparisons increase with their parameters in the displayed range. Consequently every competitive canonical cap has $C > C_{\mathrm{corner}}$. For $\alpha \ge \alpha_0$, increasing the angle increases its sine and decreases its cosine, so this same width gives $3C\sin\alpha > 1+\cos\alpha$. The width cut is therefore exactly what the corner geometry asks for.

Reducing to one support domain. For $b \le u$, write

$$\begin{aligned} x_0 &= \frac{u-\kappa b}{1-\kappa}, \\ Q_C(u,b) &= \int_C^{2C} \max\{0, x-u, \kappa(x-b)\}\,dx \\ &= \kappa g_C(b) + (1-\kappa) g_C(x_0). \end{aligned} \tag{E.9}$$

The identity follows from $b \le u \le x_0$: the three successive active pieces are zero, $\kappa(x-b)$, and $x-u$, with any pieces outside $[C,2C]$ automatically clipped by g_C.

Define

$$\begin{aligned} N(u,v,b) &= \int_{-C}^{C} \max\{0, \min(u-K-x, v-K+x), \kappa(b-d-x)\}\,dx, \\ F(u,v,b) &= 2C - g_C(v) - Q_C(u,b) - N(u,v,b). \end{aligned} \tag{E.10}$$

In either high-endpoint alternative the left exterior majorant in (E.4) is $\min(1, 1+v+x)$, and the right exterior majorant is $1-\max(0, x-u, \kappa(x-b))$. Thus $P_\alpha \le F$.

Suppose first that the high endpoint is on the right and $v < C$. Increase v to C, keeping u, b fixed. This is a variation of the relaxation; it need not preserve realizability by a cap. The charged left exterior reward increases by C times the support increment. Subtracting the terminal line leaves a positive asymmetric tent. Its complete-area increase bounds the change of the clipped niche, and is at most the final rising-wall exposure length times that increment. Put

$$p = \frac{u - v}{2}, \quad e = b - d, \quad x_L = \frac{\kappa e - v + K}{1 + \kappa}. \tag{E.11}$$

The rising wall of that complete tent runs from x_L to p, so its length is $(p - x_L)_+$. Since $u \le 2C$, $v \le C$, and $b \ge C$,

$$p - x_L \le \frac{(3 - \kappa)C + 2\kappa d - 2K}{2(1 + \kappa)} < C. \tag{E.12}$$

The strict inequality uses $\kappa > 1/3$ and $\kappa d < K$. The exterior reward therefore pays for every finite niche increase, and F increases until $v = C$. All of (E.5) is preserved. It is therefore enough to bound F on the convex domain

$$\boxed{D_C = \{C \le v \le 2C,\ -C \le b \le 2C,\ b \le u \le 2(1 - \kappa)C + \kappa b\}.} \tag{E.13}$$

This also contains the original high-left alternative. No reflection of the partial objective has been used.
Two wall comparisons. Write

$$a = u - K, \quad c = v - K, \quad e = b - d, \quad p = (u - v)/2, \qquad Y = (1+\kappa)C - c + \kappa e.$$

If the diagonal apex is nonpositive, or lies to the left of J, the exterior losses already give $F \le 2/3$, as above. We may therefore assume $u + v \ge 2K$ and $p \in J$.

The falling diagonal line is the only concave part of the wall maximum. Separate it with the pointwise inequality

$$\max(B, L) - (B - A)_+ + (-A)_+ \le \max\{0, \min(A, B), L\}, \qquad A + B \ge 0. \tag{E.14}$$

To verify it, split at $A = B$ and $A = 0$. With $A = a - x$, $B = c + x$, and $L = \kappa(e - x)$, integration gives

$$N \ge V - (C - p)^2 + \int_{-C}^{C} (x - a)_+ \, dx, \qquad V = \int_{-C}^{C} \max(c + x, \kappa(e - x)) \, dx. \tag{E.15}$$

The negative term is the area swept out by moving the tent apex. The last term retains the falling wall's floor loss.

The useful supporting planes come from integrating a chosen affine branch over a fixed interval. For any $q \in [C, 2C]$,

$$g_C(z) \ge \int_q^{2C} (x - z) \, dx.$$

The balanced diagonal support is $C + K/2$. Use $q = C$ for the terminal ramp and $q = C + K/2$ for the diagonal ramp in (E.9). In V, use the terminal line on $[-C, K/2 - C]$ and the rising diagonal line on $[K/2 - C, C]$. These

choices give, for every position of the actual switches,

$$\begin{aligned} Q_C &\ge \frac{3C^2}{2} - Cu + \frac{K}{2}\bigl(u - \kappa b - (1-\kappa)C\bigr) - \frac{(1-\kappa)K^2}{8}, \\ V &\ge 2Cc + \frac{K}{2}\bigl((1+\kappa)C - c + \kappa e\bigr) - \frac{(1+\kappa)K^2}{8}. \end{aligned} \tag{E.16}$$

The split positions balance exterior and niche exposure. Define their algebraic center and comparison value by

$$\begin{aligned} b_* = (K+d)/2, \qquad u_* &= (1-\kappa)C + K/2 + \kappa d/2, \\ v_* &= (1+\kappa)C + K/2 - \kappa d/2, \\ F_* &= 2C - 2(C - K/2)^2 - \kappa K(C - d/2). \end{aligned} \tag{E.17}$$

These formulas only center square identities; the point need not represent a cap or have any particular ordering of its wall intersections.

First suppose $a \le C$. The apex assumptions imply $a \ge -C$, so the last integral in (E.15) is $(C-a)^2/2$. Also $g_C(v) = (2C-v)^2/2$. Adding (E.16) and completing the square gives

$$F \le F_* - \frac{(u + v - 2C - K)^2}{4} \le F_*. \tag{E.18}$$

Now suppose $a \ge C$ and $Y \ge 0$. Compatibility places x_0 in $[C, 2C]$; it also gives $Y \le 2(1+\kappa)C$. Thus retain the two exact switch triangles:

$$Q_C \ge \kappa(3C^2/2 - Cb) + \frac{(2(1-\kappa)C - u + \kappa b)^2}{2(1-\kappa)}, \qquad V = 2Cc + \frac{Y^2}{2(1+\kappa)}.$$

Set $\xi = u - u_*$, $\eta = v - v_*$, $\zeta = b - b_*$, and $T = C + K - u_*$. Substitution into (E.15), dropping its nonnegative last term, gives

$$\begin{aligned} 2(F_* - F) \ge{}& 2T\xi - T^2 + \eta^2 \\ &+ \frac{(\xi - \kappa\zeta)^2}{1-\kappa} + \frac{(\eta - \kappa\zeta)^2}{1+\kappa} - \frac{(\xi - \eta)^2}{2}. \end{aligned} \tag{E.19}$$

Here $T \ge 0$, because $C \ge K > d/2$, and $\xi \ge T$, because $u \ge C + K$. The last three terms are nonnegative by weighted Cauchy–Schwarz: the two denominator weights sum to two. The right side is nonnegative, so $F \le F_*$.

We have proved

$$\boxed{F \le F_* \quad \text{whenever } a \le C \text{ or } Y \ge 0.} \tag{E.20}$$

Only the case $a > C$, $Y < 0$ remains.

The remaining domain pays a larger diagonal tent. Suppose $u > C + K$ and $Y < 0$. Combining the latter with (E.5) gives

$$v - u > (3\kappa - 1)C + K - \kappa d > 0. \tag{E.21}$$

In particular $u, v > C + K$, so both ends of the 45-degree tent are strictly positive. Its apex belongs to $(-C, C)$, since $u, v \in (C+K, 2C]$. Dropping the outgoing support and charge only enlarges the relaxation. With $h = (u+v)/2$, exact 45-degree integration therefore gives

$$F \le 2C - 2C^2 + 2KC - (h - C)^2. \tag{E.22}$$

Moreover, (E.21), $C \ge K$, and $\kappa d < K$ imply

$$\begin{aligned} h - C &> \frac{(3\kappa - 1)C + 3K - \kappa d}{2} > \frac{(1 + 3\kappa)K}{2} \ge g, \\ g &= \frac{(1 + 3\kappa_0)K}{2}. \end{aligned} \tag{E.23}$$

Thus the gap comes from the diagonal offset and the angular endpoint. Finally, $1 + K = \sqrt{2}$ gives $2(1 + K)C - 2C^2 \le 1$, so

$$\boxed{F < 1 - g^2 < \underline{p}.} \tag{E.24}$$

For the last comparison use the lower Newton enclosure $K > k_-$ from (G.1); substituting its defining formula and the formula for $\underline{p}$ is an exact rational check.

This completes all parameter and clipping cases in D_C.

Completing the width square. Completing the square in C in (E.20) gives the unrestricted bound

$$F_* \le B(\kappa) := \frac{1}{2} + \frac{K}{2} + \frac{K}{2}\sqrt{1 + \kappa^2} - \frac{\kappa(1 - K)}{2} + \frac{\kappa^2 K^2}{8}. \tag{E.25}$$

The maximizing width for this quadratic is $C = 1/2 + K(2 - \kappa)/4$; no assumption that a cap realizes the critical supports or this width is required for the upper bound. On $[3/8, 4/5]$,

$$\begin{aligned} B'(\kappa) &= \frac{K\kappa}{2\sqrt{1 + \kappa^2}} - \frac{1 - K}{2} + \frac{\kappa K^2}{4} \\ &\le \frac{2K}{5} - \frac{1 - K}{2} + \frac{K^2}{5} = \frac{5K - 3}{10} < 0. \end{aligned} \tag{E.26}$$

Hence the largest value occurs at $\kappa = 3/8$, where

$$\begin{aligned} B(3/8) &= b(K, \sqrt{73}), \\ b(K, s) &= \frac{(2 - 3/8)^2}{8} + K\left[\frac{1 + 3/8}{2} - \frac{(3/8)^2}{4} + \frac{s}{16}\right]. \end{aligned} \tag{E.27}$$

This endpoint can be compared directly with the reference area. For $q > 0$ and $x > 0$, write $\mathcal{N}_q(x) = (x + q/x)/2$. Squaring shows $\sqrt{q} \le \mathcal{N}_q(x)$, with strict inequality unless $x = \sqrt{q}$. Use two steps from $3/2$ for $\sqrt{2}$ and one step from 9 for $\sqrt{73}$:

$$B(3/8) < b(\mathcal{N}_2(\mathcal{N}_2(3/2)) - 1, \mathcal{N}_{73}(9)) < \underline{p} < M/2. \tag{E.28}$$

The middle inequality follows by substituting the Newton formulas and Equation (2.2).

All other cases were bounded strictly below $\underline{p}$. The earlier outgoing-only theorem covers $[\pi/4, \arctan(5/4)]$ by a value below $\underline{p}$. Fixed-angle attainment and the canonical reduction therefore prove the asserted universal inequality. □

E.2. **A width cut with the actual partial barrier.**

Proposition E.2. *A canonical proper joint maximum has $C < 37/50$.*

Proof. By Corollary G.4, it suffices to exclude $37/50 \le C < 4/5$ and $0 \le h = h_L + h_R < 1/3$. Relabel the high middle endpoint as the right one, reflecting both the cap and its actual barrier if necessary. Put $a = 2C$ and let $q_\pm$ be the values of this relabeled barrier at $\pm a/2$. The endpoint equations and roof concavity give

$$e_R = \tfrac{1}{2} + \tfrac{h}{4} + (3q_+ - q_-)/4 \le 1, \qquad e_L = \tfrac{1}{2} - \tfrac{3h}{4} + (3q_- - q_+)/4 \le 1 - 3h/2.$$

Use the upper bound on e_L in the first equation, and the upper bound on e_R in the second, to obtain

$$e_R \ge \tfrac{1}{3} + \tfrac{h}{2} + \tfrac{2}{3}q_+, \qquad e_L \ge \tfrac{1}{3} - \tfrac{2h}{3} + \tfrac{2}{3}q_-.$$

Lift the low wing by h, and use the diagonal parameters u, v and the functions F_h, S, G_h of (C.5). Write $r = \sqrt{2}$, $k = r - 1$, $c_8 = \cos(\pi/8)$, $\gamma = 1 - r/2$ and $t = 2c_8 - r$. The outer deficits $w_R = a - u$, $w_L = a - v + h$ consequently satisfy

$$0 \le w_R \le \tfrac{2}{3} - \tfrac{h}{2}, \quad 0 \le w_L \le \tfrac{2}{3} + \tfrac{2h}{3}, \quad w_R + w_L \le 1 + \tfrac{h}{2}.$$

The last inequality uses $e_R + e_L \ge 1 - h/2$.

Here the three-angle calculation must retain the floor. The exterior support triangles and the two extra tent triangles are exactly those in the proof of Lemma C.5: all their angles are visited, and the actual barrier dominates their tents. A triangle whose signed peak is Z loses, when clipped by the zero floor, at most Z_-^2/r. Completing the same two squares in the exterior support deficits therefore gives

$$P_\alpha(U) \le G_h(a, u, v) + \gamma\big[(w_R - t)_+^2 + (w_L - t)_+^2\big].$$

This derivation uses no upper bound on $q_\pm$ and applies to the relabeled actual barrier. The convex function in brackets attains its maximum over the displayed polygon on the sum boundary. Its endpoints are $(2/3 - h/2, 1/3 + h)$ and $(1/3 - h/6, 2/3 + 2h/3)$. The second pair is more spread out and has the larger value; since $1/3 - h/6 < t$, that value is $(D + 2h/3)^2$, where $D = 2/3 - t$.

Set $a_0 = 37/25$, $A = 1 - k$ and $\sigma_0 = (3r - 1)a_0/2 - (4c_8 - 2)$. The supporting-plane estimate of Lemma C.7 now gives

$$P_\alpha(U) \le B(a_0) - kh/2 - h^2/8 - \tfrac{2}{9}(\sigma_0 - Ah/4)^2 + \gamma(D + 2h/3)^2,$$
$$B(a) = a - (a - k)^2/2.$$

The proper row of Lemma C.8 bounds this expression at $h = 0$: the direct tilt loss pays for moving the stationary support and the single floor square together. At zero tilt its value is the proper row of Lemma C.9, so

$$P_\alpha(U) < \underline{p} < M/2.$$

This contradicts the reference competitor. □

E.3. **The circle gap available for curvature excess.** The terminal wall must exceed the companion's unit-curvature envelope over the reserve interval. The worst point is its right endpoint. Writing the circle difference as an average of its slope exposes the convexity. A single tangent comparison pays for widening the chord and shifting its center; then only the two width endpoints need checking. The small retained margin will pay for curvature excess.

Put $q = \cos\alpha_0$, $S = \sin\alpha_0$ and $\mathcal{A}_q(y) = q^{-1}\int_{y-q/2}^{y+q/2} x/\sqrt{1-x^2}\,\mathrm{d}x$. Retain the actual endpoint gap

$$\Gamma = \min_{C\in\{1/2,37/50\}} \left[\frac{2C-q}{S} - \mathcal{A}_q(C - q^3/8)\right].$$

Proposition E.3. *For the residual data of* (8.1)*, and* $0 \le z \le w$*,*

$$a_\alpha(-C+z) - \left[\sqrt{1-(C-T_L)^2} - \sqrt{1-(C+z)^2}\right] \ge \Gamma c.$$

Proof. Write $g(x) = \sqrt{1-x^2}$ and $K(x) = x/g(x)$. The wall decreases with z and the circle difference increases, so it suffices to take $z = w = c - d$. Since $T_R \ge 0$ and $T_L \le d$, the gap there is at least

$$\frac{c}{s}(2C - c) - [g(C-d) - g(C+c-d)].$$

Also $d \ge \delta = c/(1+s)$, since $H \le 1$. The bracket decreases with d. Put

$$\eta = \delta - c/2 = \frac{c^3}{2(1+s)^2} \ge c^3/8, \qquad \mathcal{A}_q(y) = \frac{1}{q}\int_{y-q/2}^{y+q/2} K(x)\,\mathrm{d}x.$$

The gap divided by c is at least $(2C-c)/s - \mathcal{A}_c(C-\eta)$. On all these intervals K is increasing and convex. Hence the average increases with its center and width and is convex in its center. For width monotonicity, pair the positive and negative integration variables after rescaling to $[-1/2, 1/2]$ and apply convexity. A linear function of C minus a fixed-width average is concave, so its minimum on $[1/2, 37/50]$ occurs at an endpoint.

The terminal-angle cut gives $c \le q := \cos\alpha_0$. Put

$$q^2 = \frac{\kappa_0^2}{1+\kappa_0^2}, \qquad S^2 = \frac{1}{1+\kappa_0^2}, \qquad y_r = C - r^3/8.$$

We compare every chord with the chord of width q. All endpoints lie below $37/50 + 3/16$; squaring gives $K \le 3$ there. When the center increases from x

to y, the left circle height decreases and the right height drops by at most $3(y-x)$. The height-difference formula for the average therefore gives

$$\mathcal{A}_q(y) - \mathcal{A}_q(x) \le \frac{3}{q}(y-x).$$

Widening the chord and then paying for its center shift yields

$$\mathcal{A}_c(y_c) \le \mathcal{A}_q(y_c) \le \mathcal{A}_q(y_q) + \frac{3(q^3-c^3)}{8q} \le \mathcal{A}_q(y_q) + \frac{9q}{8}(q-c).$$

The tangent to the unit circle at (q,S) says $qc+Ss \le 1$. Hence

$$\begin{aligned}\frac{2C-c}{s} &\ge \frac{(2C-c)S}{1-qc}\\ &= \frac{2C-q}{S} + \frac{(q-c)(1-2Cq)}{S(1-qc)}\\ &\ge \frac{2C-q}{S} + (q-c)(1-2Cq).\end{aligned}$$

Its gain pays for shifting the center, since $1-2Cq > 9q/8$ for $C \le 3/4$ and $q < 3/8$. Thus the whole cosine range reduces to one comparison:

$$\frac{2C-c}{s} - \mathcal{A}_c(C-\eta) \ge \frac{2C-q}{S} - \mathcal{A}_q(C-q^3/8).$$

The last expression is concave in C, so its minimum is Γ. We need only check that this gap pays for the cubic curvature overlap. Use one root procedure for both endpoints: let

$$U(v) = \mathcal{N}_v^{\circ 4}(1), \qquad L(v) = v/U(v).$$

Then $L(v) \le \sqrt{v} \le U(v)$, by the Newton square identity. Put $q_+ = U(q^2)$, $q_- = L(q^2)$ and $S_+ = U(S^2)$, with $S_- = L(S^2)$. At either width endpoint set $y_+ = C - (q^2/8)q_+$. Using q_+ moves the left coordinate left and the right coordinate right. Their circle heights therefore give

$$\begin{aligned}&\frac{2C-q}{S} - \mathcal{A}_q(C-q^3/8)\\ &\qquad \ge \frac{2C-q_+}{S_+} - \frac{U(1-(y_+-q_+/2)^2) - L(1-(y_++q_+/2)^2)}{q_-}.\end{aligned}$$

Substitution of the four Newton updates at $C = 1/2, 37/50$ gives

$$\Gamma > \frac{(4/5)(2/3)^3(3/8)^2}{6(2/3)S_-} > 0. \tag{E.29}$$

The quantity on the right is exactly the cubic overlap price required below. The error estimate below is strict, so this available gap suffices. □

E.4. **Reflected supports and the terminal impulse.** Read the companion support backwards from the terminal angle. The missing angular interval gives a free rotation, and the terminal facet gives one upward impulse. The first zero of $e - Dr + r^2/2$ is $D - \sqrt{D^2 - 2e}$. For $S > 2$, an energy bound $Q + 1 \le S$ gives excess height at most $S - 2$; solving $\ell/2 + \ell^2/4 = S - 2$ gives duration $\ell = \sqrt{4S - 7} - 1$. Comparing the sum of these two times accounts for changes that advance the crossing but lengthen the decay. Write $C_* = 37/50$ and $k_* = 3/8$ for the previously proved width and terminal-slope cuts.

Lemma E.4 (Crossing and decay comparison)**.** *For fixed D, e, η, write*

$$A = \sqrt{D^2 - 2e}, \qquad S = \sqrt{D^2 - e(1-e) + \eta}, \qquad R = \sqrt{4S - 7},$$

and put $W(D, e, \eta) = D - A + R - 1$. Suppose

$$2 - 2^{-6} \le D \le D_* \le 3C_* + k_*/4, \qquad 0 \le e \le e_* \le 1, \qquad 0 \le \eta \le k_*^2/4.$$

If $D_*^2 - 1/4 + \eta \ge 4$, then $W(D, e, \eta) \le W(D_*, e_*, \eta)$. In particular, for $2 \le D \le 111/50$ and $0 \le e \le 3/4$,

$$W(D, e, 0) < 2/5 + 1/3 = 11/15.$$

Proof. First increase the width while holding e, η fixed. Throughout the box, squaring gives $A \ge 11/8$, $S \le 7/3$, and $3/4 \le R \le 8/5$. For a width increase $\Delta \ge 0$, differences of squares give

$$A(D + \Delta) - A(D) \le 2\Delta, \qquad S(D + \Delta) - S(D) \ge \frac{4}{5}\Delta,$$

and

$$R(D + \Delta) - R(D) = \frac{4[S(D + \Delta) - S(D)]}{R(D + \Delta) + R(D)} \ge \frac{5}{4}[S(D + \Delta) - S(D)] \ge \Delta.$$

Thus the decay gain pays for the change of the first-zero time: $\Delta - 2\Delta + \Delta = 0$ suffices.

At width D_* every energy is at least four, because $e(1 - e) \le 1/4$. Hence $S \ge 2$ and $R \ge 1$ for every height. Also $S^2 - A^2 = e + e^2 + \eta \ge 0$, so $S \ge A$. Compare heights $e \le E$ at this fixed width and write lower-case a, s, r for the roots at e, and A, S, R for those at E. The same difference-of-squares identities give

$$W(D_*, E, \eta) - W(D_*, e, \eta) = \frac{2(E - e)}{a + A} + \frac{4(E - e)(E + e - 1)}{(s + S)(r + R)} \ge 0.$$

Indeed $(s + S)(r + R) \ge 2(a + A)$ and $E + e - 1 \ge -1$. This proves the corner comparison without a height subdivision. At $(D_*, e_*) = (111/50, 3/4)$, substitution in $e_* - D_* t + t^2/2$ gives a first zero below $2/5$. The energy is below $(2 + (1/3)/2 + (1/3)^2/4)^2$, giving $R - 1 < 1/3$. These two checks prove the final assertion. □

Lemma E.5 (One angular stopping budget)**.** *Use the zero-slope span and pressure bounds* $D_0 = 3C_*$ *and* $e_0 = 1/2 + (3/4)(1/3) = 3/4$. *Put* $B_0 = 11/15$ *and*

$$b(k) = (2C_*k - 1/3)_+, \qquad B(k) = B_0 + 2k/3.$$

Suppose $0 \le k \le k_*$, $2 - 2^{-6} \le D \le D_0 + k/4$, $0 \le e \le 1$, $0 \le \eta \le k^2/4$, *and*

$$\tfrac{3}{2}(D - D_0)_+ + \tfrac{9}{10}(e - e_0 - \tfrac{3}{4}b(k))_+ \le 3k/8. \tag{E.30}$$

Then $W(D, e, \eta) \le B(k)$ *and* $R - 1 < 3/5$.

Proof. First increase D to $\bar{D} = \max(D, D_0)$, then e to $\bar{e} = \max(e, e_0)$. Since $D_0^2 - 1/4 > 4$, Lemma E.4 shows that these changes increase W. Put $\Delta D = \bar{D} - D_0$, $\Delta e = \bar{e} - e_0$. We price the change of the whole stopping time by

$$W(\bar{D}, \bar{e}, \eta) - W(D_0, e_0, 0) \le \tfrac{3}{2}\Delta D + \tfrac{9}{10}\Delta e + \tfrac{2}{5}\eta. \tag{E.31}$$

Here is the difference-of-squares calculation behind these prices. Let A_0, S_0, R_0 be the baseline roots, A, S, R the new roots, and $A_1 = \sqrt{\bar{D}^2 - 2e_0}$. Squaring gives $A, A_1 \ge 5/3$ and $A_1 \le 6\bar{D}/7$, with the analogous bound for A_0. Therefore

$$A_1 - A_0 \ge \tfrac{7}{6}\Delta D, \qquad A_1 - A \le \tfrac{3}{5}\Delta e.$$

For the decay, $S \ge S_0$, $R \ge R_0$, and

$$(R - R_0)(R + R_0)(S + S_0) = 4[(\bar{D} + D_0)\Delta D + (\bar{e} + e_0 - 1)\Delta e + \eta].$$

The baseline root product satisfies

$$S_0R_0 \ge \tfrac{3}{5}(2D_0 + k_*/4), \qquad S_0R_0 \ge 5/2.$$

These inequalities are checked by the same Newton enclosures below. Using $(R + R_0)(S + S_0) \ge 4S_0R_0$ and $\bar{e} + e_0 - 1 \le 3/4$, the last identity gives

$$R - R_0 \le \tfrac{5}{3}\Delta D + \tfrac{3}{10}\Delta e + \tfrac{2}{5}\eta.$$

Add the crossing and decay changes to obtain (E.31).

Now $\Delta D = (D - D_0)_+$ and $\Delta e \le \frac{3}{4}b(k) + (e - e_0 - \frac{3}{4}b(k))_+$. The parameter budget consequently gives

$$W(D, e, \eta) \le W_0 + J(k), \qquad W_0 = W(D_0, e_0, 0),$$
$$J(k) = 3k/8 + \tfrac{9}{10}\tfrac{3}{4}b(k) + \tfrac{2}{5}k^2/4.$$

The function J is convex. Thus it suffices to compare $W_0 + J(k)$ with the line $B(k)$ at 0 and k_*. Both comparisons use one evaluation of the baseline. Write $\mathcal{U}_x(q) = \mathcal{N}_q^{\circ 3}(x)$ and $\mathcal{L}_x(q) = q/\mathcal{U}_x(q)$ for three Newton steps and their reciprocal lower bound. They give

$$W_0 \le \widehat{W}_0 := D_0 - \mathcal{L}_2(D_0^2 - 3/2) + \mathcal{U}_1(4\mathcal{U}_2(D_0^2 - 3/16) - 7) - 1. \tag{E.32}$$

Inserting the Newton formulas verifies

$$\widehat{W}_0 < B_0, \qquad \widehat{W}_0 + J(k_*) < B_0 + 2k_*/3.$$

Convexity now gives the deadline throughout $[0, k_*]$. Finally the common energy bound is

$$D^2 - e(1-e) + \eta \le (D_0 + k_*/4)^2 + k_*^2/4 < [2 + (3/5)/2 + (3/5)^2/4]^2.$$

The defining equation for the decay length gives $R - 1 < 3/5$. □

Proposition E.6. *For a residual canonical proper joint maximum, the error in* (8.4) *satisfies* $\mathcal{E}_\theta < \Gamma c$.

Proof. For a rising middle use the left-wing companion surrogate of Lemma 8.7. For a descending middle use the right-wing first-support surrogate: it equals the actual first support on every visited angle and removes the unused middle atom. Saturation gives its unused tail at $(C, 1 - h_R)$. In every case the companion unused tail is $G(t) = 2C \sin t + e_L \cos t$ for $\alpha < t < L$. Set $\varepsilon = L - \alpha$, $k = \cot \alpha$, $\delta = c/(1+s)$, and reflect the support equations, without reflecting their spatial objective:

$$P(r) = -q(L-r), \quad Q(r) = -p(L-r), \quad U(r) = v(L-r), \quad V(r) = u(L-r).$$

The curvature and shadow laws give

$$\begin{gathered} P' = U - 1 - Q, \quad Q' = V - 1 + P, \quad P \le 1,\ Q \ge -1, \\ 0 \le U \le \kappa(Q), \quad 0 \le V \le \kappa(P), \\ P, Q > 0 \Rightarrow U = 0,\ V \le 1/2, \qquad P, Q < 0 \Rightarrow V = 0,\ U \le 1/2. \end{gathered} \tag{E.33}$$

In the final removed interval of the rising surrogate both curvatures are zero directly; no arm law is needed there. The initial values are $(P(0), Q(0)) = (e, D - 1)$, where

$$(e, D) = \begin{cases} (e_L, 3C + T_R), & \text{rising or horizontal middle,} \\ (e_L + h_R, 3C), & \text{descending middle.} \end{cases}$$

Both densities vanish on $(0, \varepsilon)$. At ε, P is continuous and Q jumps upwards by $j = \tau/s \le \delta$. There is no companion terminal atom or later visited atom. The free initial trajectory is

$$P = 1 + (e-1)\cos r - D \sin r, \qquad Q + 1 = D \cos r + (e-1) \sin r.$$

We retain the endpoint pressure and outgoing wall together. Let H be the actual terminal tail height, $h = 1 - H$, and $d_\alpha = (1 - Hs)/c$ its actual inset. The pressure law gives

$$e \le \tfrac{1}{2} + \tfrac{5}{4} h + \tfrac{3}{4} \max\{1/3, k(2C + T_R - d_\alpha)\}.$$

Here the ordinary endpoint box is below $1/3$ because $C < C_*$ implies $\sqrt{1 - C^2} > 2/3$. Also $\delta \le k/2$ and $1 - s \le k^2/2$; both follow from $s^2 + c^2 = 1$ and $c = ks$. For rising middle, $H = 1$ and $T_R \le \delta/2$, so

$$D \le D_0 + k/4, \qquad e \le e_0 + \tfrac{3}{4} b(k) \le 1.$$

For descending middle, $T_R = 0$ and $k d_\alpha = k\delta + h$. Thus the height loss also lowers the outgoing wall, and

$$e \le \max\{3/4 + 5h/4,\ 1/2 + h/2 + (3/2) C_* k\} < 1, \qquad 0 \le h < 1 - s \le k^2/2.$$

The last comparison follows at $k = k_*$; both expressions increase with h and k. In both orientations $D > 3/2$ and $e \geq 0$.

Put $\mathcal{H} = (P - 1/2)^2 + (Q + 1)^2$. Expansion at the impulse gives

$$\mathcal{H}(\varepsilon+) - \mathcal{H}(0) = (1 - e)(1 - s) - Dc + 2j[Ds - (1 - e)c] + j^2.$$

Since $Ds - (1 - e)c > 1$, this increases with $j \geq 0$. At $j = \delta$ it equals

$$\delta^2[e - (1 - e)s - Dc] < \delta^2. \tag{E.34}$$

So the geometric impulse allowance is $\eta = \delta^2 \leq k^2/4$. After the impulse, in the initial $P, Q > 0$ sector, $\mathcal{H}' = 2(Q + 1)(V - 1/2) \leq 0$. At the first P-zero, or at ε if that zero occurred earlier,

$$Q + 1 \leq \sqrt{Z}, \qquad Z = D^2 - e(1 - e) + \eta.$$

The initial Q component remains positive across the unused gap. Every later positive Q component has amplitude at most $1/8$: it starts at $Q = 0, P \leq 1$; while $P > 0$, the shadow laws give $P' \leq -1$, $Q' \leq P - 1/2$, so its total positive increase is at most $\int_0^{1/2}(1/2 - r)\,\mathrm{d}r = 1/8$. Once $P \leq 1/2$, Q' is nonpositive, and once $P \leq 0$ it is strictly negative. P cannot rise while $Q > 0$. Thus only the initial component can have $U > 1$, since that requires $Q > 1$ by (E.33).

After $P \leq 0$ and while $Q > 1$, write ρ for elapsed time. Then $P' \leq -1$, hence $P \leq -\rho$ and, for $\rho \leq 1$, $Q' \leq -1/2 - \rho/2$. For $-1 \leq P \leq 0$ use $V \leq (1 - P)/2$; for $P < -1$ use $V \leq -P$, giving $Q' \leq -1$. Consequently

$$Q - 1 \leq \sqrt{Z} - 2 - \rho/2 - \rho^2/4, \qquad \ell = \sqrt{1 + 4(\sqrt{Z} - 2)_+} - 1. \tag{E.35}$$

Once $0 < Q \leq 1$, one has $P' \leq -(1+Q)/2$ and $Q' \leq -1/2$, so this component cannot return to $Q > 1$. While $P, Q > 0$, $Q' \geq -1$ and the impulse is upward; therefore $P(r) \leq e - Dr + r^2/2$. The last possible excess time is at most $\max(\varepsilon, D - \sqrt{D^2 - 2e}) + \ell$.

Both orientations satisfy the single parameter budget (E.30). For rising middle only the span term is needed: $\frac{3}{2}(D - D_0)_+ \leq 3k/8$. For descending middle $D \leq D_0$ and the pressure bound gives $(e - e_0 - \frac{3}{4}b(k))_+ \leq 5k^2/8$. Its priced cost is at most $9k^2/16 \leq 3k/8$. If $Z \leq 4$ there is no excess after the zero or impulse. Otherwise $D > 2 - 2^{-6}$, so Lemma E.5 bounds the total time by $B(k)$ and gives $\ell < 3/5$. The other term of the maximum is covered too: $\varepsilon = \arctan k < k$ and $k + 3/5 < B(k)$ for $0 \leq k \leq k_*$.

The decay polynomial factors as $(\ell - \rho)(1/2 + (\ell + \rho)/4)$. Its second factor is below $1/2 + (3/5)/2 = 4/5$. Since $U - 1 \leq Q - 1$ whenever $U > 1$, the excess therefore satisfies

$$(U(r) - 1)_+ \leq \tfrac{4}{5}(B(k) - r)_+, \qquad (v(t) - 1)_+ \leq \tfrac{4}{5}(t - t_0)_+, \quad t_0 = L - B(k).$$

The prefix ends at $\theta = \arcsin(C - T_L) \leq \arcsin C$. The common cosine cut $\cos B_0 > C_*$ gives $\theta < L - B_0$, hence $(\theta - t_0)_+ \leq 2k/3$. Its kernel is

$\sin(\theta - t)/\cos\theta$, bounded by $(\theta - t)/\cos\theta$. Thus

$$\mathcal{E}_\theta \le \frac{4/5}{\cos\theta}\int_0^\theta (t - t_0)_+(\theta - t)\,\mathrm{d}t = \frac{(4/5)(\theta - t_0)_+^3}{6\cos\theta}. \tag{E.36}$$

Finally $\cos\theta > 2/3$ and $c = ks \ge S_- k$. By (E.29),

$$(4/5)(2/3)^3 k_*^2 < \Gamma\, 6(2/3)S_-.$$

Thus the cubic overlap is less than Γc throughout $0 < k < k_*$. This is precisely the circle gap retained above. □

Appendix F. The horizontal middle: width and curvature estimates

Throughout this appendix $L = \pi/2$. For a centered height-one cap write $I = [-2C, 2C]$, $J = [-C, C]$, $A = f_U$ and $n = n_L(U)$. A horizontal canonical middle has $A = 1$ on J, since the top face meets J by Lemma 4.5. We prove the following input to the full-turn argument.

Lemma F.1 (Horizontal unit curvature)**.** *Every canonical full maximum with horizontal middle satisfies*

$$\frac{1}{2} < C < C_H,$$

where $C_H = \sqrt{17}/6$ is the endpoint-energy threshold when the axis heights are one half. Its top face is exactly $J \times \{1\}$, and its two regular curvature densities satisfy $0 \le u, v \le 1$ almost everywhere. There are unique $0 < \beta < \gamma < L$ such that

$$\begin{array}{llll} p > 0 & (0, \beta), & p < 0 & (\beta, L), \\ q > 0 & (0, \gamma), & q < 0 & (\gamma, L). \end{array}$$

The preliminary estimates already give the range needed here. The zero-tilt case of Lemma G.1 excludes $C \le 1/2$, and Corollary C.10 excludes $C \ge 37/50$. Thus throughout the proof

$$\frac{1}{2} < C < \frac{37}{50}.$$

We will first bound how long curvature can exceed one. The key point is that this excess ends before a corner can cut a middle endpoint. The endpoint pressures then become exact, and the ordinary-sofa bound supplies the width estimate.

F.1. **The top face and transfer between quarters.** The top face already contains the middle interval. Any extra top length would consume part of the projection budget available to the positive niche. The following argument shows that this budget is fully used.

For a full maximum the endpoint law gives, with $n_\pm = n(\pm C)$,

$$\begin{gathered} e_R = \frac{2 + 3n_+ - n_-}{4}, \qquad e_L = \frac{2 + 3n_- - n_+}{4}, \\ E = e_R + e_L = 1 + (n_+ + n_-)/2. \end{gathered} \tag{F.1}$$

Both end heights are positive.

Lemma F.2. *If a horizontal canonical full maximum has $C \geq 1/2$, its top face is exactly $[-C, C] \times \{1\}$.*

Proof. Write the top face $[a, b] \times \{1\}$; it contains J. As $t \downarrow 0$, $F(t) \to 2C$ and $G(t) = 1 - at + o(t)$. For $a < x < 2C - 1$ the first wall tends to $+\infty$, and the second is $(x - a)t + o(t) > 0$. Thus $n(x) > 0$ there. Reflection gives positivity on $(-2C + 1, b)$. For $C \geq 1/2$ these intervals overlap or touch and cover the interior of J except possibly one point. The positive-barrier projection bound of Corollary 5.10 therefore gives

$$2C \leq 2C - (b - a - 2C).$$

Since $b - a \geq 2C$, equality holds and the endpoints are $-C, C$. Only lower semicontinuity on compact positive-barrier intervals is used in this projection bound. □

The endpoint support values now give

$$p(0) = e_R, \quad q(0) = 3C - 1, \quad p(L) = 1 - 3C, \quad q(L) = -e_L. \tag{F.2}$$

On $\{q > 0\}$ the curvature and shadow laws imply $p' \leq -1/2$; on $\{p < 0\}$ they imply $q' \leq -1/2$. Indeed, for $0 < q \leq 1$ the bound $u \leq (1 + q)/2$ gives $p' \leq -(1 + q)/2$; for $q > 1$, $u \leq q$ gives $p' \leq -1$. For $-1 \leq p < 0$, $v \leq (1 - p)/2$ gives $q' \leq (p - 1)/2$; for $p < -1$, $v \leq -p$ gives $q' \leq -1$. The positive-positive sector also has

$$p' = -1 - q \leq -1, \qquad q' \leq p - 1/2.$$

Consequently any positive-q component beginning at $t_A > 0$ has $q \leq 1/8$: until p becomes negative, $p(t_A + s) \leq 1 - s$ and the increase of q is at most $\int_0^{1/2} (1/2 - s)\, ds = 1/8$; after that crossing q decreases. At a crossing $p = 0$ with $q > 0$, $p' < 0$, so the crossing on such a component is unique.

Lemma F.3 (Endpoint energy)**.** *If $9C^2 \leq 4 + e_R(1 - e_R)$, then $u \leq 1$ almost everywhere. The reflected criterion with e_L implies $v \leq 1$.*

Proof. On the initial positive-positive sector,

$$\mathcal{E} = (p - 1/2)^2 + (q + 1)^2, \qquad \mathcal{E}' = 2(q + 1)(v - 1/2) \leq 0.$$

If p crosses zero on the initial positive-q component, at that crossing $(q+1)^2 \leq 9C^2 - e_R(1 - e_R) \leq 4$. Thereafter q decreases on that component, and later positive components have $q \leq 1/8$. If the initial component ends before the crossing, its positive-positive part has $u = 0$. Thus whenever $q > 1$, one has $u = 0$; elsewhere $-1 \leq q \leq 1$ and $u \leq \kappa(q) \leq 1$. □

Lemma F.4 (One controlled quarter controls both)**.** *For a horizontal canonical full maximum with $C > 1/2$, either $u \leq 1$ or $v \leq 1$ almost everywhere implies both inequalities.*

Proof. The two quarters share the same niche. If one curvature density stays below one, its wall moves monotonically and makes part of that niche visible. The source law on this visible part then constrains the other density. We show that an excursion above one would force a growing energy to finish below its starting value.

Reflect so that $v \le 1$, and suppose $u > 1$ on a set of positive measure. Such a point has $q > 1$ and belongs to the initial positive-q component after its $p = 0$ crossing. This component ends at $\tau < L$. Let b_0 be its unique subsequent $q = 1$ crossing; then $p < 0 < q < 1$ on (b_0, τ). Every future positive component has $q \le 1/8$, so $u \le 1$ on (b_0, L).

For the inner corner $c = (F-1)u(t) + (G-1)u^{\perp}(t)$ put

$$B = c + pu^{\perp}(t), \qquad D = c - qu(t).$$

Differentiation gives $B' = (u-1)u^{\perp}(t)$ and $D' = (1-v)u(t)$. The top endpoints give $D(0) = (-C, 0)$ and $B(L) = (C, 0)$. Thus D_x, D_y are globally nondecreasing, and B_x is nondecreasing after b_0. On (b_0, τ),

$$-C \le D_x < c_x < B_x \le C, \qquad c_y = D_y + q \sin t > 0. \tag{F.3}$$

The D graph has positive height wherever its arc density $1 - v$ is nonzero: the derivative of D_y vanishes almost everywhere on its zero level set. Its contributing points lie strictly inside J.

The wall parameter derivatives are

$$\partial_t a_t(x) = \frac{x - B_x(t)}{\sin^2 t}, \qquad \partial_t b_t(x) = \frac{x - D_x(t)}{\cos^2 t}.$$

Global monotonicity of D_x makes $D(t)$ a maximum of the entire second-wall family, with positive companion gap $q/\sin t$. Unless D is constant on a parameter interval, which has zero arc contribution, its active source parameter is unique. At $c(t)$ every earlier second wall is strictly lower, because $c_x(t) - D_x(s) \ge q(t)\cos t > 0$ for $s < t$; every later first wall is strictly lower, because $c_x(t) - B_x(s) \le p(t)\sin t < 0$ for $s > t$. The corner likewise has unique global source parameter and strictly decreasing abscissa. For $K = [t_0, t_1] \subset (b_0, \tau)$ the two graph images are disjoint, since $D_x(t_1) < c_x(t_1)$.

We can therefore add their finite source fluxes. On positive-height compact subarcs, active finite source parameters converge to the unique limiting parameter: any contrary subsequence would supply a different maximizing parameter by uniform wall convergence. The angles stay away from the axes, so graph slopes are bounded. With increasing abscissa the finite vector flux is

$$(\mathrm{d}x, \mathrm{d}y) = -u^{\perp}(t)\,\mathrm{d}\nu_i^f + u(t)\,\mathrm{d}\nu_i^g.$$

Its first component is Lebesgue measure, and its second converges by uniform roof convergence and the common slope bound. Solving the two-direction system gives $\mathrm{d}\nu^g = \cos t\,\mathrm{d}x + \sin t\,\mathrm{d}y$. The D graph therefore contributes $(1-v)\mathrm{d}t$ to this source, and the reverse-oriented corner contributes $-p\mathrm{d}t$.

Exhaustion removes constant and zero-height pieces. Source equality from Theorem 5.8 yields

$$2v \geq 1 - p \quad \text{almost everywhere on } (b_0, \tau).$$

Since $v \leq 1$, $p \geq -1$ there. The upper curvature bound gives the reverse inequality $v \leq (1-p)/2$, hence equality. Set $x = 1 - p$, $y = 1 + q$. Then

$$x' = 1 + q - u \geq y/2, \qquad y' = -x/2, \qquad (x^2 + y^2)' \geq 0.$$

At b_0, $y = 2$ and $x > 1$, so this energy exceeds five; at τ, $y = 1$ and $x \leq 2$, so it is at most five. This contradiction proves the transfer. □

F.2. **Curvature excess ends before endpoint contact.** The box bounds the endpoint niches by

$$0 \leq n_\pm \leq H(C) := 1 - \sqrt{1 - C^2}.$$

Indeed, $a_t(C) \leq 1 + (C\cos t - 1)/\sin t$, whose maximum is $H(C)$; reflection gives the other endpoint. By (F.1), either axis height e satisfies

$$\frac{1}{2} - \frac{H(C)}{4} \leq e \leq \frac{1}{2} + \frac{3H(C)}{4}. \tag{F.4}$$

If $C \leq 2/3$, then $9C^2 \leq 4 \leq 4 + e(1-e)$ because $0 \leq e \leq 1$. The endpoint-energy lemma already gives both unit bounds in this range. We therefore consider $C > 2/3$.

Lemma F.5 (Dry middle endpoints)**.** *If $2/3 < C < 37/50$, then $n(-C) = n(C) = 0$ and $e_R = e_L = 1/2$.*

Proof. Fix the right endpoint $e = e_R$. As in the endpoint-energy proof, any $u > 1$ must occur on the initial positive-q component, after its unique $p = 0$ crossing t_A. Before that crossing, $q' \geq -1$ and $p' = -1 - q$, so

$$p(t) \leq e - 3Ct + t^2/2.$$

The circle identity gives a useful uniform bound on the pressure. For $C \leq 4/5$, $\sqrt{1-C^2} \geq 3/5$, hence

$$H(C) = \frac{C^2}{1 + \sqrt{1-C^2}} \leq \frac{5}{8}C^2, \qquad e \leq \frac{1}{2} + \frac{15}{32}C^2.$$

The concave quadratic

$$\frac{6}{5}C - \frac{2}{25} - \frac{1}{2} - \frac{15}{32}C^2$$

is positive at both endpoints of $[2/3, 4/5]$, so it is positive throughout. Thus p must cross zero by $t = 2/5$, unless the positive-q component has already ended; that case has no excess.

We next bound the duration after the crossing. Since $C < 37/50$, $H(C) < 1/3$ and (F.4) gives $1/4 \leq e \leq 3/4$. The initial energy therefore satisfies

$$9C^2 - e(1-e) \leq 9(37/50)^2 - 3/16 < \left(2 + \frac{(1/3)^2 + 2/3}{4}\right)^2.$$

At the crossing, this implies $q(t_A) - 1 \le 7/36$. While $q > 1$, $p' \le -1$ and

$$p(t_A + \tau) \le -\tau, \qquad q' \le -(1+\tau)/2 \quad (0 \le \tau \le 1).$$

For $p \ge -1$ the second bound follows from $v \le (1-p)/2$; for $p < -1$ the stronger bound $q' \le -1$ suffices. Consequently

$$(u(t_A + \tau) - 1)_+ \le \left(\frac{7}{36} - \frac{\tau}{2} - \frac{\tau^2}{4}\right)_+. \tag{F.5}$$

The right side vanishes at $\tau = 1/3$. Later positive-q components have $q \le 1/8$, as proved above. Thus every possible curvature excess has ended by

$$\frac{2}{5} + \frac{1}{3} = \frac{11}{15}.$$

These are coarse crossing and duration bounds for the whole width range. We need only that their sum precedes endpoint contact. The quadratic bound at $t/2$ gives $\cos t = 2\cos^2(t/2) - 1 \ge 2(1 - t^2/8)^2 - 1$ at $t = 11/15$. This exceeds $37/50$, hence $11/15 < \arccos C$.

The exact top endpoint data $F(L) = 1$, $F'(L) = -C$ give

$$a_t(C) = \frac{1}{\sin t} \int_t^L \sin(s - t)(u(s) - 1)\, \mathrm{d}s.$$

For $t \ge \arccos C$ the integrand is nonpositive, so $a_t(C) \le 0$. For earlier angles, the wall increases: if its supporting point has abscissa $x_t \le 2C$, then

$$\partial_t a_t(C) = \frac{C - x_t + \cos t}{\sin^2 t} \ge 0 \qquad (t \le \arccos C).$$

It is therefore bounded above by the nonpositive wall at $\arccos C$. Every visited angle has $a_t(C) \le 0$, proving $n(C) = 0$. Reflection gives $n(-C) = 0$, and the endpoint-pressure identities give $e_R = e_L = 1/2$. □

F.3. **Circle confinement and the ordinary survivor.** The next circle inequality keeps the corners inside the middle window. We state it with a tilt parameter because the same argument will be used in Appendix H.

Lemma F.6 (Circle margin)**.** *If $s, c > 0$, $s^2 + c^2 = 1$, $1/2 \le C \le 37/50$, and $h \le 1/3$, then*

$$c - s - C(1 - 3s^2) - hsc > 0.$$

Proof. Put $D = 37/50$, $H = 1/3$. When $s^2 \le 1/3$, $c \ge 4/5$, so the circle identity gives

$$c = 1 - \frac{s^2}{1 + c} \ge 1 - \frac{5}{9}s^2.$$

Also the tangent to s^3 at $s = 1/3$ lies below that cubic:

$$s^3 - \frac{s}{3} + \frac{2}{27} = (s - 1/3)^2(s + 2/3) \ge 0.$$

Substituting these two bounds into the margin gives the quadratic as^2+bs+d, where

$$a = 3D - 5/9, \quad b = -(1+H) + 5H/27, \quad d = 1 - D - 10H/243.$$

Here $a > 0$ and $4ad - b^2 > 0$. Thus completing the square proves positivity for every s.

When $s^2 \geq 1/3$, decrease C to $1/2$ and increase h to $2/5$. The resulting margin factors as

$$c\left[1 - \frac{2}{5}s - \frac{c(3s+1)}{2(1+s)}\right].$$

The 3–4–5 tangent inequality $3s + 4c \leq 5$ bounds the bracket below by

$$\frac{15 - 36s + 29s^2}{40(1+s)}.$$

Its numerator is positive because $36^2 < 4 \cdot 29 \cdot 15$. □

Continue to assume $C > 2/3$. For the horizontal cap, put $s = \sin t$, $c = \cos t$. The box and left top endpoint imply $F \leq 2Cc + s$ and $G \geq Cs + c$, whence

$$c_x \leq C(2 - 3s^2) - c + s < C$$

by Lemma F.6 with $h = 0$. Reflection gives $-C < c_x$. Each exterior wing therefore sees only one affine wall from every positive quadrant. Its niche is convex. The box gives $n(\pm 2C) = 0$, and Lemma F.5 gives $n(\pm C) = 0$. Convexity and nonnegativity make both wings dry. On J, Corollary 5.9 gives $n \leq E/2 = 1/2 < A$. Thus $n \leq A$ on I, and Lemma A.2 makes the actual survivor an ordinary sofa.

The wing-length identity and the chord bound, now with both axis heights equal to one half, give

$$P_L(U) = L_{\text{wing}}/2 \geq \sqrt{C^2 + 1/4}.$$

The survivor has area $P_L(U) + 2C$, since the wings are dry. The ordinary-sofa bound therefore gives

$$2C + \sqrt{C^2 + 1/4} \leq G_0 := 2.2199. \tag{F.6}$$

The 3–4–5 supporting direction bounds the square root below by $4C/5 + 3/10$. At $C = C_H = \sqrt{17}/6$ this yields

$$2C_H + \sqrt{C_H^2 + 1/4} \geq \frac{14}{5}\frac{\sqrt{17}}{6} + \frac{3}{10} > G_0.$$

The last comparison follows by squaring $\sqrt{17} > (G_0 - 3/10)\,15/7$; all terms are positive. The left side of (F.6) increases with C, so $C < C_H$.

Proof of Lemma F.1. The preliminary range gives $1/2 < C < 37/50$ and Lemma F.2 gives the exact top face. If $C \leq 2/3$, both endpoint-energy

criteria hold directly. If $C > 2/3$, the preceding argument gives dry endpoints, $e_R = e_L = 1/2$, and $C < C_H = \sqrt{17}/6$. Hence

$$9C^2 < 17/4 = 4 + e_R(1 - e_R) = 4 + e_L(1 - e_L),$$

so Lemmas F.3 and F.4 give $u, v \le 1$. In either case $C < C_H$; use $2/3 < C_H$ in the first case.

For the final sign statement, B_x, D_x are nondecreasing and

$$B_x - D_x = q\cos t - p\sin t \ge 4C - 2 > 0,$$

because $B_x \ge B_x(0) = 2C - 1$ and $D_x \le D_x(L) = 1 - 2C$. At any zero of p, this strict inequality gives $q > 0$, so $p' = u - 1 - q < 0$ nearby; a return through zero is impossible. At any zero of q, it gives $p < 0$, so $q' = v - 1 + p < 0$ nearby, likewise excluding a return. The endpoint signs in (F.2) give one crossing each. They cannot coincide, and at the p crossing $q > 0$, so their order is $\beta < \gamma$. □

Appendix G. Preliminary width and tilt estimates

The estimates here use only the tents at $\pi/8, \pi/4, 3\pi/8$. They apply to a full niche and to any larger partial barrier. In a partial-turn comparison a horizontal reflection also reflects the actual barrier; it need not produce another partial-turn objective.

Throughout this appendix put

$$r = \sqrt{2}, \quad k = r - 1, \quad c = \cos(\pi/8), \quad s = \sin(\pi/8).$$

For the diagonal offset, let $\mathcal{N}_q(x) = (x + q/x)/2$ and set

$$b = \mathcal{N}_2(\mathcal{N}_2(3/2)), \qquad k_- = 2/b - 1, \qquad k_+ = b - 1.$$

The reciprocal of a square-root upper bound supplies the lower bound. Useful exact identities and bounds are

$$k^2 = 1 - 2k, \quad s/c = k, \quad k + 1/k = 2r, \quad 1/s + 1/c = 4c, \quad 1/s - 1/c = 4s,$$
$$k_- < k < k_+, \qquad \frac{2}{5} < k < \frac{29}{70}, \qquad s > \frac{3}{8}. \tag{G.1}$$

G.1. **Caps below the unit-width boundary.** The extra-tail height is supplied by the wall geometry. Put

$$N = 1/c = \sqrt{1 + k^2}, \qquad \eta = 1 - N + k^2 = N(N - 1), \qquad t_0 = k - \eta,$$
$$m_* = (N + \eta/k)/2 = (1 + k)(1 - N/2),$$
$$a_* = 1 - k^2/4, \qquad h_* = k^2/4, \qquad D_* = (k/3)^2/r.$$

Here η is the available height of the nearby companion wall at a diagonal zero, and m_* is exactly the midpoint needed for the companion support inequality. Choosing $t_0 = k - \eta$ makes the omitted-tip length a diagonal multiple of η, so its area refund combines directly with the two triangle payments. The remaining reserves use k^2.

Lemma G.1 (Short three-tent bound)**.** *Let U have height one and projection $[-a,a]$, with $0<a\le 1$. Suppose its roof A is affine on $J=[-a/2,a/2]$, with $A(-a/2)=1-h$, $A(a/2)=1$, $0\le h<1$. If a nonnegative barrier B dominates the three actual tents, then*

$$\int_{[-a,a]\setminus J} A-\int_J B<\underline{p}<M/2.$$

Proof. Raise the left exterior roof by h, put the middle roof at height one, and retain the right roof. The resulting roof $\widehat{A}$ is concave: its left slopes are nonnegative and its right slopes are nonpositive. It defines a height-one cap $\widehat{U}$, and its exterior area exceeds the original by $ah/2$.

The first-quarter supports are unchanged, since the old point $(a/2,1)$ dominates every raised left point in those directions. If $\widehat{G}(t)=h_{\widehat{U}}(t+\pi/2)$, then

$$h_U(t+\pi/2)=\max\{\widehat{G}(t)-h\cos t,\ \cos t-(a/2)\sin t\}.$$

The wall of the second term is nonpositive on J. Thus the positive two-wall tent of U is obtained from that of $\widehat{U}$ by lowering its companion wall by h.

Put

$$u=rh_{\widehat{U}}(\pi/4)-1,\quad v=rh_{\widehat{U}}(3\pi/4)-1,\quad M_0=(u+v)/2.$$

The two middle points give $a/2\le u,v\le a$. The lifted cap is below $A_0(x)=\min(1,1+u-x,1+v+x)$. Write $D=\int_{[-a,a]\setminus J}(A_0-\widehat{A})\ge 0$. The diagonal tent is $(\min(u-k-x,v-h-k+x))_+$. Suppose the proposed score P is at least $\underline{p}$. The rectangle bound $P\le a(1-h/2)$ gives $a\ge\underline{p}$ and $h\le 2(1-\underline{p})$. The reference budget in Equation (2.2) implies $h<a/2$ and $a-h>k$, so the tent's apex lies in J. Exact integration gives

$$\begin{aligned}P\le F_h(a,u,v)-D,\qquad F_h=a&-\frac{(a-u)^2+(a-v)^2}{2}-\frac{ah}{2}\\&-(M_0-h/2-k)_+^2\\&+\frac{(u-k-a/2)_+^2+(v-h-k-a/2)_+^2}{2}.\end{aligned}\tag{G.2}$$

In this range, F_h is concave on $[a/2,a]^2$. In the positive-apex region its Hessian has diagonal entries $-3/2$ plus the corresponding clipping indicator, and off-diagonal entries $-1/2$. In the other region the clipping terms vanish and the Hessian is $-\mathrm{Id}$. The first derivatives agree across the interfaces. Its unclipped critical point is $u=v=(a+k+h/2)/2$. Thus, with $B_0(a)=a-(a-k)^2/2$,

$$F_h\le B_0(a)-kh/2-h^2/8,\qquad B_0(1)=2k.\tag{G.3}$$

The boundary comparisons required below are

$$\begin{aligned} B_0(a_*) &< \underline{p}, \\ 2k - kh_*/2 - h_*^2/8 &< \underline{p}, \\ 2k - \tfrac{2}{3}(t_0 - k/2)^2 &< \underline{p}, \\ 1 - (1-m_*)^2 - (m_* - k)^2 &< \underline{p}, \\ a_*/2 - t_0 - h_*/4 > k/3, \qquad & 2k - \underline{p} < D_*. \end{aligned} \tag{G.4}$$

They use the same automatically generated enclosures as the reference comparison. Enclose k by the bounds in Equation (G.1); for N, apply two Newton steps from one to an upper bound for $N^2 = 2 - 2k$. Substitute these bounds and the formula for $\underline{p}$, reducing squares by $k^2 + 2k = 1$ and $N^2 + 2k = 2$. This reduces the six comparisons to rational arithmetic.

Since B_0 increases up to $1 + k$, the first two comparisons force $a > a_*$ and $h < h_*$. Also $D \le 2k - P < D_*$, so

$$a > a_*, \qquad h < h_*, \qquad D < D_*. \tag{G.5}$$

Maximizing Equation (G.2) subject to $u \ge a/2 + t_0 + h/4$ gives the constrained critical point

$$u = a/2 + t_0 + h/4, \qquad v = a/2 + (2k - t_0)/3 + h/4.$$

Its v derivative is zero and its u derivative is $(2k - 4t_0)/3 < 0$. The value is at most the third line of Equation (G.4). Exchanging the coordinates gives

$$u - a/2 < t_0 + h/4, \qquad v - a/2 < t_0 + h/4.$$

The bounds in Equation (G.5) place the comparison points in the square, remove both clips, and make the actual apex positive. With $m = M_0 - h/4$, the relaxation is then

$$F_h = a - (a - m)^2 - (m - k)^2 - \frac{(u - v)^2}{4} - kh/2 - h^2/8.$$

If $m \le m_*$, subtract its value from the fourth line of Equation (G.4). The difference is a sum of squares and the nonnegative terms $(2m_* - 1)(1 - a)$ and $2(1 + k - 2m_*)(m_* - m)$. This contradicts the high score. Hence $m > m_*$.

We now pay for two genuine extra tents. Define the lifted support deficits

$$\delta_L = c + sv - h_{\widehat{U}}(5\pi/8), \qquad \delta_R = c + su - h_{\widehat{U}}(3\pi/8).$$

They are nonnegative and at most $s(v - a/2), s(u - a/2)$. A support cut at $(-v, 1)$ has horizontal legs δ_L/s and $\delta_L/(c - s)$. Its whole charged triangle has area $K\delta_L^2$, where $K = 1/k$. If the latter leg exceeded $a - v$, the cut with $\delta_0 = (c - s)(a - v)$ would already give $D \ge (a - v)^2/r$. But Equation (G.4) gives

$$a - v > a/2 - t_0 - h/4 > k/3, \qquad (a - v)^2/r > D_*.$$

This contradicts Equation (G.5). The right cut is identical. They lie in separate wings, so

$$D \ge K(\delta_L^2 + \delta_R^2). \tag{G.6}$$

Put

$$\Gamma = Kc^2 = \frac{4+3r}{4}, \qquad \eta_L = \eta - (1-k)h,$$
$$E_L = (\eta_L - \delta_L/c)_+, \qquad E_R = (\eta - \delta_R/c)_+.$$

The definitions give the exact companion identity

$$1 - 1/c + k^2 = \eta, \qquad 1/c + K\eta = 2m_*.$$

At the left zero $x_0 = k - v + h$ of the diagonal tent, the actual $\pi/8$ companion wall has height at least E_L. A point attaining u has $x + y = 1 + u$ and $y \le 1$, and lies on the unchanged right wing. Its first-wall lower bound is $1 - 1/s + (u - x)/k$. This dominates the additional ascending wall up to $x_0 + E_L/(1-k)$, because

$$2M_0 - h > 2m_* - h/2 \ge 1/c + KE_L.$$

The two inner tips lie before the diagonal apex. The useful identity is

$$m_* - \big(k + \eta/(1-k)\big) = N - 1 > h_*/4 > h/4.$$

In the reflected coordinate, an attaining point for v lowered by h supplies the other companion wall. Its condition is $2M_0 - kh > 2m_* = 1/c + K\eta \ge 1/c + KE_R$. We obtain two disjoint additional triangles of full areas $\Gamma E_L^2, \Gamma E_R^2$. Each omitted exterior tip has horizontal length at most $L_0 + h/4$, where

$$L_0 = t_0 - k + \eta/k = \eta(1-k)/k = (1+k)\eta.$$

Thus

$$\int_J B \ge N_{45} + \Gamma(E_L^2 + E_R^2) - k(L_0 + h/4)^2,$$

where N_{45} is the diagonal-tent integral used above. This remains valid when one of the peaks is zero.

For $\eta_i, \delta \ge 0$, completing a square gives

$$K\delta^2 + \Gamma(\eta_i - \delta/c)_+^2 \ge \Gamma\eta_i^2/2.$$

Combining this with Equation (G.6) gives

$$P \le 2k - kh/2 - h^2/8 - \frac{\Gamma}{2}\big[(\eta - (1-k)h)^2 + \eta^2\big] + k(L_0 + h/4)^2.$$

Its linear and quadratic coefficients in h are nonpositive, so

$$P \le 2k - \Gamma\eta^2 + kL_0^2 = 2k - (\Gamma - 2k)\eta^2 < \underline{p} < M/2.$$

The strict comparison uses the same generated enclosures as Equation (G.4). This contradicts the high score and proves the exclusion at the natural unit-width boundary. □

G.2. **Wide caps and the endpoint equations.** For a wide cap, endpoint stationarity forces the diagonal supports outward. That raises the middle tent. The nearby hallway positions then either cut two exterior triangles or raise the niche farther. We first use the diagonal balance to confine the parameters, and then pay for these two extra triangles.

Lemma G.2 (Wide three-tent bound)**.** *Let $8/5 \le a < 3$, $0 \le h < 1/2$, and let U, A, J, B satisfy the geometric hypotheses of Lemma G.1, without its upper width restriction. Put $q_\pm = B(\pm a/2)$, $e_R = A(a)$, and $e_L = A(-a)$. Suppose*

$$e_R = \tfrac{1}{2} + \tfrac{h}{4} + \tfrac{3q_+ - q_-}{4}, \qquad e_L = \tfrac{1}{2} - \tfrac{3h}{4} + \tfrac{3q_- - q_+}{4}. \tag{G.7}$$

Then the score is less than $\underline{p}$.

Proof. Here use the original supports $u = rh_U(\pi/4) - 1$, $v = rh_U(3\pi/4) - 1$, $m = (u+v)/2$, $w_R = a - u$, and $w_L = a - v$. The affine roof extension is $L_h(x) = 1 - h/2 + (h/a)x$. Concavity and the height bound imply

$$a/2 \le u \le a, \quad a/2 - h \le v \le a - 3h/2, \quad u \ge a + e_R - 1, \quad v \ge a + e_L - 1.$$

The majorant $\min(1, L_h, 1 + u - x, 1 + v + x)$ has exterior area

$$O_0 = a - \frac{5}{8}ah - \frac{1}{2}\left[w_R^2 + \frac{(w_L - 3h/2)^2}{1 - h/a}\right].$$

The endpoint equations give $e_R + e_L = 1 - h/2 + (q_+ + q_-)/2 \ge 1 - h/2$. Thus

$$w_R + w_L \le 1 + h/2, \qquad m \ge a - 1/2 - h/4. \tag{G.8}$$

The diagonal tent has its apex in J, and its integral is

$$N_{45} = (m-k)^2 - \tfrac{1}{2}[(a/2 - k - w_R)_+^2 + (a/2 - k - w_L)_+^2].$$

Write $b_0 = a/2 - k$. For $t \ge b_0$ with $3/2 - t/a \ge 0$,

$$w^2 - (b_0 - w)_+^2 \ge 2tw - t^2,$$

$$\frac{(w - 3h/2)^2}{1 - h/a} - (b_0 - w)_+^2 \ge 2tw - t^2 + (-3t + t^2/a)h.$$

The second follows from the exact remainder $[w - t - h(3/2 - t/a)]^2/(1 - h/a)$; if $w < b_0$, this remainder is at least $(b_0 - w)^2$. Hence

$$P \le a - 2t(a-m) + t^2 - (m-k)^2 - \left[\tfrac{5}{8}a - \tfrac{3}{2}t + \frac{t^2}{2a}\right]h. \tag{G.9}$$

For $a \le 1 + 2k$ take $t = 1/2$. This yields

$$P \le m + 1/4 - (m-k)^2 - D(a)h, \qquad D(a) = 5a/8 - 3/4 + 1/(8a) \ge 21/64.$$

We need the two nearby-angle peaks to fit in J, so estimate their span directly. A hypothetical score at least $\underline{p}$ first forces $a \le 1 + 2k$: otherwise take $t = b_0$ in Equation (G.9) and use Equation (G.8) to obtain $P \le (5/2)k - 1/4 < 41/50 < \underline{p}$.

At the first wide width put

$$a_0 = 8/5, \qquad m_0 = a_0 - 1/2, \qquad d_0 = D(a_0),$$

and let the width price, height price, and available area be

$$p_a = 2(m_0-k)-1, \qquad p_h = d_0-p_a/4, \qquad b = m_0+1/4-(m_0-k)^2-41/50.$$

The midpoint tangent and endpoint law give these prices. Indeed, completing the midpoint square gives

$$m + 1/4 - (m-k)^2 = m_0 + 1/4 - (m_0-k)^2 - p_a(m-m_0) - (m-m_0)^2.$$

Since $p_a > 0$ and $m - m_0 \geq a - a_0 - h/4$, the diagonal area bound and $P \geq \underline{p} > 41/50$ give one joint budget:

$$p_a(a-a_0) + p_h\, h \leq b. \tag{G.10}$$

Let $\lambda = 1 - k/2$. The same root enclosures as before give

$$p_h > 0, \qquad p_a \geq p_h\lambda, \qquad b < p_h(rN - \lambda a_0).$$

For a direct check use $k^2 + 2k = 1$, $k < 29/70$, and $N > 27/25$; the last comparison is already strict with N replaced by $27/25$. Thus the width price pays for its contribution to the peak span, and Equation (G.10) yields

$$a \leq 1 + 2k, \qquad \lambda a + h < rN = 4s. \tag{G.11}$$

Put $z = h/a$ and $q = (v + h/2)/(1 - z)$, so $a/2 \leq q \leq a$. The near-vertical deficits of the majorant are

$$\delta_L = c(1 - h/2) + (s - cz)q - h_U(5\pi/8), \qquad \delta_R = c + su - h_U(3\pi/8).$$

The middle points show $\delta_L \leq (s - cz)(q - a/2)$ and $\delta_R \leq s(u - a/2)$, so the inner half of each support cut fits between the middle endpoint and the diagonal contact. Since $z < k$, their total exterior payment is at least

$$\frac{\delta_L^2 + \delta_R^2}{2kc^2} = r(\delta_L^2 + \delta_R^2).$$

Only these inner halves are needed; their outer legs need no fitting check. The extra left deficit from tilt is

$$\tau = (c + sv - c(1 - h/2) - (s - cz)q)/c = (1 - k)h(1/2 + q/a) \leq \tfrac{3}{2}(1 - k)h.$$

The $\pi/8$ tent has ascending wall $b_R + kx$ and descending companion lower bound $B_R - x/k$, where

$$b_R = 1 - 1/c + kv - \delta_L/c - \tau, \qquad B_R = e_R + a/k - 1/s.$$

In the reflected coordinate the corresponding $\pi/8$-type tent at $3\pi/8$ has $b_L = 1 - 1/c + ku - \delta_R/c$ and $B_L = e_L + a/k - 1/s$. Their crossovers with the descending diagonal baselines are

$$x_L^R = (u - k - b_R)/r, \quad x_L^L = (v - k - b_L)/r,$$

and their additional peaks are

$$Z_R = (B_R + b_R)/2 - u + k, \qquad Z_L = (B_L + b_L)/2 - v + k.$$

The crossover offsets from the two diagonal apices are $km+2s-1+(\delta_L/c+\tau)/r$ and $km+2s-1+\delta_R/(rc)$, which are positive by Equation (G.8): $m \geq a-1/2-h/4 > 3/4$. The middle endpoint support bounds place the tentative peaks in J provided

$$e_R + h + \lambda a \leq 1 + 4s, \qquad e_L + \lambda a \leq 1 + 4s.$$

Both follow at once from $e_R, e_L \leq 1$, $h \geq 0$, and Equation (G.11). The same diagonal budget therefore places both extra triangles without separate width or height cuts.

Truncate each half-triangle at its diagonal baseline zero. Set

$$F_R = r(u - k - x_L^R), \quad F_L = r(v - k - x_L^L),$$
$$\widehat{Z}_Q = \min(Z_Q, F_Q), \quad \ell_Q = (Z_Q - F_Q)_+ \quad (Q = R, L).$$

The two retained triangles are disjoint, lie in J, and add niche area at least $[(\widehat{Z}_R)_+^2 + (\widehat{Z}_L)_+^2]/(2r)$. Their truncation lengths obey

$$2\ell_R \leq [e_R + 2rw_R + h - (3k/2)a - 1 - 4s + 2rk]_+,$$
$$2\ell_L \leq [e_L + 2rw_L - (3k/2)a - 1 - 4s + 2rk]_+.$$

To bound these expressions we need no box estimate for a partial niche. Solving Equation (G.7) using $e_R \leq 1$, $e_L \leq 1 - 3h/2$, and $q_\pm \geq 0$ gives directly

$$e_R \geq 1/3 + h/2 + (2/3)q_+, \qquad e_L \geq 1/3 - 2h/3 + (2/3)q_-.$$

In particular $w_R \leq 2/3 - h/4$ and $w_L \leq 2/3 + 3h/4$. Using also $e_Q \leq 1 - w_Q$, $a \geq 8/5$, and

$$(2r-1)\tfrac{2}{3} - \tfrac{12}{5}k - 4s + 2rk < 0$$

in the displayed truncation bounds gives

$$\ell_R \leq (5-2r)h/8, \quad \ell_L \leq 3(2r-1)h/8, \quad \ell_R + \ell_L \leq h.$$

The constant's sign follows from $r < 10/7$, $k < 3/7$, $s > 3/8$.

Adding the two peak formulas, using $e_R + e_L \geq 1 - h/2$, $\tau \leq 3(1-k)h/2$, and the last truncation bound, gives

$$\widehat{Z}_R + \widehat{Z}_L \geq Q(a, m, h) - (\delta_L + \delta_R)/(2c),$$

where the exact demand is

$$Q = (2+k)a + (k-2)m + \tfrac{3}{2} + 2k - (3+k)N - (2 - 3k/4)h.$$

There is no need to round its coefficients. Put $D_0 = r(\delta_L^2 + \delta_R^2)$ and $G_0 = [(\widehat{Z}_R)_+^2 + (\widehat{Z}_L)_+^2]/(2r)$. Weighted Cauchy–Schwarz gives

$$Q_+^2 \leq (4 + 5k)(D_0 + G_0).$$

Indeed each retained peak contributes reciprocal price $2r$, and each normalized support deficit contributes $k/2$; their sum is $4r + k = 4 + 5k$. Consequently

$$P \leq m + 1/4 - (m-k)^2 - d_0 h - \frac{Q_+^2}{4+5k}, \qquad d_0 = D(a_0).$$

Choose the common price to pay exactly for the tilt:

$$t = \frac{d_0}{2(2 - 3k/4)}.$$

The inequality $Q_+^2 \geq 2(4 + 5k)tQ - (4 + 5k)^2t^2$ cancels the height term. Completing the remaining midpoint square and using $a \geq a_0$ gives

$$P \leq k + \tfrac{1}{2} - 2t\big[(2 + k)a_0 + \tfrac{3}{2} - \tfrac{3k}{2} - (3 + k)N\big] + (9 - k)t^2.$$

The existing Newton bounds for k and $N \leq 13/12$ compare this expression directly with $\underline{p}$. Thus one price and one square suffice; no constrained midpoint optimization remains. This contradiction proves the lemma. □

G.3. **The intermediate tilt bound.** The inclined middle chord also bounds the continuation of the low wing. Pay its triangular area loss together with the diagonal niche: one common price cancels the unknown support coordinates. Completing the price square gives a bound coupling width and tilt.

Lemma G.3 (One-angle tilt estimate)**.** *For $1 \leq a \leq 8/5$, $0 \leq h < 1/2$, and the same middle roof, a barrier dominating the actual diagonal tent gives*

$$P \leq a - \tfrac{5}{8}ah - \frac{2(a - 3h/4 - k)^2}{4 - h/a} = 1 - \frac{(a - r)^2}{2} - \frac{2a}{4a - h}(k + h/2)^2. \quad \text{(G.12)}$$

A score at least $\underline{p}$ consequently forces

$$h < 1/3, \qquad (a - r)^2 + \frac{4a}{4a - h}(k + h/2)^2 \leq 2(1 - \underline{p}).$$

Proof. Use the lifted cap and its diagonal supports u, v from the short proof. Concavity of the original roof imposes $v \leq a - h/2$. In addition to the horizontal majorant, the lifted left wing lies below $1 + h/2 + (h/a)x$. Its cut removes exterior area

$$D_{\text{tilt}} = \frac{h(v - a/2)^2}{2(a - h)}.$$

Indeed the first triangle has area $(h/a)(v - a/2)^2/2$, and the second has this value times $(h/a)/(1 - h/a)$; both are charged because $v \leq a - h/2$. The score is therefore bounded by $F_h(a, u, v) - D_{\text{tilt}}$. We can pay this loss together with the diagonal niche using one price. Put

$$w_R = a - u, \qquad w_L = a - v + h, \qquad m = (u + v - h)/2.$$

Completing the ceiling square gives the same perspective losses as in Equation (G.9). Apply its two clipped-square inequalities with a common price $t \geq a/2 - k$, and also use

$$(m - k)_+^2 \geq 2t(m - k) - t^2 \qquad (t \geq 0).$$

The two support coordinates cancel. We obtain

$$P \leq a - \frac{5}{8}ah - 2t(a - 3h/4 - k) + \frac{4 - h/a}{2}t^2. \quad \text{(G.13)}$$

Complete this single price square and choose

$$t = \frac{2(a - 3h/4 - k)}{4 - h/a}.$$

This price is admissible: $0 \le t \le a/2$, and

$$t - (a/2 - k) = \frac{2ak - h(a + k)}{4a - h} \ge 0.$$

For the numerator use $h < 1/2$, $a \ge 1$, and $k > 1/3$:

$$2ak - \frac{a + k}{2} = (a - 1)(2k - 1/2) + (3k - 1)/2 \ge 0.$$

The clipped perspective inequalities need only $t/a \le 3/2$, already supplied by $t \le a/2$. Substitution in Equation (G.13) proves the first formula in Equation (G.12), without optimizing either support coordinate. The second follows from $k^2 + 2k = 1$.

Compare directly with the reference area. Dropping the nonpositive width square and using $a \le 8/5$ gives

$$P \le 1 - \frac{16/5}{32/5 - h}(k + h/2)^2.$$

If $h \ge 1/3$, the last loss increases with h and is greater than its value at $1/3$ with k replaced by k_-. The generated root bounds in Lemma 2.1 give the single comparison

$$1 - \frac{16/5}{32/5 - 1/3}(k_- + 1/6)^2 < \underline{p}.$$

This proves $h < 1/3$. Keeping the width square in Equation (G.12) gives the stated joint area budget.

□

Corollary G.4 (Canonical range)**.** *Every canonical full maximum satisfies*

$$1/2 < C < \frac{4}{5}, \qquad |f_U(C) - f_U(-C)| < \frac{1}{3}, \quad I_U = [-2C, 2C].$$

The same conclusions hold for every canonical proper joint maximum whose angle exceeds $\arctan(8/3)$*.*

Proof. The reference competitor and Lemma 2.1 give score $M/2 > \underline{p}$. Width coercivity gives $a = 2C < 3$. The positive-pressure and top-localization results give height one, an affine middle roof, and lower middle height greater than $1/2$. Orient its higher middle endpoint to the right. In the proper case reflect the actual barrier as well. It remains nonnegative and dominates all three tents, since $\arctan(8/3) > 3\pi/8$. The endpoint equations are precisely Equation (G.7) after relabeling. Apply Lemmas G.1 and G.2 to obtain $1 < a < 8/5$, then Lemma G.3 to obtain the height-drop bound. □

Appendix H. Excluding a tilted full-turn maximum

Throughout this appendix U is a canonical full-turn maximum. Reflect horizontally when necessary, and write

$$I=[-2C,2C],\quad J=[-C,C],\quad f_U(-C)=1-h,\quad f_U(C)=1,\quad h>0.$$

The preliminary cuts and Corollary C.10 give $1/2<C<37/50$ and $0<h<1/3$. Write $n=n_L(U)$, $n_\pm=n(\pm C)$ and $e_R=f_U(2C)$, $e_L=f_U(-2C)$. The actual endpoint and wing identities are

$$\begin{gathered}e_R=\tfrac12+\tfrac{h}{4}+(3n_+-n_-)/4,\quad e_L=\tfrac12-\tfrac{3h}{4}+(3n_--n_+)/4,\\ E=e_R+e_L=1-h/2+(n_++n_-)/2,\quad \max_J n\le E/2,\quad 2P_L(U)=L_{\mathrm w}.\end{gathered}\tag{H.1}$$

Let F be the first support and G the low-left-wing companion support, with the uncharged middle facet removed, as in the tilted support construction. Use p,q,u,v for their velocities and regular densities in (5.6).

H.1. **A connected ordinary survivor.** The ordinary-sofa bound can be used once the niche stays below the roof. We prove this by keeping every positive corner inside the middle window. Each exterior wing then sees only one affine wall family, so its niche is convex and cannot cross the concave roof.

Lemma H.1. *Every attached quadrant with positive apex height has its apex abscissa in $(-C,C)$. The full niche is convex on each exterior wing, and $n\le f_U$ on I. Consequently the survivor $S=\{(x,y):x\in I,\ n(x)\le y\le f_U(x)\}$ is an ordinary sofa.*

Proof. Lift the low wing by h and make the middle roof one, obtaining $\widehat U$. Write $s=\sin t$, $c=\cos t$ and let $\widehat F,\widehat G$ be its two supports. For a horizontal-middle cap the box and top endpoints imply

$$\widehat F\le 2Cc+s,\quad \widehat G\ge Cs+c,\quad \widehat z_x\le C(2-3s^2)-c+s.$$

By Lemma F.6 with $h=0$, $C(1-3s^2)<c-s$ throughout the preliminary width range. Reflection gives $-C\le\widehat z_x\le C$.

Put $\delta=\arctan(h/(2C))$. For $t\ge\delta$ the actual second support is on the low wing, and $F=\widehat F$, $G=\widehat G-hc$. Thus $z_x=\widehat z_x+hsc>-C$ in the open angular interval. To prove its upper bound it suffices to show

$$c-s-C(1-3s^2)-hsc>0.\tag{H.2}$$

This is precisely Lemma F.6: the unit circle's quadratic chord bound handles $s^2\le1/3$, and its 3–4–5 tangent handles $s^2\ge1/3$. For $0<t<\delta$ the actual second support is $(C,1)$, and the apex height is at most $1-c-s+Csc=sc[C-2/(1+c+s)]<0$. Such quadrants have no positive roof.

Every positive quadrant uses its descending first wall on the right wing and its increasing second wall on the left wing. The niche is therefore a maximum of affine functions and zero on each wing, hence convex there; the box gives $n(\pm2C)=0$. The endpoint box gives $n_\pm\le2/5$. By (H.1),

$\max_J n \le 7/10 - h/4 < 1 - h \le f_U$. On a wing $f_U - n$ is concave and nonnegative at both endpoints, so the same inequality holds throughout I. Apply Lemma A.2. □

In particular, with $G_0 = 2.2199$, convexity of the wing niche and the stationary chord bounds give

$$P_L(U)+C(3-E-3h/2) \le |S| \le G_0, \qquad P_L(U) \ge \sqrt{C^2 + (1 - h/2 - E/2)^2}. \tag{H.3}$$

Indeed the middle cap area is $2C - Ch$, while the exterior niche is at most $C(n_+ + n_-)/2 = C(E - 1 + h/2)$. The two wing chords have vectors $(C, 1 - e_R)$ and $(C, 1 - h - e_L)$; their average length is at least the length of their average vector.

H.2. **The forward excess ends before endpoint tangency.** A positive niche maximum at the high middle endpoint must be tangent at an angle $t \ge \arccos C$. We show that the first curvature excess ends before this angle. The backward support formula then charges the endpoint niche to the top overhang alone.

Put $J_* = (11/15)[(1/3)^2/4 + (1/3)^3/6]$, the integrated envelope budget.

Lemma H.2. *If $2/3 < C < 37/50$, then $u(t) \le 1$ for almost every $t \ge 11/15$. Moreover, if the top face is $[C, C + T] \times \{1\}$, then*

$$n_+ \le \frac{CT}{\sqrt{1 - C^2}}, \qquad J_F := \int_0^L (u - 1)_+ \sin t \, \mathrm{d}t \le J_*, \qquad \int_C^{2C} n \le (T + J_F) n_+/2. \tag{H.4}$$

Proof. Write $D = 3C$, $B = e_R - 1 + h$ and $\gamma_0 = \arccos(1 - h)$. The endpoint box $H(C) = 1 - \sqrt{1 - C^2} < 1/3$ and the sharper low-wing box $n_- \le (H(C) - h)_+$ imply

$$-1/2 + 5h/4 - (1/3 - h)_+/4 \le B \le -1/4 + 5h/4, \qquad -7/12 < B < 9/50.$$

Only the initial positive-q component can have $u > 1$. As in the reflected calculation in Proposition E.6, later positive components have $q \le 1/8$. Before its first p-zero β the same-sign law gives $u = 0$. If $q(\beta) \le 1$ no excess occurs. Otherwise put $\eta = q(\beta) - 1 > 0$. On its only possible excess episode, at elapsed time $r \le 1$,

$$p \le -r, \quad q' \le -(1 + r)/2, \quad (u(\beta + r) - 1)_+ \le (\eta - r/2 - r^2/4)_+.$$

The duration is at most $\ell(\eta) = \sqrt{1 + 4\eta} - 1$. All bounds below give $\ell < 1$, as required.

If $\beta \le \gamma_0$, the initial-floor solution in (C.12) gives $(q(\beta) + 1)^2 = D^2 + B^2 - 1 > 4$. Thus $B < 0$, since $(111/50)^2 + (9/50)^2 < 5$. Also $D \sin \beta = 1 + B \cos \beta \le 1$, so $\beta < \pi/6$. Finally $(q(\beta) + 1)^2 < (111/50)^2 + (7/12)^2 - 1 < (2 + 1/12)^2$, so $\eta < 1/12$, $\ell < 1/6$, and the episode ends before $(\pi + 1)/6 < 11/15$.

If $\gamma_0 < \beta$, then $1 + B(1-h) - D\sqrt{h(2-h)} > 0$. This forces $h < 1/5$. The cutoff is $1 - 4/5$, from the same 3–4–5 circle direction used in the area comparison. For $1/5 \le h \le 1/3$, the circle identity gives

$$h(2-h) - [(1+h)/2]^2 = (1-h)(5h-1)/4 \ge 0.$$

Thus $\sqrt{h(2-h)} \ge (1+h)/2$. Substituting this and $B \le -1/4+5h/4$, $D \ge 2$ makes the floor momentum nonpositive. For $h < 1/5$ the sharper pressure interval is $-7/12 + 3h/2 \le B \le -1/4 + 5h/4 \le 0$. The energy used in Lemma C.11 gives

$$(q(\beta)+1)^2 \le Q(D,B,h) = D^2 + B^2 + B(1-h) - D\sqrt{h(2-h)}.$$

The upper endpoint pressure at zero tilt is $3/4$. We compare with that trajectory. For $0 \le h \le 1/5$, squaring gives $\sqrt{h(2-h)} \ge 3h$. The test Q increases with B on the displayed interval, and its polynomial part at the upper endpoint is

$$(-1/4+5h/4)^2+(-1/4+5h/4)(1-h) = -3/16+7h/8+5h^2/16 \le -3/16+h.$$

Since $D \ge 2$, the floor therefore removes at least $5h$ of energy:

$$(2+\eta)^2 + 5h \le D^2 - 3/16. \tag{H.5}$$

For this zero-tilt comparison, put

$$A_0 = D - \sqrt{D^2 - 3/2}, \qquad Y_0 = \sqrt{D^2 - 3/16}, \qquad Z_0 = \sqrt{4Y_0 - 7}.$$

The endpoint pressure bound gives $0 \le 3/4 + 5h/4 - D\beta + \beta^2/2$. As $\sqrt{D^2-3/2} \ge 3/2$, squaring shows

$$\beta \le D - \sqrt{D^2 - 3/2 - 5h/2} \le A_0 + h.$$

Thus tilt delays the first zero by at most h. Its energy loss shortens the excess by at least the same amount. Indeed $2 + \eta \le Y_0 < D < 9/4$, so the sum of these two roots is below five. By (H.5),

$$Y_0 - (2+\eta) = \frac{Y_0^2 - (2+\eta)^2}{Y_0 + (2+\eta)} \ge h.$$

Similarly $1 + \ell(\eta) \le Z_0 < 3/2$. Their sum is below four, giving

$$Z_0 - [1 + \ell(\eta)] = \frac{4[Y_0 - (2+\eta)]}{Z_0 + 1 + \ell(\eta)} \ge h.$$

The delay and shortening cancel:

$$\beta + \ell(\eta) \le A_0 + Z_0 - 1 = W(D, 3/4, 0) < 11/15$$

by Lemma E.4. The same energy bound gives $\eta < 19/100$ and $\ell < 1/3$ throughout the range.

The quadratic bound at the half-angle and the double-angle identity give

$$\cos(11/15) \ge 2\left[1 - \frac{1}{2}\left(\frac{11}{30}\right)^2\right]^2 - 1 > \frac{37}{50}.$$

At a positive endpoint-active angle at $x = C$, confinement gives strict companion slack, so stationarity has $B_x(t) = C$. The outer first-support abscissa is at most $2C$, hence $\cos t \le C$ and $t \ge \arccos C > 11/15$. The backward support formula then gives

$$a_t(C) = T\cot t + \frac{1}{\sin t}\int_t^L \sin(r-t)(u(r)-1)\,\mathrm{d}r \le T\cot t \le \frac{CT}{\sqrt{1-C^2}}.$$

Also $(F(t)-1)/\cos t \le C + T + J_F$. Thus the convex right-wing niche vanishes by $C + T + J_F$ and has area at most $(T + J_F)n_+/2$. In every excess case above $\eta < 19/100$ and $\ell < 1/3$. The quadratic envelope has integral less than $11/324$, and occurs below $11/15$. Consequently $J_F \le J_* = (11/15)(11/324)$. □

H.3. **The low endpoint is dry and the top overhang is positive.** The increasing companion-wall envelope has one zero. Its position determines both the dry part of the middle interval and the available source projection. Matching that projection to the exterior wings identifies the top overhang. Comparing the same zero with walls from the low middle point and the outer endpoint bounds the overhang.

Lemma H.3. *For $2/3 < C < 37/50$, the top overhang satisfies $T > 0$ and*

$$n_- = 0, \qquad T \le \sqrt{h(2-h)}, \qquad T \le \sqrt{1-e_L^2} - C, \qquad T < 1 - C.$$

The late-unit and leakage conclusions of Lemma H.2 hold as well.

Proof. Let $x_1 = 1 - 2C$ and consider the second-wall envelope $\Psi(x) = \sup_{0<t<L}(G(t) - 1 + x\sin t)/\cos t$ for $x < x_1$. It is finite, convex and nondecreasing. Its endpoint-angle limits are $-h$ and $-\infty$. It is negative for $x \le -2C$ and approaches a positive value at x_1, since the outer endpoint $(-2C, e_L)$ has $e_L > 0$. It has a unique zero $x_0 < x_1$: a zero maximum is attained at an interior angle with strictly positive affine slope, which gives strictly negative values to its left and positive values to its right. For $x \le x_1$, the pinned top point $(C, 1)$ gives

$$F(t) - 1 - x\cos t \ge (3C-1)\cos t + \sin t - 1 \ge \cos t + \sin t - 1 > 0.$$

For $x_1 < x < C$, angles near L have first wall $(C+T-x)(L-t)+o(L-t) > 0$ and second wall tending to $+\infty$. The extra high-wing companion walls are nonpositive on J. Hence $\{x \in (-C, C) : n(x) > 0\} = (\max(-C, x_0), C)$.

In this geometry the finite source projection identity has an exact limit. On a compact interval to the left of x_0, all surrogate second walls have a uniform negative gap. The high-point second walls also have a negative gap outside arbitrarily small endpoint angle neighborhoods. The coarse source bounds make their remaining mass and horizontal projection $O(\delta) + o_i(1)$ for an angle cutoff δ. On compact subsets of the positive interval, uniform barrier convergence gives the opposite inclusion. Let the approximation

index tend to infinity and then $\delta \to 0$. The finite projection identity and source balance yield

$$2C - T = C - \max(-C, x_0), \qquad T = (C + x_0)_+.$$

This argument controls shallow finite graph pieces explicitly.

Suppose $T = 0$. Reflect the local support equations as $P = -q(L - r)$, $Q = -p(L - r)$, $U = v(L - r)$, $V = u(L - r)$. Before the removed middle normal they have initial data $(P(0), Q(0)) = (e_L, 3C - 1)$ and the same arm and shadow laws. Beyond that normal $U = 0$ directly. The box and endpoint equations give $3/20 < e_L < 3/4$. Let $D = 3C$ and $e = e_L$. At the first P-zero, the energy calculation without an impulse gives $(Q + 1)^2 \le D^2 - e(1 - e)$ and a crossing time at most $D - \sqrt{D^2 - 2e}$. The same quadratic excess envelope then applies. The crossing and decay are bounded together by $W(D, e, 0)$. By Lemma E.4, the whole interval $0 \le e \le 3/4$ compares with $D = 111/50$, $e = 3/4$, where the sum is below $11/15$. Also $e(1 - e) \ge 1/8$ on $[3/20, 3/4]$, so the excess height is below $1/5$ and its duration below $1/3$. Later positive-Q components have amplitude at most $1/8$. Thus $v(t) \le 1$ whenever $L - t \ge 11/15$.

If $\Psi(-C) \ge 0$, its maximum is attained at a proper angle t and has shifted companion abscissa $D_x(t) = -C$. The outer supporting abscissa is $D_x - \sin t \ge -2C$, so $\sin t \le C$ and $L - t \ge \arccos C > 11/15$. The entire prefix $[0, t]$ then has $v \le 1$; its support formula from $G(0) = 1 - h$, $G'(0) = C$ gives

$$\frac{G(t) - 1 - C \sin t}{\cos t} = 1 - h - \sec t + \frac{1}{\cos t} \int_0^t v(r) \sin(t - r)\, \mathrm{d}r \le -h < 0.$$

This contradiction shows $\Psi(-C) < 0$, hence $x_0 > -C$, contrary to $T = (C + x_0)_+ = 0$. Therefore $T > 0$, $x_0 = -C + T$ and $n_- = 0$.

The outer endpoint supplies a nonnegative second wall at $x = -2C + \sqrt{1 - e_L^2}$; the low middle point supplies one at $x = -C + \sqrt{h(2 - h)}$. Comparing each with the unique zero x_0 gives the two square-root bounds (if the test point is at least x_1, use $x_0 < x_1$). Finally $x_0 < x_1$ implies $T < 1 - C$. □

H.4. **The triangle payment above the radius width.** A dry low endpoint sharpens both endpoint heights. The resulting floor payment decreases with tilt, so its zero-tilt value bounds every height. This excludes widths above the boundary used by the initial energy test.

Lemma H.4. *No canonical full-turn maximum with a dry low middle endpoint has $C_{\mathrm{rad}} \le C < 37/50$ and $0 < h < 1/3$, where C_{rad} is defined in (C.10).*

Proof. Put $z = n_+$ and $n_- = 0$. Then $e_R = 1/2 + h/4 + 3z/4$ and $e_L = 1/2 - 3h/4 - z/4$, with $z < 1/3$. Use the lifted three-angle parameters of (C.5). Put $a = 2C$, $r = \sqrt{2}$, $k = r - 1$, $c_8 = \cos(\pi/8)$, $A = 1 - k$, $\gamma = 1 - r/2$, and

$t = 2c_8 - r$. The outer deficits satisfy $w_R \le 1/2 - h/4$ and $w_L < 7/12 + 3h/4$. The triangle and floor calculation gives

$$P_L(U) \le G_h(a, u, v) + L_{\mathrm{a}}(h),$$
$$L_{\mathrm{a}}(h) = \gamma[(D - h/4)^2 + (D + 1/12 + 3h/4)^2], \qquad D = 1/2 - t.$$

A signed square bounds its positive-part square even if the endpoint depth lies below the floor threshold. Thus this payment needs no height subdivision.

The triangles fit throughout the stated range. Indeed the pressure sum gives

$$m - h/2 \ge a - 1/2 - h/4 > a_{\mathrm{rad}} - 1/2 - (1/3)/4 > 3/4.$$

Together with $k > 2/5$ and $2\sin(\pi/8) > 3/4$, this places both inner crossings past the diagonal apex. Their outer zeros remain inside J, since $C < 4/5$. These are the same geometric checks as in Lemma C.5.

Use the shared supporting plane at the Newton comparison width a_{rad}. Its clipping and width-derivative inequalities follow from the Newton formulas in (C.8) and (C.10), throughout $h \le 1/3$. Consequently

$$G_h(a, u, v) \le B(a_{\mathrm{rad}}) - kh/2 - h^2/8 - \tfrac{2}{9}[s_0(a_{\mathrm{rad}}) - Ah/4]^2.$$

The dry-endpoint row of Lemma C.8 bounds the right side plus L_{a} at $h = 0$. Its value is the dry-endpoint row of Lemma C.9, which is below $\underline{p}$. This contradicts maximality. □

H.5. **The last small-height alternative.** If either initial-floor energy test fails, the width must increase. For a small tilt, the resulting gain in ordinary survivor area exceeds the losses from the tilt and the short leaked niche. We compare those quantities using chord lengths in a fixed unit direction.

Lemma H.5. *No canonical full-turn maximum has* $2/3 < C < 37/50$ *and* $0 < h \le h_0$*, with* h_0 *defined by* (C.9)*.*

Proof. Put $P = P_L(U)$ and $\mathcal{S} = S$, the ordinary survivor of Lemma H.1. By Theorem A.1,

$$|\mathcal{S}| \le G_0 := 2.2199. \tag{H.6}$$

Exact endpoint and leakage information. Write the top face as $[C, C+T] \times \{1\}$, and put

$$z = n(C), \qquad s = \sqrt{h(2-h)}, \quad K = \frac{C}{\sqrt{1 - C^2}},$$
$$J_F = \int_0^{\pi/2} (u(t) - 1)_+ \sin t \, \mathrm{d}t.$$

Lemma H.3 and the actual pressure equations give $T > 0$ and $n(-C) = 0$. The triangle comparison in Lemma H.4 also gives $C < C_{\mathrm{rad}}$.

$$e_R = \frac{1}{2} + \frac{h}{4} + \frac{3z}{4}, \qquad e_L = \frac{1}{2} - \frac{3h}{4} - \frac{z}{4}. \tag{H.7}$$

The pinned low middle point and the endpoint estimate (H.4) imply

$$T \le s, \qquad 0 \le z \le KT, \qquad K \le K_+ := \mathcal{N}_5(2)/2. \tag{H.8}$$

To see $T \le s$, use the signed second wall envelope at its zero $x_{\text{zero}} = -C+T$. The low middle point $(-C, 1-h)$ gives the lower wall $1 - h - \sec t + T\tan t$. Its value at $\sin t = T$ is $1 - h - \sqrt{1-T^2}$, and cannot be positive at an envelope zero. Thus $T^2 \le h(2-h)$. We have $T \le C < 1$, so this angle is proper. For the projection bound, the radius-width cut gives $9C^2 < 5$, so $C < \mathcal{N}_5(2)/3$ and $\sqrt{1-C^2} > 2/3$. Dividing gives exactly $K < K_+$; the same radius supplies both bounds.

The early-end proof bounds all possible first-source excess by one initial envelope

$$(u(\alpha+\tau)-1)_+ \le \left(\varepsilon - \frac{\tau}{2} - \frac{\tau^2}{4}\right)_+, \qquad \varepsilon < \frac{19}{100}.$$

Take $d_0 = 1/3$. The parabola with initial height $d_0/2 + d_0^2/4 > 19/100$ vanishes at d_0, so it bounds both the duration and the area of the excess:

$$\int (u-1)_+ \, dt < \int_0^{d_0} \left(\frac{d_0}{2} + \frac{d_0^2}{4} - \frac{\tau}{2} - \frac{\tau^2}{4}\right) d\tau = \frac{d_0^2}{4} + \frac{d_0^3}{6}.$$

All excess occurs before angle $11/15$, by the early-end estimate. Using $\sin t \le t$ therefore gives

$$J_F \le J_* := \frac{11}{15}\left(\frac{d_0^2}{4} + \frac{d_0^3}{6}\right). \tag{H.9}$$

Lemma H.2 and convexity of the actual niche on the right wing give

$$N_{\text{out}} \le \frac{(T + J_F)z}{2}. \tag{H.10}$$

The left wing contributes no outside niche: its niche roof is convex with zero values at both endpoints. The middle cap area is $2C - Ch$. Thus

$$|\mathcal{S}| \ge P + C(2-h) - \frac{(T+J_F)z}{2}. \tag{H.11}$$

A 3–4–5 lower bound for the wing lengths. The initial-energy test singles out the width

$$C_0 = \frac{\sqrt{17}}{6}.$$

The chord slope near this width is comparable to $1/(2C_0)$. The unit direction of a 3–4–5 triangle has a nearby slope $3/4$ and gives an affine lower bound for both chord lengths. Its coefficients are exact:

$$\lambda = \frac{4}{5}, \qquad m = \frac{3}{20}, \qquad a = \frac{1}{10}. \tag{H.12}$$

Put $S_0 = (14/5)C_0 + 3/10$.

The two charged-wing vertical drops are

$$r_R = 1 - e_R = \frac{1}{2} - \frac{h}{4} - \frac{3z}{4}, \qquad r_L = 1 - h - e_L = \frac{1}{2} - \frac{h}{4} + \frac{z}{4}.$$

The top overhang has length T, while the nonhorizontal right wing has horizontal displacement $C - T$. The stationary length identity and the two chord bounds yield

$$P \geq \frac{1}{2}\left[T + \sqrt{(C-T)^2 + r_R^2} + \sqrt{C^2 + r_L^2}\right]. \tag{H.13}$$

Cauchy–Schwarz in the chosen unit direction gives

$$\sqrt{x^2 + y^2} \geq \frac{4}{5}x + \frac{3}{5}y.$$

Applying it to both chords gives $P \geq 4C/5 + 3/10 - 3(h+z)/20 + T/10$. Together with (H.11), this becomes

$$|\mathcal{S}| \geq S_0 + \frac{14}{5}(C - C_0) - Ch - \frac{3}{20}(h+z) + \frac{T}{10} - \frac{(T + J_F)z}{2}. \tag{H.14}$$

Use $C \leq C_+ := \mathcal{N}_5(2)/3$, $T \geq z/K_+$, and $J_F \leq J_*$. Collecting the height and niche losses gives

$$D := C_+ + \frac{3}{20}, \qquad E := \frac{3}{20} + \frac{J_*}{2} - \frac{1/10}{K_+}. \tag{H.15}$$

Thus the common lower bound is

$$|\mathcal{S}| \geq S_0 + \frac{14}{5}(C - C_0) - Dh - Ez - \frac{Tz}{2}. \tag{H.16}$$

One Newton step from 4 gives an upper bound $\mathcal{N}_{17}(4)$ for $\sqrt{17}$, and its reciprocal gives the lower bound $17/\mathcal{N}_{17}(4)$. Hence

$$S_0 > \frac{14}{5}\frac{17}{6\mathcal{N}_{17}(4)} + \frac{3}{10} > G_0. \tag{H.17}$$

Both tests use these same area prices. The balance-height bracket $h_0 \leq h_+$ in (C.9) gives

$$\kappa = \frac{1}{2 - h_+}, \qquad r = \mathcal{N}_{h_+(2-h_+)}(1/3).$$

The identity $s^2 = h(2-h)$ gives $h \leq \kappa s^2$ and $0 < s < r$. These bounds are generated from the energy balance, using its canonical interval endpoint as the Newton seed. In particular $z \leq K_+ s$ and $h + z < 2/3$.

The floor-energy test. If the floor-energy test fails, completing its square forces an increase in C. That width gain pays for the height and niche losses in (H.16).

Put

$$d = 3C, \qquad B = e_R - 1 + h = -\frac{1}{2} + \frac{5h + 3z}{4}.$$

The second condition of Lemma C.11 is

$$Q := d^2 + B^2 + B(1-h) - ds \le 4.$$

Suppose it fails. The exact square completion is

$$Q = \left(3C - \frac{s}{2}\right)^2 + \frac{9}{16}(h+z)^2 - \frac{1}{4},$$

so

$$\left(3C - \frac{s}{2}\right)^2 + \frac{9}{16}(h+z)^2 > \frac{17}{4}. \tag{H.18}$$

Write $w = h+z$. The common bounds above give $w < 2/3$, and $3C - s/2 > 0$. Rationalizing the difference of the two positive square roots gives

$$\frac{\sqrt{17}}{2} - \sqrt{\frac{17}{4} - \frac{9}{16}w^2} = \frac{9w^2/16}{\sqrt{17}/2 + \sqrt{17/4 - 9w^2/16}} < \frac{9}{64}w^2.$$

The denominator exceeds four because $w < 2/3$. Therefore (H.18) implies

$$C - C_0 > \frac{s}{6} - \frac{3}{64}(h+z)^2. \tag{H.19}$$

Insert (H.19) into (H.16) and use $z \le K_+ s$ and $T \le s$. This gives

$$|\mathcal{S}| - S_0 \ge As - Dh - \frac{K_+}{2}s^2 - Q_0\left(h + K_+ s\right)^2, \tag{H.20}$$

where the coefficients come directly from the preceding bounds:

$$A = \frac{14/5}{6} - K_+ E, \qquad Q_0 = \frac{14}{5}\frac{3}{64}.$$

The common height and sine bounds give

$$h + K_+ s \le (K_+ + \kappa r)\, s.$$

All losses in (H.20) are therefore bounded by one multiple of s^2. Put

$$L_r = D\kappa + \frac{K_+}{2} + Q_0\left(K_+ + \kappa r\right)^2.$$

The area gain is at least $s(A - L_r s)$, whose bracket is smallest at $s = r$. Substituting the defining formulas gives

$$A - L_r r > 0. \tag{H.21}$$

Thus $|\mathcal{S}| > S_0 > G_0$, contradicting (H.6).

The radius failure gives a joint pressure budget. Suppose the other energy test fails:

$$9C^2 + B^2 > 5. \tag{H.22}$$

The pressure has the useful form

$$B = -\frac{1}{2} + \frac{5}{4}\left(h + \frac{3}{5}z\right).$$

The common bounds $h \le h_0$, $z \le K_+ r$ put $-1/2 < B < 0$. Recall that $b_0 = -1/2 + 5h_0/4$ and $9C_{\mathrm{rad}}^2 = 5 - b_0^2$. Since $C < C_{\mathrm{rad}}$, a failed radius test forces $B^2 > b_0^2$, hence $B < b_0$. The pressure equation therefore gives

$$\boxed{h + \frac{3}{5}z \le h_0.} \tag{H.23}$$

This is the available loss budget, with both terms retained together. From the defining area prices, $E + r/2 < (3/5)D$. Since $T \le r$, the whole loss obeys

$$Dh + Ez + Tz/2 \le D\left(h + \frac{3}{5}z\right) \le Dh_0. \tag{H.24}$$

Also $|B| < 1/2$, so (H.22) implies

$$C > C_1 := \frac{\sqrt{5 - 1/4}}{3} = \frac{\sqrt{19}}{6}. \tag{H.25}$$

This width gain pays for the pressure budget. Rationalizing the two circle radii gives

$$C_1 - C_0 = \frac{1}{3(\sqrt{19} + \sqrt{17})} > \frac{1}{27},$$

because each root is less than $9/2$. Finally $(14/5)/27 > Dh_0$, by the definitions of D, h_0. The shared area bound (H.16) now gives

$$|\mathcal{S}| > S_0 + \frac{14}{5}(C_1 - C_0) - Dh_0 > S_0 > G_0. \tag{H.26}$$

This contradicts (H.6). No separate bound on z or subdivision of the small-height interval is needed. Both energy conditions in Lemma C.11 therefore hold. That lemma gives $u \le 1$ on the whole first quarter, and Proposition 6.2 excludes this tilted maximum. □

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